ALMA Protoplanetary Disk Targets

Sorted summary of all processed targets

Generated: 2026-10-26 11:11
Source folder: /Volumes/SSD2/MyJulia/Alma/alma_plots
Targets: 81 (detailed pages: 42, filter :useful, max 200)
Sorted by: scientific value (resolved disks first, then significant features, then size in beams)
Pipeline: alma_target_explorer_julia

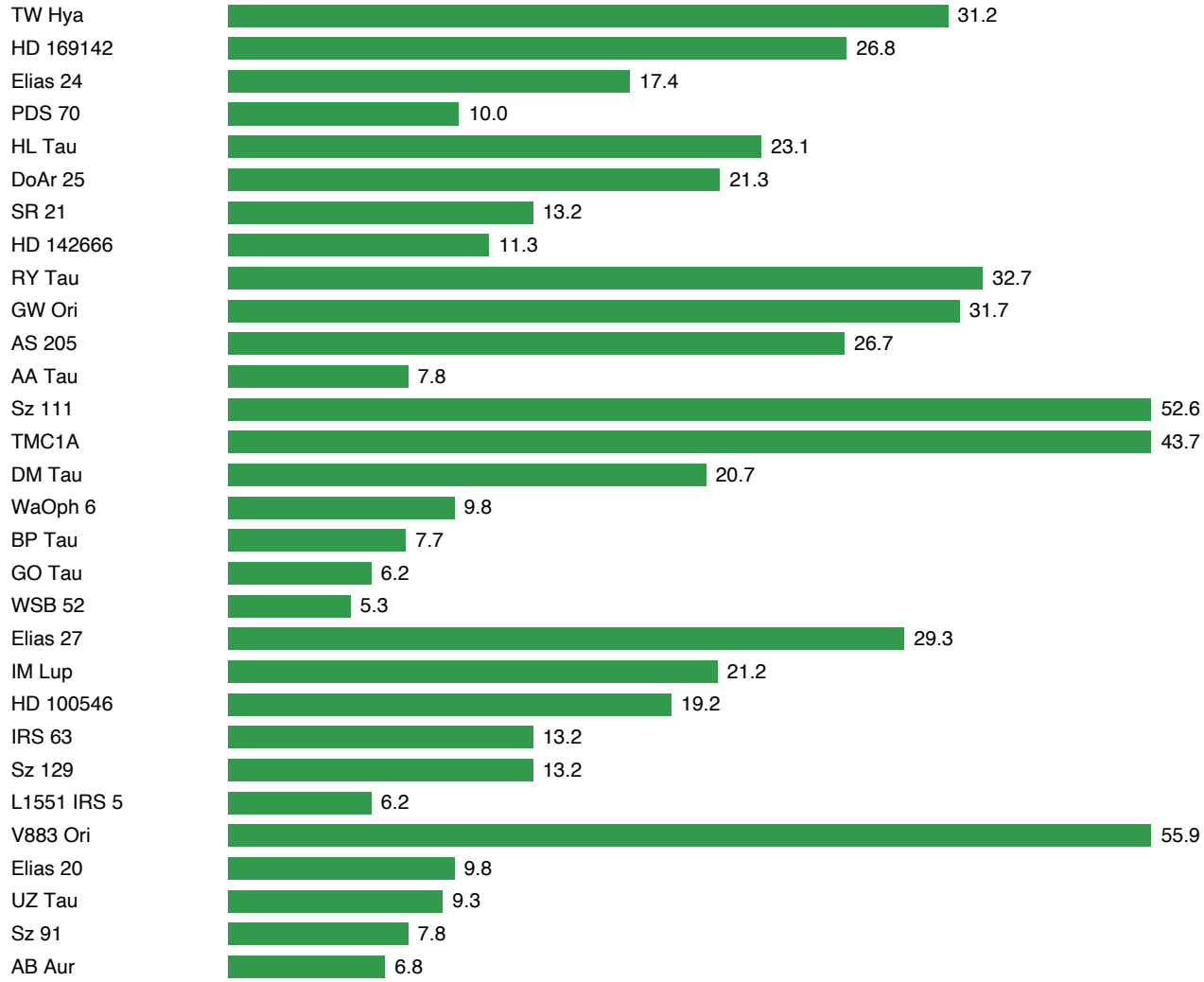
1. Overview (sorted)

Status: Resolved = 90% of the flux spans >= 5 beams; Barely resolved = 2-5 beams; Unresolved = < 2 beams or flagged as a point source; Wrong image? = the FITS file name does not match the target. Sig.R / Sig.G = rings / gaps detected at >= 3 sigma. Pub.rec = published features recovered / testable.

#	Target	d[pc]	Band	Beam[au]	R50[au]	R90[au]	Status	Sig.R	Sig.G	Pub.rec	Chk
1	TW Hya	60	7	2x1	34	64	Resolved	2	4	3/5	
2	HD 169142	114	7	4x4	57	112	Resolved	2	4	1/4	
3	Elias 24	136	6	7x4	35	116	Resolved	3	3	4/4	
4	PDS 70	112	7	9x8	74	86	Resolved	3	3	–	
5	HL Tau	140	7	4x4	59	104	Resolved	2	3	8/13	
6	DoAr 25	138	6	6x4	65	135	Resolved	2	3	4/4	!
7	SR 21	138	6	5x4	56	64	Resolved	2	3	–	
8	HD 142666	148	6	5x3	32	53	Resolved	2	3	4/6	
9	RY Tau	138	6	5x3	47	167	Resolved	2	2	3/3	
10	GW Ori	402	6	10x8	164	311	Resolved	2	2	–	
11	AS 205	128	6	5x3	33	130	Resolved	2	2	–	
12	AA Tau	136	6	13x8	58	105	Resolved	2	2	2/3	
13	Sz 111	158	6	5x4	66	254	Resolved	1	2	–	
14	TMC1A	140	6	5x4	22	223	Resolved	2	1	–	
15	DM Tau	145	6	4x3	32	81	Resolved	1	2	–	
16	WaOph 6	123	6	9x4	42	87	Resolved	2	1	0/2	
17	BP Tau	128	6	4x3	24	35	Resolved	1	2	–	
18	GO Tau	144	6	18x13	40	112	Resolved	1	2	4/4	
19	WSB 52	135	6	6x4	19	32	Resolved	2	1	–	
20	Elias 27	116	6	5x3	52	140	Resolved	1	1	1/2	
21	IM Lup	158	6	5x4	32	102	Resolved	1	1	0/1	!
22	HD 100546	108	7	5x3	34	96	Resolved	1	1	–	
23	IRS 63	144	6	6x4	37	79	Resolved	1	1	–	!
24	Sz 129	160	6	4x3	26	48	Resolved	0	2	2/3	
25	L1551 IRS 5	140	4	8x6	33	49	Resolved	1	1	–	
26	V883 Ori	388	6	12x10	68	655	Resolved	1	0	–	
27	Elias 20	138	6	5x4	27	50	Resolved	1	0	2/4	
28	UZ Tau	131	4	8x5	39	76	Resolved	0	1	1/2	
29	Sz 91	159	4	14x11	86	107	Resolved	1	0	–	
30	AB Aur	156	6	16x12	85	109	Resolved	1	0	0/1	
31	HD 135344B	135	4	10x6	53	94	Resolved	0	0	–	
32	MWC 480	162	3	24x15	61	166	Resolved	0	0	2/2	
33	V1094 Sco	153	8	40x38	134	256	Resolved	0	0	–	
34	IQ Tau	131	6	19x12	49	95	Resolved	0	0	0/2	
35	DoAr 33	140	6	5x3	17	25	Barely resolved	2	1	–	
36	T Cha	107	3	13x7	35	56	Barely resolved	1	1	–	
37	LkCa 15	157	6	45x30	95	159	Barely resolved	0	1	–	!
38	XZ Tau	140	6	8x4	22	26	Barely resolved	0	1	–	
39	FT Tau	127	6	16x14	29	48	Barely resolved	0	1	–	
40	RX J1852.3-3	146	7	27x21	55	77	Barely resolved	0	1	–	
41	EX Lup	155	4	9x4	21	25	Barely resolved	1	0	–	
42	WL 17	137	7	15x14	17	31	Barely resolved	1	0	–	
43	GM Aur	159	7	48x31	91	195	Barely resolved	0	0	–	
44	HD 34282	309	7	59x40	147	224	Barely resolved	0	0	–	
45	DL Tau	159	6	47x31	91	163	Barely resolved	0	0	–	
46	HD 97048	185	6	73x32	146	257	Barely resolved	0	0	–	!
47	L1489 IRS	140	6	33x22	52	108	Barely resolved	0	0	–	
48	GY 91	140	3	14x12	23	43	Barely resolved	0	0	–	
49	DN Tau	128	6	21x16	35	61	Barely resolved	0	0	0/1	
50	V4046 Sgr	72	7	21x12	39	61	Barely resolved	0	0	–	
51	IRS 43	137	6	34x24	24	93	Barely resolved	0	0	–	
52	TWA 7	34	7	4x4	6	11	Barely resolved	0	0	–	
53	HD 100453	104	6	23x14	34	55	Barely resolved	0	0	–	
54	HK Tau	133	4	16x5	17	37	Barely resolved	0	0	–	!
55	IP Tau	130	6	18x14	29	42	Barely resolved	0	0	–	
56	CY Tau	128	6	30x20	35	67	Barely resolved	0	0	–	

57	Elias 29	137	7	15x14	10	17	Unresolved	1	0	–	
58	CQ Tau	162	6	44x38	61	85	Unresolved	0	0	–	
59	V892 Tau	134	6	31x22	37	57	Unresolved	0	0	–	
60	DR Tau	195	7	45x36	34	73	Unresolved	0	0	–	
61	HD 143006	165	6	80x64	65	108	Unresolved	0	0	–	
62	DH Tau	135	6	27x12	17	30	Unresolved	0	0	–	
63	HD 145718	152	6	132x109	72	125	Unresolved	0	0	–	!
64	1RXS J172919	63	7	50x48	27	42	Unresolved	0	0	–	
65	2M1207	64	7	22x19	11	18	Unresolved	0	0	–	
66	HD 142527	157	7	28x20	135	300	Wrong image?	1	6	–	!
67	V1247 Ori	400	7	19x12	184	344	Wrong image?	2	1	–	!
68	DG Tau	121	6	5x4	25	51	Wrong image?	1	1	–	!
69	L1527 IRS	140	4	17x5	32	74	Wrong image?	1	0	–	!
70	IRAS 04302+2	160	4	14x6	122	212	Wrong image?	0	0	–	!
71	GG Tau A	150	7	22x15	229	289	Wrong image?	0	0	–	!
72	SR 24S	114	3	10x9	42	60	Wrong image?	0	0	–	!
73	RX J1604.3-2	150	6	36x30	94	124	Wrong image?	0	0	–	!
74	MWC 758	160	7	38x25	76	108	Wrong image?	0	0	–	!
75	RX J1615.3-3	156	9	43x31	58	113	Wrong image?	0	0	–	!
76	Oph IRS 48	134	3	13x11	14	26	Wrong image?	0	0	–	!
77	SU Aur	158	7	36x23	28	51	Wrong image?	0	0	–	!
78	UX Tau A	140	3	39x15	38	52	Wrong image?	0	0	–	!
79	Sz 114	162	3	64x47	45	77	Wrong image?	0	0	–	!
80	Sz 98	156	3	62x45	43	74	Wrong image?	0	0	–	!
81	HD 141569	111	7	47x38	31	47	Wrong image?	0	0	–	!

Disk size: radius holding 90% of the flux, in beams (capped at 40) - top 40 of 81 targets



HD 135344B
8.9
MWC 480
6.9
V1094 Sco
6.4
IQ Tau
5.0
DoAr 33
4.8
T Cha
4.4
LkCa 15
3.6
XZ Tau
3.3
FT Tau
2.9
RX J1852.3-370
2.9
Barely resolved
Wrong image?
Unresolved
Resolved

2. Statistics

Targets	81
Detailed pages	42 (filter :useful, max 200)
Resolved	34
Barely resolved	22
Unresolved	9
Wrong image?	16
ALMA bands used	Band 3: 8, Band 4: 8, Band 6: 42, Band 7: 21, Band 8: 1, Band 9: 1
Median beam (usable)	9.4 au
Total >= 3 sigma rings	56
Total >= 3 sigma gaps	67
Coarser fallback used	46
Negative bowl (filtering)	22

Data-integrity flags (check before using these results)

- DoAr 25: Catalogue entry is for 'ODISEA_C4_039', not 'DoAr 25': published values (R68, flux, M_dust, ring radii) may belong to another source.

- IM Lup: Catalogue entry is for 'Sz82', not 'IM Lup': published values (R68, flux, M_dust, ring radii) may belong to another source.

- IRS 63: Catalogue entry is for 'ODISEA_C4_130', not 'IRS 63': published values (R68, flux, M_dust, ring radii) may belong to another source.

- LkCa 15: Catalogue entry is for 'J04391779+2221034', not 'LkCa 15': published values (R68, flux, M_dust, ring radii) may belong to another source.

- HD 97048: Catalogue entry is for 'J11080329-7739174', not 'HD 97048': published values (R68, flux, M_dust, ring radii) may belong to another source.

- HK Tau: Catalogue entry is for 'HKTauA', not 'HK Tau': published values (R68, flux, M_dust, ring radii) may belong to another source.

- HD 145718: Catalogue entry is for 'J16131158-2229066', not 'HD 145718': published values (R68, flux, M_dust, ring radii) may belong to another source.

- HD 142527: FITS image 'member.uid___A002_X7368a9_X8.cont.clean.pbcor.fits' does not appear to contain the target name 'HD 142527'. The package holds several sources and the wrong one may have been picked.

- V1247 Ori: FITS image 'member.uid___A001_X2de_Xfd.target_spw0_mfsb7.cl01.pbcor.fits' does not appear to contain the target name 'V1247 Ori'. The package holds several sources and the wrong one may have been picked.

- DG Tau: FITS image 'member.uid___A001_X2d6_X2b2.calibrated.ms.cont.clean.pbcor.fits' does not appear to contain the target name 'DG Tau'. The package holds several sources and the wrong one may have been picked.

- L1527 IRS: FITS image 'member.uid___A001_X1465_X4f0.L1527_sci.spw25_27_29_31.cont.I.tt0.pbcor.fits' does not appear to contain the target name 'L1527 IRS'. The package holds several sources and the wrong one may have been picked.

- IRAS 04302+2247: FITS image 'member.uid___A001_X126d_X104.uid___A002_Xc4d618_X6812.IRAS_04302p2247_sci.spw0_1_2_3.cont.I.manual.image.pbcor.fits' does not appear to contain the target name 'IRAS 04302+2247'. The package holds several sources and the wrong one may have been picked.

- GG Tau A: FITS image 'member.uid___A001_X340_X205.ari_I.GG_Tau_sci.spw0_1_2_3_4_5_6_7_8_336058MHz.12m.cont.I.pbcor.fits' does not appear to contain the target name 'GG Tau A'. The package holds several sources and the wrong one may have been picked.

- SR 24S: FITS image 'member.uid___A001_X1296_X7b1.EM_star_SR_24A_sci.spw19.mfs.I.manual.cont.image.tt0.pbcor.fits' does not appear to contain the target name 'SR 24S'. The package holds several sources and the wrong one may have been picked.

- RX J1604.3-2130A: FITS image 'member.uid___A001_X147_Xc4.ari_I._rx_j1604.3-2130__sci.spw0_1_2_3_227063MHz.12m.cont.I.pbcor.fits' does not appear to contain the target name 'RX J1604.3-2130A'. The package holds several sources and the wrong one may have been picked.

- MWC 758: FITS image 'member.uid___A002_X5d7935_X306.calibrated_final_cont.pbcor.fits' does not appear to contain the target name 'MWC 758'. The package holds several sources and the wrong one may have been picked.

- RX J1615.3-3255: FITS image 'RXJ_Xd6.cal.AVG.CONTIN.image.fits' does not appear to contain the target name 'RX J1615.3-3255'. The package holds several sources and the wrong one may have been picked.

- Oph IRS 48: FITS image 'member.uid___A001_X2d2_Xd6.calibrated_final_cont_IRS_48.pbcor.fits' does not appear to contain the target name 'Oph IRS 48'. The package holds several sources and the wrong one may have been picked.

- SU Aur: FITS image 'member.uid___A001_X13c_X4c.calibrated_final_cont_CX_TAU.pbcor.fits' does not appear to contain the target name 'SU Aur'. The package holds several sources and the wrong one may have been picked.

- UX Tau A: FITS image 'member.uid___A001_X892_X33.ari_l.UXTau_sci.spw0_1_2_3_102496MHz.12m.cont.I.tt0.pbcor.fits' does not appear to contain the target name 'UX Tau A'. The package holds several sources and the wrong one may have been picked.

- Sz 114: FITS image 'member.uid___A001_X87d_X5d7.MY_Lup_sci.spw25_27_29_31.cont.I.tt0.pbcor.fits' does not appear to contain the target name 'Sz 114'. The package holds several sources and the wrong one may have been picked.

- Sz 98: FITS image 'member.uid___A001_X87d_X5d7.MY_Lup_sci.spw25_27_29_31.cont.I.tt0.pbcor.fits' does not appear to contain the target name 'Sz 98'. The package holds several sources and the wrong one may have been picked.

- HD 141569: FITS image 'member.uid___A002_X684eb5_X23f.cont.im.image.fits' does not appear to contain the target name 'HD 141569'. The package holds several sources and the wrong one may have been picked.

Targets that need a finer / different observation

- LkCa 15: Barely resolved; beam 44.7 au (finest archive resolution 0.0281 arcsec).
- XZ Tau: Barely resolved; beam 8.1 au (finest archive resolution 0.0290 arcsec).
- FT Tau: Barely resolved; beam 16.4 au (finest archive resolution 0.0246 arcsec).
- RX J1852.3-3700: Barely resolved; beam 26.8 au (finest archive resolution 0.0908 arcsec).
- EX Lup: Barely resolved; beam 9.1 au (finest archive resolution 0.0390 arcsec).
- WL 17: Barely resolved; beam 14.9 au (finest archive resolution 0.0310 arcsec).
- GM Aur: Barely resolved; beam 47.5 au (finest archive resolution 0.0243 arcsec).
- HD 34282: Barely resolved; beam 59.0 au (finest archive resolution 0.0432 arcsec).
- DL Tau: Barely resolved; beam 47.0 au (finest archive resolution 0.1056 arcsec).
- HD 97048: Barely resolved; beam 72.8 au (finest archive resolution 0.0237 arcsec).
- L1489 IRS: Barely resolved; beam 33.2 au (finest archive resolution 0.0398 arcsec).
- DN Tau: Barely resolved; beam 21.0 au (finest archive resolution 0.0167 arcsec).
- V4046 Sgr: Barely resolved; beam 20.8 au (finest archive resolution 0.0470 arcsec).
- IRS 43: Barely resolved; beam 34.3 au (finest archive resolution 0.0458 arcsec).
- HD 100453: Barely resolved; beam 22.7 au (finest archive resolution 0.0209 arcsec).
- CY Tau: Barely resolved; beam 29.5 au (finest archive resolution 0.0419 arcsec).
- Elias 29: Unresolved; beam 15.0 au (finest archive resolution 0.0344 arcsec).
- CQ Tau: Unresolved; beam 44.3 au (finest archive resolution 0.0563 arcsec).
- V892 Tau: Unresolved; beam 31.0 au (finest archive resolution 0.0614 arcsec).
- DR Tau: Unresolved; beam 44.6 au (finest archive resolution 0.0224 arcsec).
- HD 143006: Unresolved; beam 80.3 au (finest archive resolution 0.0257 arcsec).
- DH Tau: Unresolved; beam 26.8 au (finest archive resolution 0.0723 arcsec).
- HD 145718: Unresolved; beam 131.7 au (finest archive resolution 0.7284 arcsec).
- 1RXS J172919.1-501454: Unresolved; beam 50.2 au (finest archive resolution 0.8109 arcsec).
- 2M1207: Unresolved; beam 22.2 au (finest archive resolution 0.0698 arcsec).

... and 16 more (see ALMA_flagged_targets.csv).

1. TW Hya

Resolved

Well resolved (90% of the flux within 31.2 beams of 2.0 au). 2 ring(s) [34, 44 au] and 4 gap(s) [8, 26, 42, 60 au] detected at >= 3 sigma. Recovered 3 of 5 testable published features.

Target

Coordinates (J2000)	11:01:51.91 -34:42:17.0 (RA 165.46627, Dec -34.70473 deg, J2000)
Distance	60.1 pc
Class	T Tauri analogue
Versus TW Hya	Reference: nearest, face-on, ~8-10 Myr, compact

Observation used

Project / band	2015.1.00686.S / Band 7 (350.9 GHz)
Beam	0.0341 x 0.0236 arcsec = 2.0 x 1.4 au
Archive resolution	0.0163 arcsec (1.0 au
Largest scale	0.38 arcsec (23 au)
Exposure / release	7910 s (2.20 h) / 2017-01-16
Package	2015.1.00686.S_uid___A001_X2d6_X18_001_of_001.tar (7.25 GB)
FITS image	member.uid___A001_X2d6_X18.ari_l.TW_Hya_sci.spw0_1_2_3_350892MHz.12m.cont.I.pbcor.fits
Archive observations	1 total; finest 0.0163 arcsec; used #1
Stellar parameters	Stellar parameters (Andrews+12 / Teague (approx.); verify): L* = 0.28 Lsun, M* = 0.80 Msun

Geometry and radial profile

Inclination / PA	7.0 / 155.0 deg
Geometry source	catalogue (literature) values.
Peak of mean profile	2.06 mJy/beam at r = 2.6 au
R50 / R90	34 / 64 au (16.7 / 31.2 beams)
Outer edge (>= 3 sigma)	91 au
Integrated flux	1468.8 mJy (n/a of catalogue expectation)
Image noise / dyn. range	0.0593 mJy/beam / 41
Shape	the emission is centrally peaked (the maximum lies at 3 au, within ~1.5 beams of the star), like TW Hya: a smooth, steeply falling profile on which any rings are shoulders and dips.

Substructure (baseline-subtracted profile)

type	radius	residual		modulation	significance	
gap	r= 7.7 au	resid=-0.0562	mJy/b	modul=-3.8%	signif= -5.5 sigma	SIGNIFICANT
ring	r= 17.9 au	resid=0.0277	mJy/b	modul=+4.0%	signif= +2.8 sigma	noise-level
gap	r= 25.7 au	resid=-0.107	mJy/b	modul=-24.6%	signif= -21.4 sigma	SIGNIFICANT
ring	r= 33.5 au	resid=0.072	mJy/b	modul=+17.4%	signif= +9.7 sigma	SIGNIFICANT
gap	r= 41.9 au	resid=-0.0644	mJy/b	modul=-17.5%	signif= -17.8 sigma	SIGNIFICANT
ring	r= 44.5 au	resid=0.0519	mJy/b	modul=+16.0%	signif= +8.1 sigma	SIGNIFICANT
gap	r= 59.9 au	resid=-0.0234	mJy/b	modul=-31.8%	signif= -5.9 sigma	SIGNIFICANT

Sequence	G 8 au -> R 18 au -> G 26 au -> R 34 au -> G 42 au -> R 45 au -> G 60 au
Published features	recovered 3 of 5 testable

Gaussian ring fit

Model / chi2_red tapered power law + 5 Gaussian ring(s); chi2_red = 2.09

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed	
1	r=2.8+/-0.4	amp=1.18+/-0.5	S/N=2.4	FWHM_int=4.7 +/- 1.5	core	<3sig
2	r=8.7+/-0.6	amp=0.856+/-0.15	S/N=5.8	FWHM_int=7.9 +/- 3.7	-	
3	r=16.5+/-2.3	amp=0.418+/-0.21	S/N=2.0	FWHM_int=8.7 +/- 2.8	det	<3sig
4	r=33.6+/-0.7	amp=0.282+/-0.029	S/N=9.8	FWHM_int=11.6 +/- 1.4	det	
5	r=44.9+/-1.3	amp=0.161+/-0.024	S/N=6.7	FWHM_int=12.3 +/- 1.7	det	

Brightness temperature and optical depth

type	radius	brightness T	dust T	ratio	optical depth	scale height
gap	r= 7.7 au	T_b= 25.1 K	T_d= 32.7 K	T_b/T_d= 0.77	tau= 1.45	H/r=0.036
ring	r= 8.7 au	T_b= 24.6 K	T_d= 30.7 K	T_b/T_d= 0.80	tau= 1.61	H/r=0.037
gap	r= 25.7 au	T_b= 10.4 K	T_d= 17.7 K	T_b/T_d= 0.59	tau= 0.88	H/r=0.048
ring	r= 33.6 au	T_b= 12.6 K	T_d= 15.7 K	T_b/T_d= 0.80	tau= 1.62	H/r=0.051

```
gap   r=  41.9 au   T_b=  10.2 K   T_d=  13.9 K   T_b/T_d= 0.73   tau= 1.31   H/r=0.055
ring  r=  44.9 au   T_b=  11.0 K   T_d=  13.6 K   T_b/T_d= 0.81   tau= 1.64   H/r=0.055
```

Azimuthal asymmetry

```
ring radius     m=1 amplitude                 m=2 amplitude
r=    8.7 au   m=1:    6.1% (S/N 6.6 )   m=2:   12.6% (S/N 13.7)
r=   33.6 au   m=1:   10.8% (S/N 9.2 )   m=2:   43.8% (S/N 37.1)
r=   44.9 au   m=1:   32.0% (S/N 18.4)   m=2:   32.0% (S/N 18.4)
```

Gap-opening planet masses

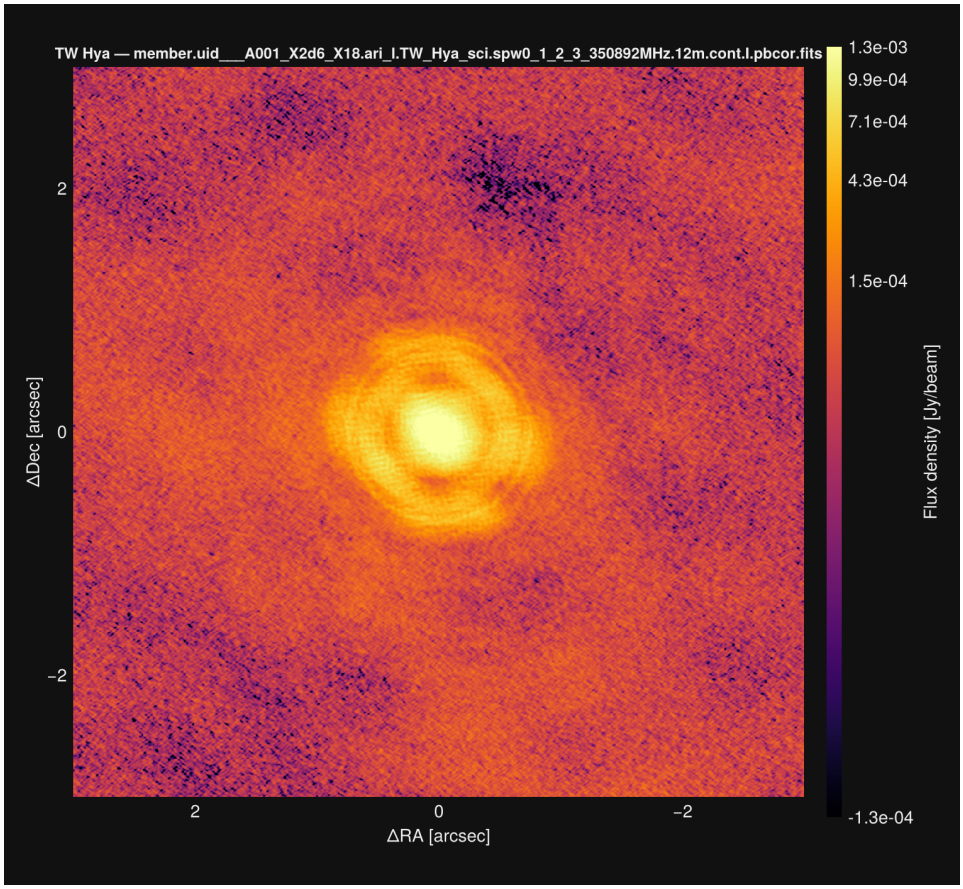
```
gap summary
gap r_c=25.5 au  h/r=0.048  depth=1.44 |5.3 5.0 0.207 | - - | 0.83 Mnep 0.26 Mnep shallow (< 2.2): width relation for
gap r_c=42.0 au  h/r=0.054  depth=1.16 |2.9 2.3 0.068 | - - | 0.69 Mnep 0.22 Mnep asymmetric gap: half-widths 1.8 au
```

Not evaluated: 8 au: no local maximum outward of the gap minimum at 8.7 au (the profile does not rise there)

Pipeline warnings and notes

- the feature(s) at 9, 45 au are shoulders (the raw-profile peak finder finds no peak or dip within 1 beam): an amplitude measured against a shoulder baseline can be large without a real azimuthal structure; the percentages of the local flux are the safer numbers.

Continuum image



2. HD 169142

Resolved

Well resolved (90% of the flux within 26.8 beams of 4.2 au). 2 ring(s) [26, 45 au] and 4 gap(s) [14, 34, 50, 71 au] detected at >= 3 sigma. Recovered 1 of 4 testable published features. Image holds 92-139% of the expected catalogue flux (consistent). A finer observation (0.0296 arcsec) exists but the package exceeded the download limit; finer structure may exist.

Target

Coordinates (J2000)	18:24:29.78 -29:46:49.4 (RA 276.12410, Dec -29.78040 deg, J2000)
Distance	114.0 pc
Class	Herbig
Versus TW Hya	Herbig, but near face-on with rings: closest of its group; Bosschaart+26: RD, R68 67 au
Catalogue	Bosschaart+26 (HD169142): class RD (ring disc); R68 67 au; F(1.33 mm) 197 mJy; M_dust 27 Earth masses; M* 2.00 Msun; distance 114 pc; this image: 68%-flux radius 78 au

Observation used

Project / band	2015.1.00806.S / Band 7 (338.2 GHz)
Beam	0.0365 x 0.0308 arcsec = 4.2 x 3.5 au
Archive resolution	0.0307 arcsec (3.5 au
Largest scale	0.43 arcsec (49 au)
Exposure / release	1615 s (0.45 h) / 2019-02-22
Package	2015.1.00806.S_uid___A001_X2de_X159_001_of_001.tar (1.28 GB)
FITS image	member.uid___A001_X2de_X159.ari_l.hd_169142_sci.spw0_1_2_3_338185MHz.12m.cont.l.pbcor.fits
Archive observations	30 total; finest 0.0296 arcsec; used #2 (fallback: finer packages too large)
Stellar parameters	Stellar parameters (A&A 2023 aa43601-22 Table 1 (d = 114 pc); Fedele+17 give L* = 10, M* = 1.65; VERIFY; verify): L* = 8.00 Lsun, M* = 1.65 Msun

Geometry and radial profile

Inclination / PA	13.0 / 5.0 deg
Geometry source	catalogue (literature) values.
Peak of mean profile	1.73 mJy/beam at r = 26.1 au
R50 / R90	57 / 112 au (13.8 / 26.8 beams)
Outer edge (>= 3 sigma)	132 au
Integrated flux	614.5 mJy (92-139% of catalogue expectation)
Image noise / dyn. range	0.146 mJy/beam / 14
Shape	the profile is ring-like, not centrally peaked. The maximum is at 26 au and the innermost bin has only 48% of it, which points to a central cavity: the mean rises to half its peak at ~20 au.

Substructure (baseline-subtracted profile)

type	radius	residual	modulation	significance	
gap	r= 14.3 au	resid=-0.253 mJy/b	modul=-29.3%	signif= -8.0 sigma	SIGNIFICANT
ring	r= 25.7 au	resid=0.674 mJy/b	modul=+63.6%	signif= +20.8 sigma	SIGNIFICANT
gap	r= 34.4 au	resid=-0.336 mJy/b	modul=-49.0%	signif= -14.6 sigma	SIGNIFICANT
ring	r= 44.8 au	resid=0.0757 mJy/b	modul=+75.0%	signif= +3.1 sigma	SIGNIFICANT
gap	r= 49.6 au	resid=-0.0406 mJy/b	modul=-27.4%	signif= -4.2 sigma	SIGNIFICANT
ring	r= 63.8 au	resid=0.0601 mJy/b	modul=+28.8%	signif= +2.9 sigma	noise-level
gap	r= 71.0 au	resid=-0.0604 mJy/b	modul=-34.2%	signif= -4.2 sigma	SIGNIFICANT

Sequence	G 14 au -> R 26 au -> G 34 au -> R 45 au -> G 50 au -> R 64 au -> G 71 au
Published features	recovered 1 of 4 testable

Gaussian ring fit

Model / chi2_red tapered power law + 5 Gaussian ring(s); chi2_red = 0.97

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed	
1	r=11.2+/-1.1	amp=0.502+/-0.16	S/N=3.1	FWHM_int=14.6 +/- 4.8	det	
2	r=25.5+/-0.2	amp=1.44+/-0.087	S/N=16.5	FWHM_int=8.7 +/- 0.5	det	
3	r=40.7+/-1.1	amp=0.0871+/-0.047	S/N=1.8	FWHM_int=< 12.4 (unresolved)	det	<3sig
4	r=61.4+/-0.7	amp=0.116+/-0.018	S/N=6.3	FWHM_int=9.1 +/- 2.3	det	
5	r=80.2+/-17.1	amp=0.0836+/-0.056	S/N=1.5	FWHM_int=70.2 +/- 29.2	lit	<3sig

Brightness temperature and optical depth

type	radius	brightness T	dust T	ratio	optical depth	scale height
ring	r= 11.2 au	T_b= 14.0 K	T_d= 61.8 K	T_b/T_d= 0.23	tau= 0.26	H/r=0.042
gap	r= 14.3 au	T_b= 13.4 K	T_d= 56.9 K	T_b/T_d= 0.24	tau= 0.27	H/r=0.043
ring	r= 25.5 au	T_b= 23.7 K	T_d= 41.0 K	T_b/T_d= 0.58	tau= 0.86	H/r=0.051
gap	r= 34.4 au	T_b= 9.2 K	T_d= 35.7 K	T_b/T_d= 0.26	tau= 0.30	H/r=0.055
ring	r= 61.4 au	T_b= 7.7 K	T_d= 26.7 K	T_b/T_d= 0.29	tau= 0.34	H/r=0.063

Azimuthal asymmetry

ring radius	m=1 amplitude	m=2 amplitude
r= 11.2 au	m=1: 24.0% (S/N 4.3)	m=2: 6.5% (S/N 1.2)
r= 25.5 au	m=1: 9.0% (S/N 5.5)	m=2: 2.7% (S/N 1.6)
r= 44.8 au	m=1: 10.6% (S/N 0.3)	m=2: 40.8% (S/N 1.1)
r= 61.4 au	m=1: 33.7% (S/N 2.6)	m=2: 12.2% (S/N 0.9)

Gap-opening planet masses

gap summary
gap r_c=13.6 au h/r=0.043 depth=2.39 |16.1 15.6 1.184 | 3.9–4.1 Mjup 1.2–1.3 Mjup | 2.4 Mnep 0.74 Mnep inner ring p

Not evaluated: 34 au: no local maximum outward of the gap minimum at 34.4 au (the profile does not rise there)

Pipeline warnings and notes

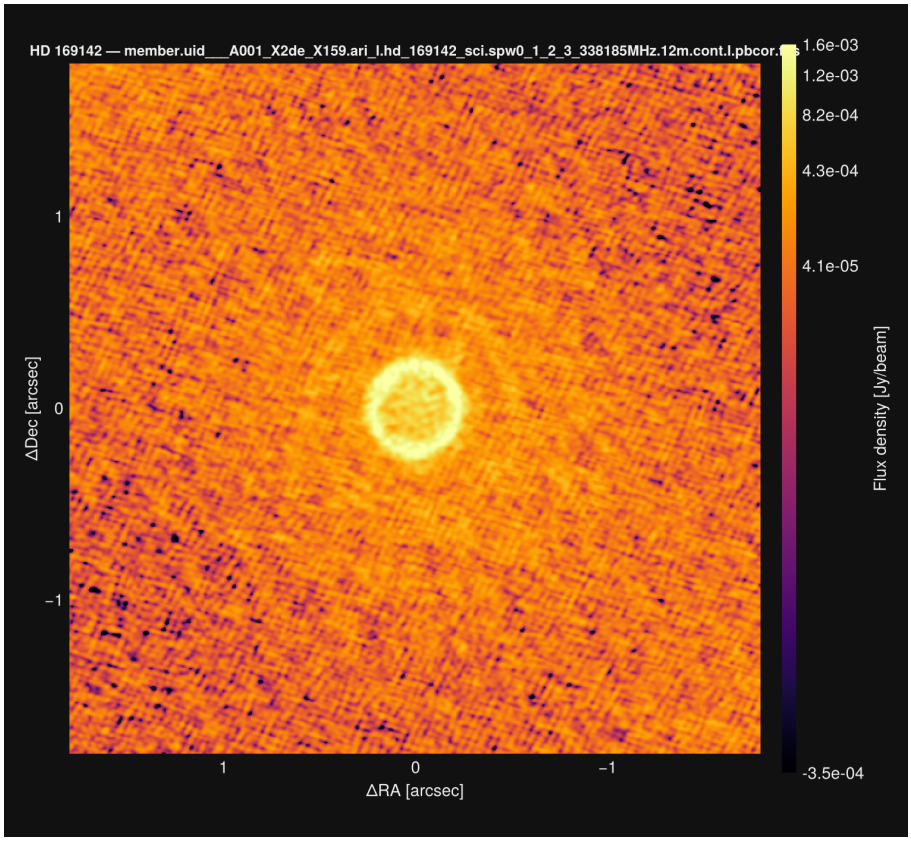
- 1 ring(s) detected in the profile at 3 sigma or more (45 au) are covered only by a fitted component with amplitude S/N < 3; the fit is otherwise consistent with the profile, so it is used, and for the asymmetry and temperature steps those rings are taken from the detection (position and residual amplitude) instead of the weak component.

- at 45, 61 au the local azimuthal-mean flux is below 2x the image rms per beam (0.146 mJy/beam): the ring is detected only by averaging over many beams, and the S/N and false-alarm values above assume white noise at the beam scale. Correlated residuals (sidelobes, large-scale noise, weak background sources) are not in that noise and can make a dipole or m=2 pattern of this size, so these asymmetries are unconfirmed; a...

- the feature(s) at 11, 45, 61 au are shoulders (the raw-profile peak finder finds no peak or dip within 1 beam): an amplitude measured against a shoulder baseline can be large without a real azimuthal structure; the percentages of the local flux are the safer numbers.

- the notebook fell back from observation #1 to #2 because the finer packages exceeded the size limit.

Continuum image



3. Elias 24

Resolved

Well resolved (90% of the flux within 17.4 beams of 6.7 au). 3 ring(s) [22, 39, 77 au] and 3 gap(s) [32, 58, 90 au] detected at >= 3 sigma. Recovered 4 of 4 testable published features. Image holds 43-46% of the expected catalogue flux. Negative bowl present: missing short spacings bias sizes, contrasts and temperatures.

Target

Coordinates (J2000)	16:26:24.07 -24:16:13.4 (RA 246.60030, Dec -24.27040 deg, J2000)
Distance	136.0 pc
Class	T Tauri analogue
Versus TW Hya	Closest analogue: axisymmetric rings/gaps; Bosschaart+26: RD, R68 136 au
Catalogue	Bosschaart+26 (Elias24): class RD (ring disc); R68 136 au; F(1.33 mm) 361 mJy; M_dust 70 Earth masses; M* 0.78 Msun; distance 136 pc; this image: 68%-flux radius 75 au

Observation used

Project / band	2016.1.00484.L / Band 6 (238.9 GHz)
Beam	0.0490 x 0.0290 arcsec = 6.7 x 3.9 au
Archive resolution	0.0222 arcsec (3.0 au
Largest scale	0.64 arcsec (87 au)
Exposure / release	4064 s (1.13 h) / 2018-10-16
Package	2016.1.00484.L_uid___A001_X8c5_X4e_001_of_001.tar (0.49 GB)
FITS image	member.uid___A001_X8c5_X4e.ari_l.Elias_24_sci.spw0_1_2_3_238891MHz.12m.cont.l.pbcor.fits
Archive observations	3 total; finest 0.0222 arcsec; used #1
Stellar parameters	Stellar parameters (Andrews+18 (approx.); verify): L* = 6.00 Lsun, M* = 0.78 Msun

Geometry and radial profile

Inclination / PA	29.0 / 45.7 deg
Geometry source	catalogue (literature) values.
Peak of mean profile	3.87 mJy/beam at r = 1.7 au
R50 / R90	35 / 116 au (5.3 / 17.4 beams)
Outer edge (>= 3 sigma)	49 au
Integrated flux	184.6 mJy (43-46% of catalogue expectation)
Image noise / dyn. range	0.0243 mJy/beam / 45
Shape	the emission is centrally peaked (the maximum lies at 2 au, within ~1.5 beams of the star), like TW Hya: a smooth, steeply falling profile on which any rings are shoulders and dips.

Substructure (baseline-subtracted profile)

type	radius	residual	modulation	significance	
ring	r= 21.8 au	resid=0.168 mJy/b	modul=+21.4%	signif= +25.1 sigma	SIGNIFICANT
gap	r= 31.9 au	resid=0.0759 mJy/b	modul=+12.8%	signif= -11.5 sigma	SIGNIFICANT
ring	r= 38.6 au	resid=0.139 mJy/b	modul=+32.1%	signif= +27.8 sigma	SIGNIFICANT
gap	r= 57.7 au	resid=-0.111 mJy/b	modul=n/a	signif= -66.7 sigma	SIGNIFICANT
ring	r= 76.6 au	resid=0.157 mJy/b	modul=+184.8%	signif= +43.9 sigma	SIGNIFICANT
gap	r= 89.8 au	resid=-0.0581 mJy/b	modul=-57.7%	signif= -21.1 sigma	SIGNIFICANT

Sequence	R 22 au -> G 32 au -> R 39 au -> G 58 au -> R 77 au -> G 90 au
Published features	recovered 4 of 4 testable

Gaussian ring fit

Model / chi2_red tapered power law + 4 Gaussian ring(s); chi2_red = 75.47

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed	
1	r=24.0+/-12.5	amp=0.746+/-1.4	S/N=0.5	FWHM_int=< 50.6 (unresolved)	det	<3sig
2	r=39.2+/-2.8	amp=0.464+/-0.5	S/N=0.9	FWHM_int=8.7 +/- 5.6	det	<3sig
3	r=77.0+/-0.6	amp=0.247+/-0.041	S/N=6.1	FWHM_int=8.1 +/- 1.7	det	
4	r=116.3+/-4.2	amp=0.0581+/-0.012	S/N=5.0	FWHM_int=41.4 +/- 13.0	lit	

Brightness temperature and optical depth

type	radius	brightness T	dust T	ratio	optical depth	scale height
gap	r= 31.9 au	T_b= 15.1 K	T_d= 34.5 K	T_b/T_d= 0.44	tau= 0.58	H/r=0.075
gap	r= 57.7 au	T_b= 3.5 K	T_d= 27.9 K	T_b/T_d= 0.12	tau= 0.13	H/r=0.084
ring	r= 77.0 au	T_b= 8.1 K	T_d= 22.4 K	T_b/T_d= 0.36	tau= 0.45	H/r=0.094

gap	r= 89.8 au	T_b= 3.9 K	T_d= 20.7 K	T_b/T_d= 0.19	tau= 0.21	H/r=0.098
ring	r= 116.3 au	T_b= 4.2 K	T_d= 18.1 K	T_b/T_d= 0.23	tau= 0.26	H/r=0.104

Azimuthal asymmetry

ring radius	m=1 amplitude	m=2 amplitude
r= 21.8 au	m=1: 10.3% (S/N 2.4)	m=2: 3.7% (S/N 0.9)
r= 38.6 au	m=1: 7.0% (S/N 1.8)	m=2: 8.3% (S/N 2.1)
r= 77.0 au	m=1: 2.1% (S/N 1.6)	m=2: 18.4% (S/N 14.2)
r= 116.3 au	m=1: 6.2% (S/N 2.9)	m=2: 24.0% (S/N 11.2)

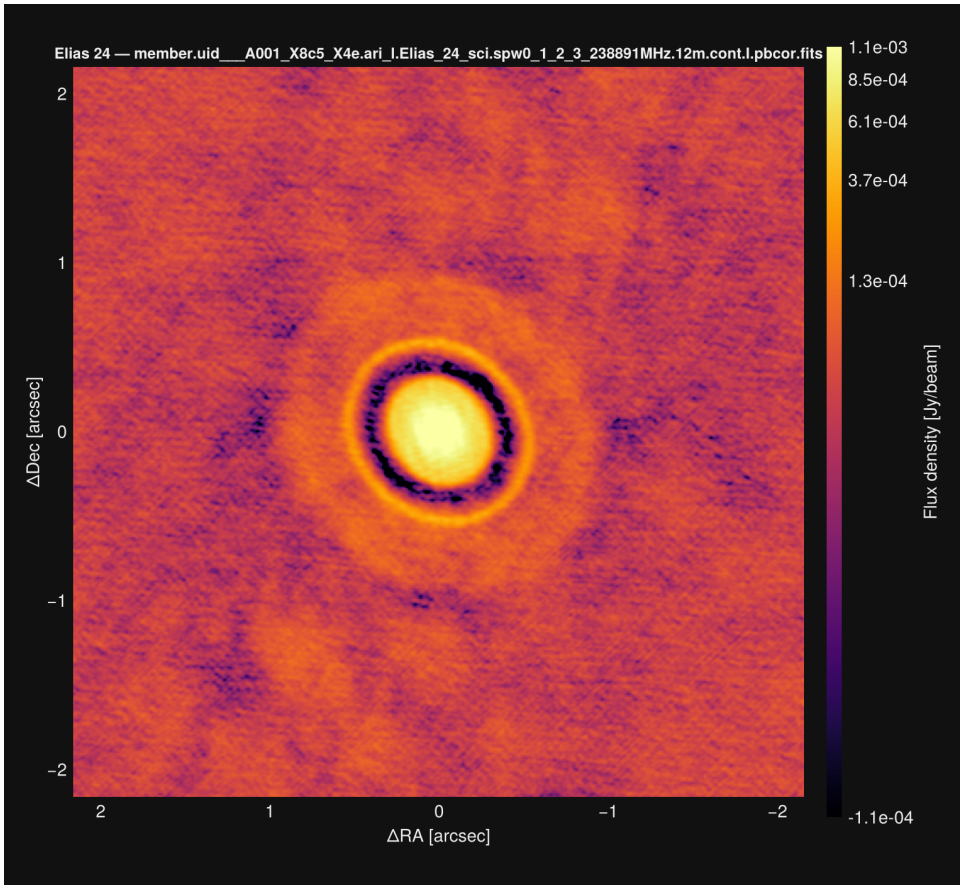
Gap-opening planet masses

Not evaluated: 32 au: no local maximum outward of the gap minimum at 31.9 au (the profile does not rise there); 90 au: gap contrast only 0.7 sigma (< 3.0)

Pipeline warnings and notes

- the smooth disk collapsed to the unresolved core (characteristic radius 14 au < 3 beams): the extended emission is shared among the ring components, so overlapping neighbours (shoulders) have degenerate widths and amplitudes; trust the isolated rings, not the blended ones.
- 2 ring(s) detected in the profile at 3 sigma or more (22, 39 au) are covered only by a fitted component with amplitude S/N < 3; the fit is otherwise consistent with the profile, so it is used, and for the asymmetry and temperature steps those rings are taken from the detection (position and residual amplitude) instead of the weak component.
- the feature(s) at 22, 39, 116 au are shoulders (the raw-profile peak finder finds no peak or dip within 1 beam): an amplitude measured against a shoulder baseline can be large without a real azimuthal structure; the percentages of the local flux are the safer numbers.
- the image is spatially filtered (flux check above): arc contrasts and asymmetry amplitudes describe only the emission that survived the missing short spacings.
- Negative bowl: the azimuthal mean is significantly negative over r = 143-200 au (18 bins, down to -10.2 sigma, -0.0256 mJy/beam = -0.7% of the peak, at 153 au), which real emission cannot produce. It is the signature of missing short spacings (spatial filtering): the extended emission is partly resolved out, so sizes, contrasts, asymmetries and brightness temperatures are biased and the smooth baseline is underestima...

Continuum image



4. PDS 70

Resolved

Well resolved (90% of the flux within 10.0 beams of 8.6 au). 3 ring(s) [32, 52, 73 au] and 3 gap(s) [44, 65, 90 au] detected at >= 3 sigma. Image holds 19-28% of the expected catalogue flux. Negative bowl present: missing short spacings bias sizes, contrasts and temperatures. A finer observation (0.0219 arcsec) exists but the package exceeded the download limit; finer structure may exist.

Target

Coordinates (J2000)	14:08:10.15 -41:23:52.4 (RA 212.04230, Dec -41.39790 deg, J2000)
Distance	112.0 pc
Class	T Tauri + cavity
Versus TW Hya	Ring, not centrally peaked; gaps = cavity physics; Bosschaart+26: TD, R68 91 au
Catalogue	Bosschaart+26 (PDS70): class TD (transition disc); R68 91 au; F(1.33 mm) 230 mJy; M_dust 12 Earth masses; M* 0.76 Msun; distance 113 pc; this image: 68%-flux radius 78 au

Observation used

Project / band	2022.1.01477.S / Band 7 (343.5 GHz)
Beam	0.0769 x 0.0670 arcsec = 8.6 x 7.5 au
Archive resolution	0.0674 arcsec (7.5 au
Largest scale	1.20 arcsec (134 au)
Exposure / release	21392 s (5.94 h) / 2025-08-20
Package	2022.1.01477.S_uid___A001_X2d1f_X187_001_of_001.tar (0.11 GB)
FITS image	member.uid___A001_X2d1f_X187.PDS70pol_sci.spw17_19_21_23.mfs.IQUV.manual.pbcor.fits
Archive observations	34 total; finest 0.0219 arcsec; used #8 (fallback: finer packages too large)
Stellar parameters	Stellar parameters (Bosschaart+26 catalogue stellar mass; verify): L* = ? Lsun, M* = 0.76 Msun

Geometry and radial profile

Inclination / PA	49.7 / 158.6 deg
Geometry source	catalogue (literature) values.
Peak of mean profile	1.38 mJy/beam at r = 73.5 au
R50 / R90	74 / 86 au (8.5 / 10.0 beams)
Outer edge (>= 3 sigma)	103 au
Integrated flux	150.6 mJy (19-28% of catalogue expectation)
Image noise / dyn. range	0.0146 mJy/beam / 46
Shape	the profile is ring-like, not centrally peaked. The maximum is at 74 au and the innermost bin has only 9% of it, which points to a central cavity: the mean rises to half its peak at ~57 au.

Substructure (baseline-subtracted profile)

type	radius	residual	modulation	significance	
ring	r= 31.5 au	resid=0.0162 mJy/b	modul=n/a	signif= +3.0 sigma	SIGNIFICANT
gap	r= 44.1 au	resid=-0.0476 mJy/b	modul=-33.0%	signif= -14.0 sigma	SIGNIFICANT
ring	r= 52.5 au	resid=0.0404 mJy/b	modul=+8.0%	signif= +9.9 sigma	SIGNIFICANT
gap	r= 65.1 au	resid=-0.0623 mJy/b	modul=-5.4%	signif= -27.4 sigma	SIGNIFICANT
ring	r= 73.0 au	resid=0.098 mJy/b	modul=+7.6%	signif= +27.6 sigma	SIGNIFICANT
gap	r= 90.3 au	resid=-0.0623 mJy/b	modul=-16.5%	signif= -22.9 sigma	SIGNIFICANT

Sequence	R 32 au -> G 44 au -> R 52 au -> G 65 au -> R 73 au -> G 90 au
Published features	none stored in the catalogue

Gaussian ring fit

Model / chi2_red tapered power law + 3 Gaussian ring(s); chi2_red = 12.01

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed	
1	r=2.8+/-5.1	amp=0.125+/-0.044	S/N=2.8	FWHM_int=< 18.6 (unresolved)	core	<3sig
2	r=54.8+/-1.2	amp=0.39+/-0.091	S/N=4.3	FWHM_int=10.5 +/- 2.5	det	
3	r=72.9+/-0.8	amp=1.37+/-0.053	S/N=26.1	FWHM_int=22.8 +/- 1.1	det	

Brightness temperature and optical depth

T_d, T_b/T_d and tau not computed: no stellar luminosity stored for this target.

Azimuthal asymmetry

ring radius	m=1 amplitude	m=2 amplitude
r= 54.8 au	m=1: 25.1% (S/N 28.4)	m=2: 15.0% (S/N 17.0)
r= 72.9 au	m=1: 8.1% (S/N 49.1)	m=2: 5.6% (S/N 33.9)

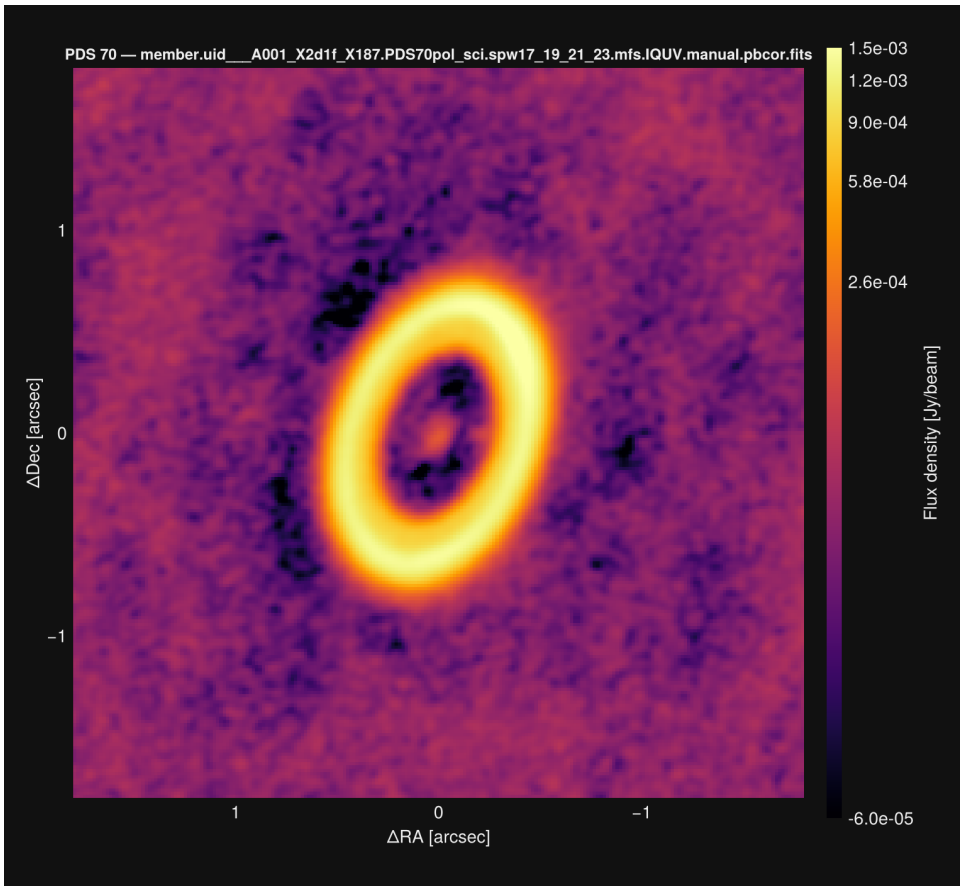
Gap-opening planet masses

Not available (needs L* and M*, which are not stored for this target; pass lstar= and mstar=).

Pipeline warnings and notes

- the feature(s) at 55 au are shoulders (the raw-profile peak finder finds no peak or dip within 1 beam): an amplitude measured against a shoulder baseline can be large without a real azimuthal structure; the percentages of the local flux are the safer numbers.
- the image is spatially filtered (flux check above): arc contrasts and asymmetry amplitudes describe only the emission that survived the missing short spacings.
- Negative bowl: the azimuthal mean is significantly negative over r = 111-166 au (14 bins, down to -9.2 sigma, -0.0251 mJy/beam = -1.8% of the peak, at 124 au), which real emission cannot produce. It is the signature of missing short spacings (spatial filtering): the extended emission is partly resolved out, so sizes, contrasts, asymmetries and brightness temperatures are biased and the smooth baseline is underestimat...
- the notebook fell back from observation #1 to #8 because the finer packages exceeded the size limit.
- gap at 44 au and ring at 52 au are 8.4 au apart (< 1 beam): probably one blended feature or noise.
- gap at 65 au and ring at 73 au are 7.9 au apart (< 1 beam): probably one blended feature or noise.

Continuum image



5. HL Tau

Resolved

Well resolved (90% of the flux within 23.1 beams of 4.5 au). 2 ring(s) [20, 46 au] and 3 gap(s) [13, 33, 65 au] detected at >= 3 sigma. Recovered 8 of 13 testable published features. Negative bowl present: missing short spacings bias sizes, contrasts and temperatures. A finer observation (0.0094 arcsec) exists but the package exceeded the download limit; finer structure may exist.

Target

Coordinates (J2000)	04:31:38.44 +18:13:57.7 (RA 67.91017, Dec +18.23269 deg, J2000)
Distance	140.0 pc
Class	Young/embedded
Versus TW Hya	Class I/II, remnant envelope; younger; inclined

Observation used

Project / band	2019.1.01051.S / Band 7 (343.5 GHz)
Beam	0.0321 x 0.0279 arcsec = 4.5 x 3.9 au
Archive resolution	0.0232 arcsec (3.2 au
Largest scale	0.80 arcsec (113 au)
Exposure / release	20115 s (5.59 h) / 2023-07-26
Package	2019.1.01051.S_uid___A001_X1465_X1bea_001_of_001.tar (0.95 GB)
FITS image	member.uid___A001_X1465_X1bea.HL_Tau_sci.spw13_15_17_19.mfs.IQUV.manual.pbcor.fits
Archive observations	66 total; finest 0.0094 arcsec; used #3 (fallback: finer packages too large)
Stellar parameters	Stellar parameters (arXiv:1804.03076 (L = 8.3 Lsun after Ruge+16 at 130 pc, about 9.6 at 140 pc; M = 1.3 Msun, literature range 0.55-1.7); VERIFY; verify): L* = 8.30 Lsun, M* = 1.30 Msun

Geometry and radial profile

Inclination / PA	46.7 / 138.0 deg
Geometry source	catalogue (literature) values.
Peak of mean profile	11.6 mJy/beam at r = 1.1 au
R50 / R90	59 / 104 au (13.2 / 23.1 beams)
Outer edge (>= 3 sigma)	129 au
Integrated flux	1445.9 mJy (n/a of catalogue expectation)
Image noise / dyn. range	0.0411 mJy/beam / 34
Shape	the emission is centrally peaked (the maximum lies at 1 au, within ~1.5 beams of the star), like TW Hya: a smooth, steeply falling profile on which any rings are shoulders and dips.

Substructure (baseline-subtracted profile)

type	radius	residual	modulation	significance	
gap	r= 13.4 au	resid=-1.58 mJy/b	modul=-39.3%	signif=-139.4 sigma	SIGNIFICANT
ring	r= 19.9 au	resid=1.21 mJy/b	modul=+48.1%	signif= +90.1 sigma	SIGNIFICANT
gap	r= 32.7 au	resid=-0.46 mJy/b	modul=-19.4%	signif= -64.9 sigma	SIGNIFICANT
ring	r= 45.9 au	resid=0.211 mJy/b	modul=+11.9%	signif= +24.1 sigma	SIGNIFICANT
gap	r= 65.2 au	resid=-0.188 mJy/b	modul=-19.3%	signif= -48.6 sigma	SIGNIFICANT

Sequence	G 13 au -> R 20 au -> G 33 au -> R 46 au -> G 65 au
Published features	recovered 8 of 13 testable

Gaussian ring fit

Model / chi2_red tapered power law + 3 Gaussian ring(s); chi2_red = 9.89

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed
1	r=20.6+/-0.5	amp=3.1+/-0.19	S/N=16.6	FWHM_int=13.0 +/- 1.4	det
2	r=41.4+/-1.0	amp=2.1+/-0.06	S/N=35.0	FWHM_int=28.5 +/- 2.1	det
3	r=85.1+/-0.9	amp=0.95+/-0.02	S/N=47.6	FWHM_int=43.8 +/- 1.3	lit

Brightness temperature and optical depth

type	radius	brightness T	dust T	ratio	optical depth	scale height
gap	r= 13.4 au	T_b= 35.8 K	T_d= 60.2 K	T_b/T_d= 0.60	tau= 0.90	H/r=0.048
gap	r= 32.7 au	T_b= 29.6 K	T_d= 37.1 K	T_b/T_d= 0.80	tau= 1.60	H/r=0.061
ring	r= 41.4 au	T_b= 31.7 K	T_d= 32.8 K	T_b/T_d= 0.96	tau= 3.35	H/r=0.065
gap	r= 65.2 au	T_b= 16.6 K	T_d= 26.0 K	T_b/T_d= 0.64	tau= 1.01	H/r=0.073
ring	r= 85.1 au	T_b= 18.9 K	T_d= 23.1 K	T_b/T_d= 0.82	tau= 1.72	H/r=0.078

Azimuthal asymmetry

ring radius	m=1 amplitude	m=2 amplitude
r= 20.6 au	m=1: 8.5% (S/N 32.5)	m=2: 1.1% (S/N 4.2)
r= 41.4 au	m=1: 6.9% (S/N 36.9)	m=2: 2.2% (S/N 11.6)
r= 85.1 au	m=1: 7.7% (S/N 33.0)	m=2: 1.5% (S/N 6.3)

Gap-opening planet masses

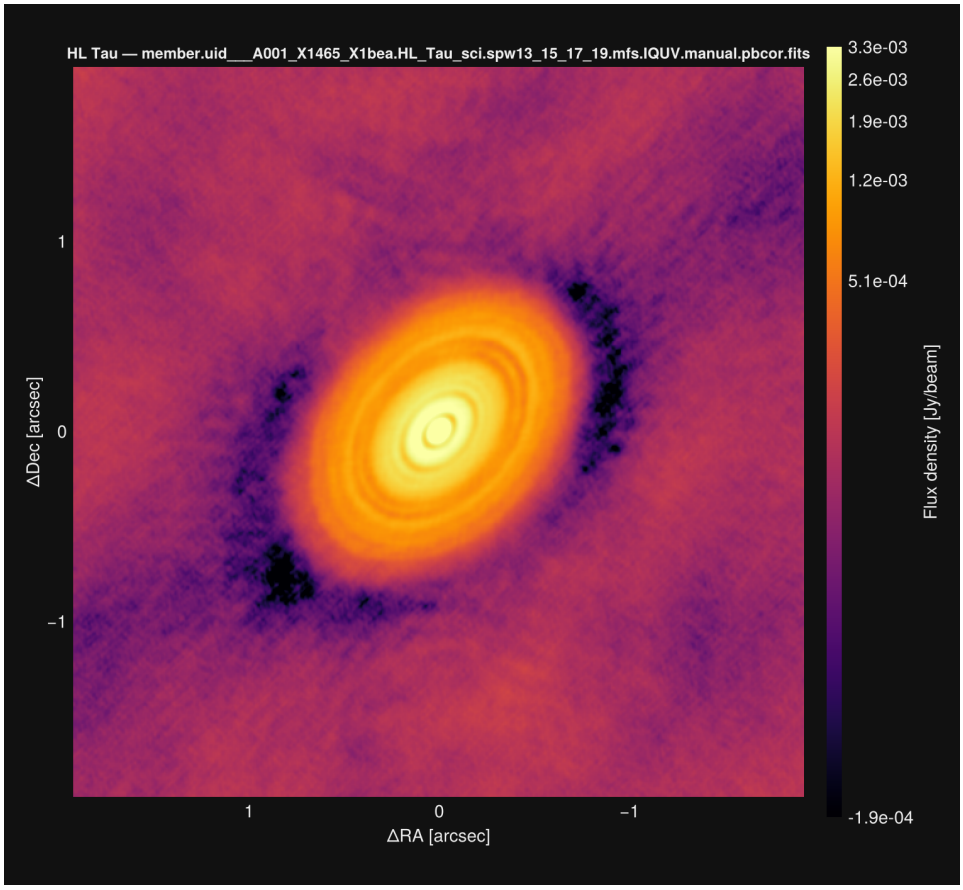
gap summary					
gap r_c=13.7 au	h/r=0.049	depth=1.47	|5.0 2.8 0.368	- - |	1.5 Mnep 0.47 Mnep asymmetric gap: half-widths 1.1 au i
gap r_c=32.8 au	h/r=0.061	depth=1.18	|4.2 0.0 0.128	- - |	1.6 Mnep 0.5 Mnep shallow (< 2.2): width relation forma

Not evaluated: 65 au: gap contrast only 2.4 sigma (< 3.0)

Pipeline warnings and notes

- the smooth disk collapsed to the unresolved core (characteristic radius 8 au < 3 beams): the extended emission is shared among the ring components, so overlapping neighbours (shoulders) have degenerate widths and amplitudes; trust the isolated rings, not the blended ones.
- Negative bowl: the azimuthal mean is significantly negative over r = 133-223 au (41 bins, down to -24.5 sigma, -0.116 mJy/beam = -1.0% of the peak, at 153 au), which real emission cannot produce. It is the signature of missing short spacings (spatial filtering): the extended emission is partly resolved out, so sizes, contrasts, asymmetries and brightness temperatures are biased and the smooth baseline is underestimat...
- the notebook fell back from observation #1 to #3 because the finer packages exceeded the size limit.

Continuum image



6. DoAr 25

Resolved

DATA CHECK: Catalogue entry is for 'ODISEA_C4_039', not 'DoAr 25': published values (R68, flux, M_dust, ring radii) may belong to another source.

Well resolved (90% of the flux within 21.3 beams of 6.3 au). 2 ring(s) [112, 134 au] and 3 gap(s) [75, 98, 127 au] detected at >= 3 sigma. Recovered 4 of 4 testable published features. Image holds 62-66% of the expected catalogue flux. Negative bowl present: missing short spacings bias sizes, contrasts and temperatures.

Target

Coordinates (J2000)	16:26:23.69 -24:43:13.9 (RA 246.59871, Dec -24.72052 deg, J2000)
Distance	138.0 pc
Class	T Tauri analogue
Versus TW Hya	Highly inclined; less suited to deprojection; Bosschaart+26: RD, R68 51 au
Catalogue	Bosschaart+26 (ODISEA_C4_039): class RD (ring disc); R68 51 au; F(1.33 mm) 239 mJy; M_dust 48 Earth masses; M* 0.95 Msun; distance 138 pc; this image: 68%-flux radius 97 au

Observation used

Project / band	2016.1.00484.L / Band 6 (239.0 GHz)
Beam	0.0460 x 0.0279 arcsec = 6.3 x 3.8 au
Archive resolution	0.0327 arcsec (4.5 au
Largest scale	0.65 arcsec (89 au)
Exposure / release	2032 s (0.56 h) / 2019-02-16
Package	2016.1.00484.L_uid___A001_X8c5_X56_001_of_001.tar (0.48 GB)
FITS image	member.uid___A001_X8c5_X56.DoAr_25_sci.spw0_1_2_3.mfs.I.manual.image.pbcor.fits
Archive observations	6 total; finest 0.0327 arcsec; used #1
Stellar parameters	Stellar parameters (Andrews+18 (approx.); verify): L* = 0.95 Lsun, M* = 0.95 Msun

Geometry and radial profile

Inclination / PA	67.4 / 110.6 deg
Geometry source	catalogue (literature) values.
Peak of mean profile	1.54 mJy/beam at r = 1.6 au
R50 / R90	65 / 135 au (10.3 / 21.3 beams)
Outer edge (>= 3 sigma)	167 au
Integrated flux	176.5 mJy (62-66% of catalogue expectation)
Image noise / dyn. range	0.021 mJy/beam / 125
Shape	the emission is centrally peaked (the maximum lies at 2 au, within ~1.5 beams of the star), like TW Hya: a smooth, steeply falling profile on which any rings are shoulders and dips.

Substructure (baseline-subtracted profile)

type	radius	residual	modulation	significance	
ring	r= 23.9 au	resid=0.0172 mJy/b	modul=+2.7%	signif= +2.1 sigma	noise-level
gap	r= 30.3 au	resid=-0.00797 mJy/b	modul=-1.4%	signif= -2.8 sigma	noise-level
ring	r= 36.6 au	resid=0.00764 mJy/b	modul=+1.6%	signif= +1.2 sigma	noise-level
gap	r= 52.5 au	resid=-0.00751 mJy/b	modul=-2.0%	signif= -2.8 sigma	noise-level
ring	r= 65.3 au	resid=0.0124 mJy/b	modul=+4.5%	signif= +2.5 sigma	noise-level
gap	r= 74.8 au	resid=-0.01 mJy/b	modul=-5.0%	signif= -4.6 sigma	SIGNIFICANT
ring	r= 87.6 au	resid=0.0113 mJy/b	modul=+11.7%	signif= +2.7 sigma	noise-level
gap	r= 97.5 au	resid=-0.0281 mJy/b	modul=-33.8%	signif= -11.2 sigma	SIGNIFICANT
ring	r= 112.3 au	resid=0.0347 mJy/b	modul=+30.1%	signif= +9.3 sigma	SIGNIFICANT
gap	r= 127.1 au	resid=-0.0221 mJy/b	modul=-21.6%	signif= -9.1 sigma	SIGNIFICANT
ring	r= 134.3 au	resid=0.0103 mJy/b	modul=+13.5%	signif= +3.0 sigma	SIGNIFICANT

Sequence	R 24 au -> G 30 au -> R 37 au -> G 53 au -> R 65 au -> G 75 au -> R 88 au -> G 98 au -> R 112 au -> G 127 au -> R 134 au
Published features	recovered 4 of 4 testable

Gaussian ring fit

Model / chi2_red	tapered power law + 5 Gaussian ring(s); chi2_red = 0.73

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed	
1	r=43.0+/-5.9	amp=0.0856+/-0.093	S/N=0.9	FWHM_int=19.0 +/- 10.8	det	<3sig
2	r=63.5+/-4.4	amp=0.143+/-0.036	S/N=3.9	FWHM_int=24.7 +/- 13.8	det	
3	r=85.6+/-3.2	amp=0.0466+/-0.047	S/N=1.0	FWHM_int=15.4 +/- 8.2	det	<3sig
4	r=111.4+/-0.4	amp=0.111+/-0.011	S/N=10.0	FWHM_int=13.9 +/- 1.2	det	
5	r=133.8+/-1.1	amp=0.0799+/-0.005	S/N=15.9	FWHM_int=34.2 +/- 2.0	det	

Brightness temperature and optical depth

type	radius	brightness T	dust T	ratio	optical depth	scale height
ring	r= 63.5 au	T_b= 9.9 K	T_d= 15.6 K	T_b/T_d= 0.63	tau= 1.00	H/r=0.064
gap	r= 97.5 au	T_b= 4.4 K	T_d= 12.5 K	T_b/T_d= 0.35	tau= 0.44	H/r=0.072
ring	r= 111.4 au	T_b= 6.6 K	T_d= 11.7 K	T_b/T_d= 0.56	tau= 0.82	H/r=0.074
gap	r= 127.1 au	T_b= 5.1 K	T_d= 11.0 K	T_b/T_d= 0.46	tau= 0.62	H/r=0.077
ring	r= 133.8 au	T_b= 5.2 K	T_d= 10.6 K	T_b/T_d= 0.49	tau= 0.68	H/r=0.078

Azimuthal asymmetry

ring radius	m=1 amplitude	m=2 amplitude
r= 63.5 au	m=1: 6.1% (S/N 3.2)	m=2: 16.0% (S/N 8.4)
r= 111.4 au	m=1: 6.5% (S/N 2.7)	m=2: 14.3% (S/N 5.9)
r= 133.8 au	m=1: 3.1% (S/N 1.6)	m=2: 13.9% (S/N 6.9)

Gap-opening planet masses

gap summary					
gap r_c=96.8 au	h/r=0.072	depth=2.72	16.8 16.0 0.173	1.8–2 Mnep 0.58–0.63 Mnep	5.3 Mnep 1.7 Mnep

Not evaluated: 127 au: gap contrast only 1.2 sigma (< 3.0)

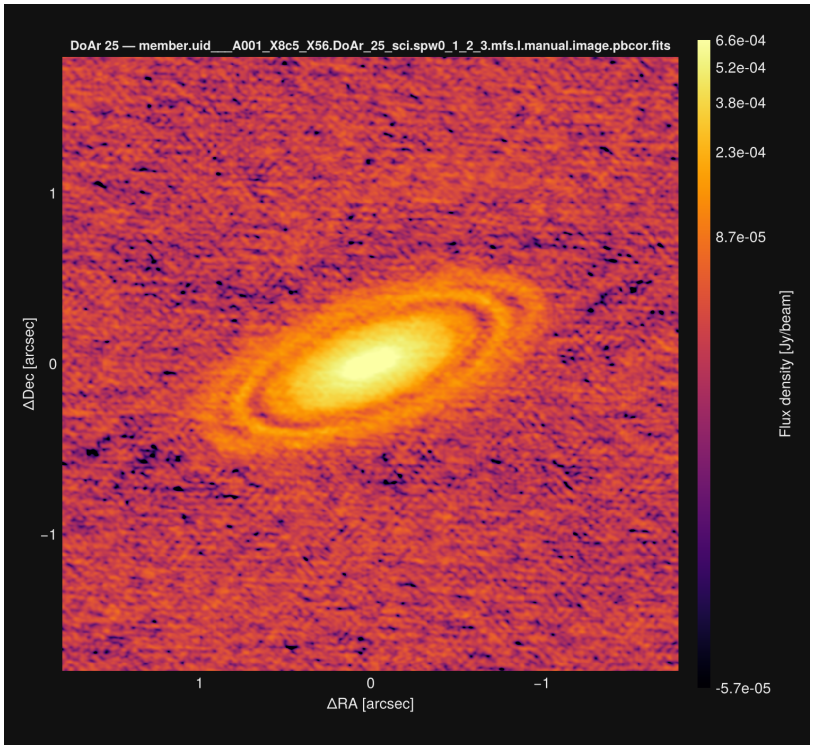
Pipeline warnings and notes

- the feature(s) at 63, 134 au are shoulders (the raw-profile peak finder finds no peak or dip within 1 beam): an amplitude measured against a shoulder baseline can be large without a real azimuthal structure; the percentages of the local flux are the safer numbers.

- the image is spatially filtered (flux check above): arc contrasts and asymmetry amplitudes describe only the emission that survived the missing short spacings.

- Negative bowl: the azimuthal mean is significantly negative over r = 202-205 au (2 bins, down to -3.9 sigma, -0.0108 mJy/beam = -0.7% of the peak, at 205 au), which real emission cannot produce. It is the signature of missing short spacings (spatial filtering): the extended emission is partly resolved out, so sizes, contrasts, asymmetries and brightness temperatures are biased and the smooth baseline is underestimate...

Continuum image



7. SR 21

Resolved

Well resolved (90% of the flux within 13.2 beams of 4.8 au). 2 ring(s) [35, 58 au] and 3 gap(s) [8, 43, 73 au] detected at >= 3 sigma. Image holds 26% of the expected catalogue flux. Negative bowl present: missing short spacings bias sizes, contrasts and temperatures.

Target

Coordinates (J2000)	16:27:10.28 -24:19:12.6 (RA 246.79282, Dec -24.32017 deg, J2000)
Distance	138.0 pc
Class	T Tauri + cavity
Versus TW Hya	Ring, not centrally peaked; gaps = cavity physics; Bosschaart+26: TD, R68 63 au
Catalogue	Bosschaart+26 (SR21): class TD (transition disc); R68 63 au; F(1.33 mm) 95 mJy; M_dust 19 Earth masses; M* 1.95 Msun; distance 138 pc; this image: 68%-flux radius 59 au

Observation used

Project / band	2018.1.00689.S / Band 6 (225.0 GHz)
Beam	0.0349 x 0.0268 arcsec = 4.8 x 3.7 au
Archive resolution	0.0205 arcsec (2.8 au
Largest scale	0.40 arcsec (55 au)
Exposure / release	2903 s (0.81 h) / 2020-11-11
Package	2018.1.00689.S_uid___A001_X133d_X140d_001_of_001.tar (0.18 GB)
FITS image	member.uid___A001_X133d_X140d.EM_star_SR_21_sci.spw19_23_25_27_29.cont.I.pbcor.fits
Archive observations	6 total; finest 0.0205 arcsec; used #1
Stellar parameters	Stellar parameters (Bosschaart+26 catalogue stellar mass; verify): L* = ? Lsun, M* = 1.95 Msun

Geometry and radial profile

Inclination / PA	18.8 / 13.1 deg
Geometry source	FITTED FROM THE IMAGE (radial-symmetry criterion), accuracy about +-5 deg.
Peak of mean profile	0.286 mJy/beam at r = 58.8 au
R50 / R90	56 / 64 au (11.7 / 13.2 beams)
Outer edge (>= 3 sigma)	71 au
Integrated flux	24.9 mJy (26% of catalogue expectation)
Image noise / dyn. range	0.02 mJy/beam / 10
Shape	the profile is ring-like, not centrally peaked. The maximum is at 59 au and the innermost bin has only -26% of it, which points to a central cavity: the mean rises to half its peak at ~52 au. Negative central values are an artifact of missing short baselines (spatial filtering), not a real flux deficit.

Substructure (baseline-subtracted profile)

type	radius	residual	modulation	significance	
gap	r= 8.0 au	resid=-0.0379 mJy/b	modul=n/a	signif= -10.6 sigma	SIGNIFICANT
ring	r= 35.0 au	resid=0.0509 mJy/b	modul=+72.8%	signif= +12.6 sigma	SIGNIFICANT
gap	r= 42.8 au	resid=-0.0787 mJy/b	modul=-58.3%	signif= -35.5 sigma	SIGNIFICANT
ring	r= 57.9 au	resid=0.136 mJy/b	modul=+90.6%	signif= +43.7 sigma	SIGNIFICANT
gap	r= 73.2 au	resid=-0.0548 mJy/b	modul=-95.3%	signif= -24.0 sigma	SIGNIFICANT

Sequence	G 8 au -> R 35 au -> G 43 au -> R 58 au -> G 73 au
Published features	none stored in the catalogue

Gaussian ring fit

Model / chi2_red tapered power law + 2 Gaussian ring(s); chi2_red = 30.32

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed	
1	r=36.2+/-1.3	amp=0.0956+/-0.016	S/N=5.9	FWHM_int=15.2 +/- 4.3		det
2	r=57.4+/-0.4	amp=0.283+/-0.019	S/N=14.7	FWHM_int=12.8 +/- 1.0		det

Brightness temperature and optical depth

T_d, T_b/T_d and tau not computed: no stellar luminosity stored for this target.

Azimuthal asymmetry

ring	radius	m=1 amplitude	m=2 amplitude
r=	36.2 au	m=1: 38.3% (S/N 15.5)	m=2: 40.6% (S/N 16.4)
r=	57.4 au	m=1: 12.5% (S/N 17.4)	m=2: 20.7% (S/N 28.9)

Gap-opening planet masses

Not available (needs L* and M*, which are not stored for this target; pass lstar= and mstar=).

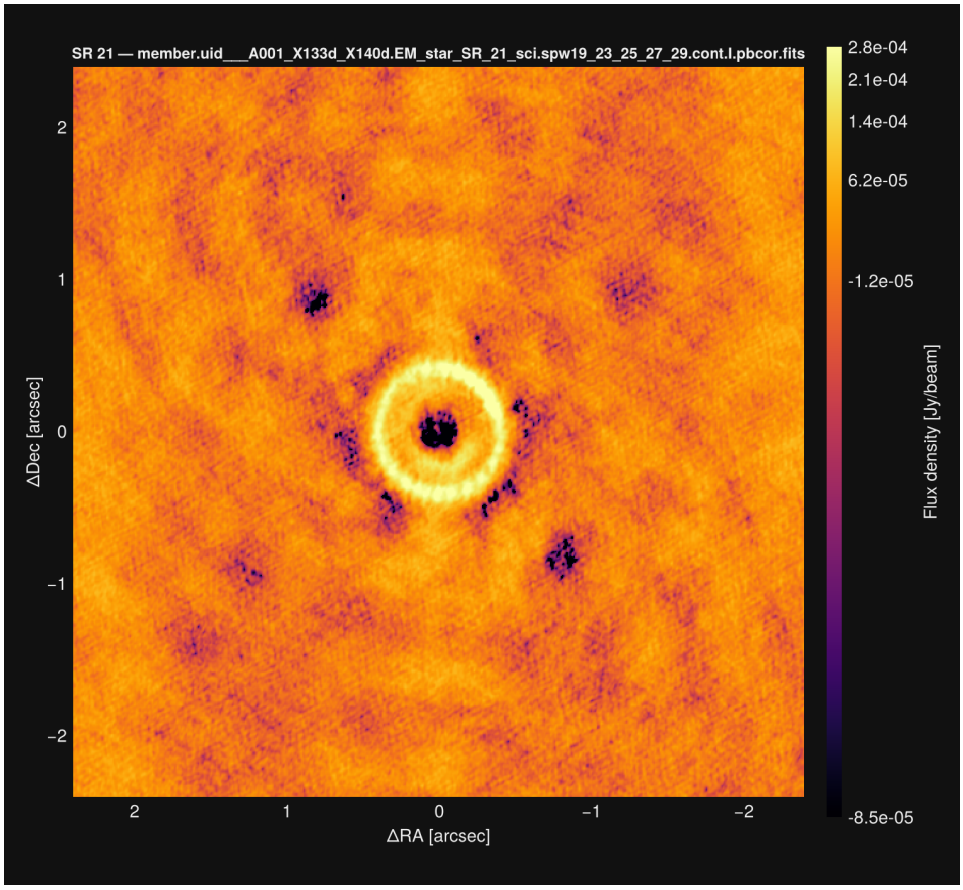
Pipeline warnings and notes

- the feature(s) at 36 au are shoulders (the raw-profile peak finder finds no peak or dip within 1 beam): an amplitude measured against a shoulder baseline can be large without a real azimuthal structure; the percentages of the local flux are the safer numbers.

- the image is spatially filtered (flux check above): arc contrasts and asymmetry amplitudes describe only the emission that survived the missing short spacings.

- Negative bowl: the azimuthal mean is significantly negative over r = 179-275 au (41 bins, down to -6.5 sigma, -0.0108 mJy/beam = -3.8% of the peak, at 210 au), which real emission cannot produce. It is the signature of missing short spacings (spatial filtering): the extended emission is partly resolved out, so sizes, contrasts, asymmetries and brightness temperatures are biased and the smooth baseline is underestimat...

Continuum image



8. HD 142666

Resolved

Well resolved (90% of the flux within 11.3 beams of 4.7 au). 2 ring(s) [21, 41 au] and 3 gap(s) [17, 34, 53 au] detected at >= 3 sigma. Recovered 4 of 6 testable published features. Image holds 78-82% of the expected catalogue flux (consistent).

Target

Coordinates (J2000)	15:56:40.02 -22:01:40.0 (RA 239.16676, Dec -22.02778 deg, J2000)
Distance	148.0 pc
Class	Herbig
Versus TW Hya	Higher-mass star; warmer, larger disk; Bosschaart+26: RD, R68 59 au
Catalogue	Bosschaart+26 (HD142666): class RD (ring disc); R68 59 au; F(1.33 mm) 130 mJy; M_dust 83.2 Earth masses; M* 1.50 Msun; distance 148 pc; this image: 68%-flux radius 41 au

Observation used

Project / band	2016.1.00484.L / Band 6 (239.0 GHz)
Beam	0.0320 x 0.0232 arcsec = 4.7 x 3.4 au
Archive resolution	0.0247 arcsec (3.7 au
Largest scale	0.59 arcsec (87 au)
Exposure / release	4064 s (1.13 h) / 2018-12-15
Package	2016.1.00484.L_uid___A001_X8c5_X92_001_of_001.tar (0.82 GB)
FITS image	member.uid___A001_X8c5_X92.HD_142666_sci.spw0_1_2_3_4_5_6_7.mfs.I.manual.image.pbcor.fits
Archive observations	12 total; finest 0.0247 arcsec; used #1
Stellar parameters	Stellar parameters (Andrews+18 Table 1 (MIST mass); verify): L* = 9.12 Lsun, M* = 1.58 Msun

Geometry and radial profile

Inclination / PA	62.2 / 162.1 deg
Geometry source	catalogue (literature) values.
Peak of mean profile	1.08 mJy/beam at r = 5.9 au
R50 / R90	32 / 53 au (6.8 / 11.3 beams)
Outer edge (>= 3 sigma)	75 au
Integrated flux	120.2 mJy (78-82% of catalogue expectation)
Image noise / dyn. range	0.0147 mJy/beam / 39
Shape	the emission is centrally peaked (the maximum lies at 6 au, within ~1.5 beams of the star), like TW Hya: a smooth, steeply falling profile on which any rings are shoulders and dips.

Substructure (baseline-subtracted profile)

type	radius	residual	modulation	significance
gap	r= 17.4 au	resid=-0.112 mJy/b	modul=-14.2%	signif= -25.4 sigma SIGNIFICANT
ring	r= 20.9 au	resid=0.0189 mJy/b	modul=+2.5%	signif= +3.6 sigma SIGNIFICANT
gap	r= 34.3 au	resid=-0.0312 mJy/b	modul=-6.4%	signif= -12.3 sigma SIGNIFICANT
ring	r= 41.4 au	resid=0.0324 mJy/b	modul=+9.3%	signif= +8.7 sigma SIGNIFICANT
gap	r= 53.3 au	resid=-0.0325 mJy/b	modul=-18.7%	signif= -11.7 sigma SIGNIFICANT

Sequence	G 17 au -> R 21 au -> G 34 au -> R 41 au -> G 53 au
Published features	recovered 4 of 6 testable

Gaussian ring fit

Model / chi2_red tapered power law + 4 Gaussian ring(s); chi2_red = 0.66

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed	
1	r=5.2+/-0.3	amp=0.697+/-0.27	S/N=2.6	FWHM_int=13.1 +/- 2.5	core	<3sig
2	r=23.6+/-1.5	amp=0.148+/-0.11	S/N=1.3	FWHM_int=10.7 +/- 4.5	det	<3sig
3	r=43.8+/-1.0	amp=0.0824+/-0.034	S/N=2.4	FWHM_int=10.7 +/- 3.4	det	<3sig
4	r=60.2+/-0.7	amp=0.0259+/-0.0095	S/N=2.7	FWHM_int=6.7 +/- 2.6	lit	<3sig

Brightness temperature and optical depth

type	radius	brightness T	dust T	ratio	optical depth	scale height
gap	r= 17.4 au	T_b= 25.3 K	T_d= 51.3 K	T_b/T_d= 0.49	tau= 0.68	H/r=0.048
gap	r= 34.3 au	T_b= 18.3 K	T_d= 36.9 K	T_b/T_d= 0.50	tau= 0.68	H/r=0.057
gap	r= 53.3 au	T_b= 8.6 K	T_d= 29.6 K	T_b/T_d= 0.29	tau= 0.34	H/r=0.064

Azimuthal asymmetry

No rings with a testable asymmetry.

Gap-opening planet masses

```
gap summary
gap r_c=17.3 au  h/r=0.048  depth=1.06 |3.3 0.0 0.190 | - - | 0.58 Mnep 0.18 Mnep asymmetric gap: half-widths 2.1 au
```

Not evaluated: 34 au: no local maximum outward of the gap minimum at 34.3 au (the profile does not rise there); 53 au: no local maximum outward of the gap minimum at 53.3 au (the profile does not rise there)

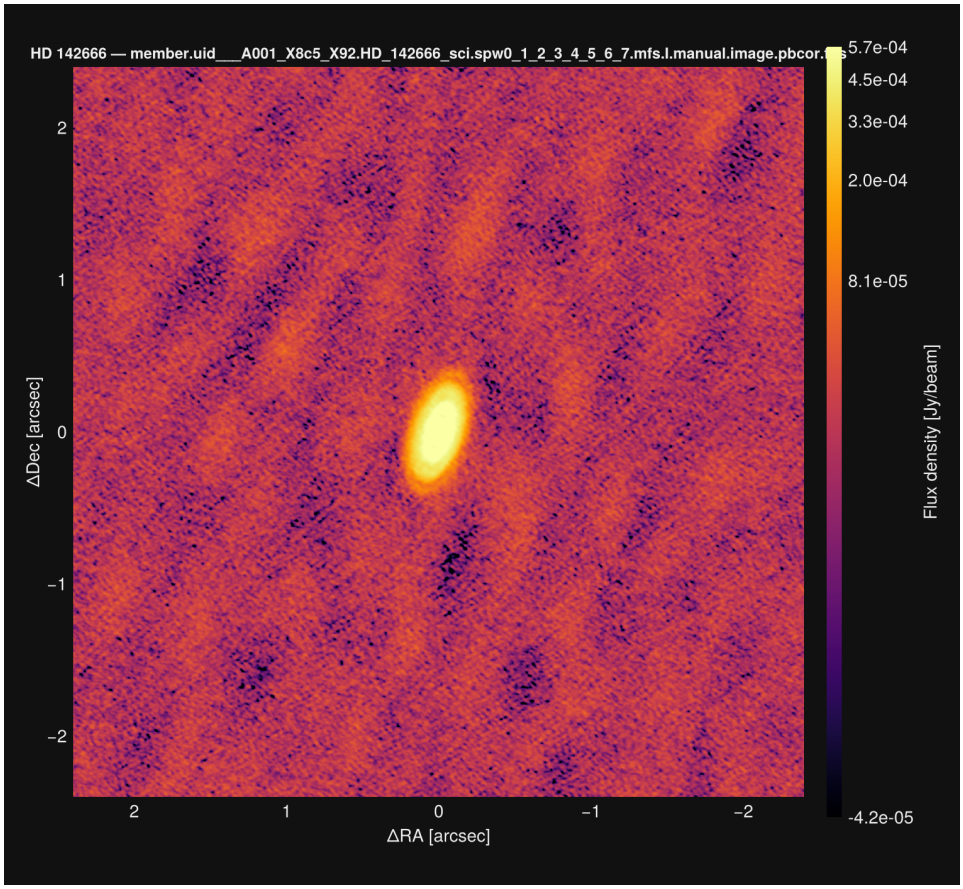
Pipeline warnings and notes

- 2 ring(s) detected in the profile at 3 sigma or more (21, 41 au) are covered only by a fitted component with amplitude S/N < 3; the fit is otherwise consistent with the profile, so it is used, and for the asymmetry and temperature steps those rings are taken from the detection (position and residual amplitude) instead of the weak component.

- the feature(s) at 21, 41 au are shoulders (the raw-profile peak finder finds no peak or dip within 1 beam): an amplitude measured against a shoulder baseline can be large without a real azimuthal structure; the percentages of the local flux are the safer numbers.

- gap at 17 au and ring at 21 au are 3.5 au apart (< 1 beam): probably one blended feature or noise.

Continuum image



9. RY Tau

Resolved

Well resolved (90% of the flux within 32.7 beams of 5.1 au). 2 ring(s) [16, 58 au] and 2 gap(s) [42, 75 au] detected at >= 3 sigma. Recovered 3 of 3 testable published features. Image holds 134% of the expected catalogue flux. Negative bowl present: missing short spacings bias sizes, contrasts and temperatures.

Target

Coordinates (J2000)	04:21:57.41 +28:26:35.6 (RA 65.48921, Dec +28.44321 deg, J2000)
Distance	138.0 pc
Class	T Tauri analogue
Versus TW Hya	Intermediate-mass (~2 Msun), jet source; bright, cavity; Bosschaart+26: TD, R68 50 au
Catalogue	Bosschaart+26 (RYTau): class TD (transition disc); R68 50 au; F(1.33 mm) 210 mJy; M_dust 39 Earth masses; M* 1.70 Msun; distance 132 pc; this image: 68%-flux radius 57 au

Observation used

Project / band	2017.1.01460.S / Band 6 (224.8 GHz)
Beam	0.0371 x 0.0198 arcsec = 5.1 x 2.7 au
Archive resolution	0.0246 arcsec (3.4 au
Largest scale	0.49 arcsec (68 au)
Exposure / release	308 s (0.09 h) / 2018-11-30
Package	2017.1.01460.S_uid___A001_X1284_X22ff_001_of_001.tar (0.04 GB)
FITS image	member.uid___A001_X1284_X22ff.ry_tau_sci.spw19_21_25_27_29.mfs.I.manual.image.pbcor.fits
Archive observations	12 total; finest 0.0246 arcsec; used #1
Stellar parameters	Stellar parameters (Long+18 Table 2; verify): L* = 12.30 Lsun, M* = 2.04 Msun

Geometry and radial profile

Inclination / PA	65.0 / 23.1 deg
Geometry source	catalogue (literature) values.
Peak of mean profile	1.11 mJy/beam at r = 16.6 au
R50 / R90	47 / 167 au (9.2 / 32.7 beams)
Outer edge (>= 3 sigma)	73 au
Integrated flux	280.2 mJy (134% of catalogue expectation)
Image noise / dyn. range	0.0817 mJy/beam / 29
Shape	the profile is ring-like, not centrally peaked. The maximum is at 17 au and the innermost bin has only 58% of it, which points to a central cavity: the mean rises to half its peak at ~1 au.

Substructure (baseline-subtracted profile)

type	radius	residual	modulation	significance
ring	r= 15.6 au	resid=0.16 mJy/b	modul=+16.7%	signif= +5.2 sigma SIGNIFICANT
gap	r= 42.2 au	resid=-0.0585 mJy/b	modul=-9.8%	signif= -6.6 sigma SIGNIFICANT
ring	r= 57.5 au	resid=0.0684 mJy/b	modul=+19.0%	signif= +4.2 sigma SIGNIFICANT
gap	r= 75.4 au	resid=-0.0426 mJy/b	modul=-62.2%	signif= -3.6 sigma SIGNIFICANT

Sequence	R 16 au -> G 42 au -> R 57 au -> G 75 au
Published features	recovered 3 of 3 testable

Gaussian ring fit

Model / chi2_red tapered power law + 2 Gaussian ring(s); chi2_red = 1.75

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed
1	r=15.4+/-0.7	amp=0.249+/-0.051	S/N=4.9	FWHM_int=6.3 +/- 2.2	det
2	r=58.0+/-0.7	amp=0.27+/-0.039	S/N=6.9	FWHM_int=18.4 +/- 2.2	det

Brightness temperature and optical depth

type	radius	brightness T	dust T	ratio	optical depth	scale height
ring	r= 15.4 au	T_b= 41.6 K	T_d= 57.2 K	T_b/T_d= 0.73	tau= 1.30	H/r=0.043
gap	r= 42.2 au	T_b= 22.6 K	T_d= 35.9 K	T_b/T_d= 0.63	tau= 0.99	H/r=0.055
ring	r= 58.0 au	T_b= 19.0 K	T_d= 30.7 K	T_b/T_d= 0.62	tau= 0.96	H/r=0.059

Azimuthal asymmetry

ring radius	m=1 amplitude	m=2 amplitude
r= 15.4 au	m=1: 71.1% (S/N 6.5)	m=2: 16.9% (S/N 1.5)
r= 58.0 au	m=1: 6.6% (S/N 1.9)	m=2: 20.9% (S/N 6.1)

Gap-opening planet masses

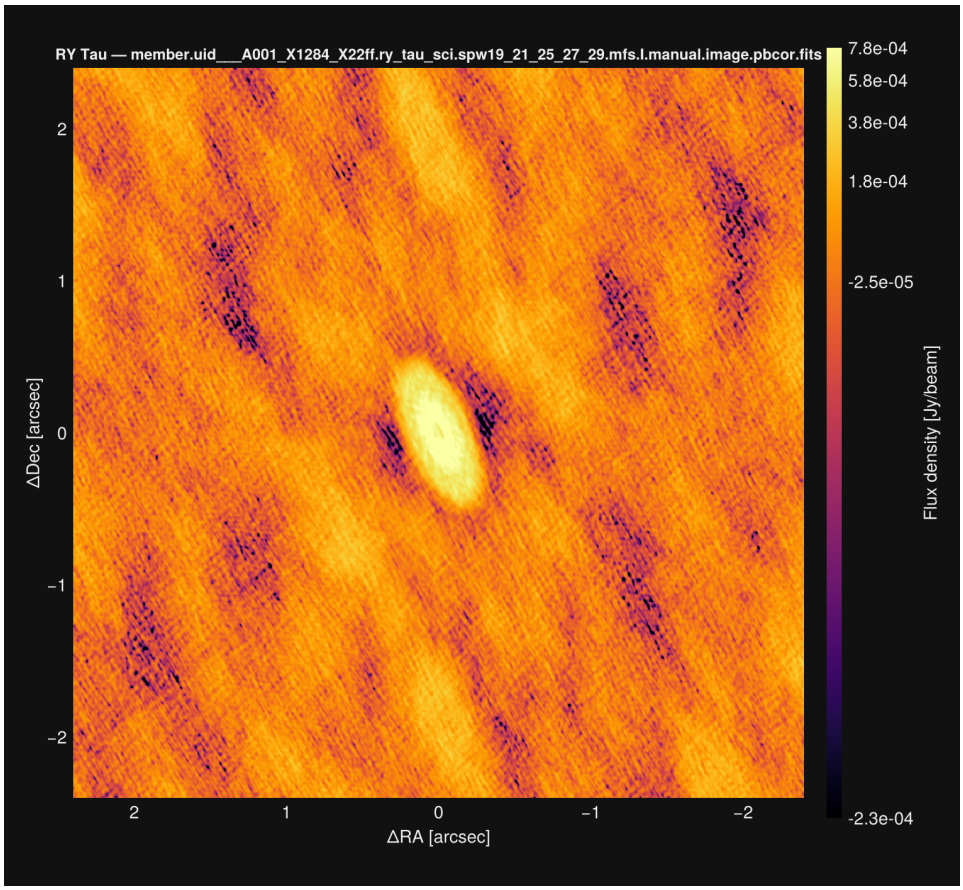
Not evaluated: 42 au: no local maximum outward of the gap minimum at 44.7 au (the profile does not rise there)

Pipeline warnings and notes

- the feature(s) at 58 au are shoulders (the raw-profile peak finder finds no peak or dip within 1 beam): an amplitude measured against a shoulder baseline can be large without a real azimuthal structure; the percentages of the local flux are the safer numbers.

- Negative bowl: the azimuthal mean is significantly negative over r = 83-114 au (13 bins, down to -6.8 sigma, -0.0873 mJy/beam = -7.9% of the peak, at 93 au), which real emission cannot produce. It is the signature of missing short spacings (spatial filtering): the extended emission is partly resolved out, so sizes, contrasts, asymmetries and brightness temperatures are biased and the smooth baseline is underestimated...

Continuum image



10. GW Ori

Resolved

Well resolved (90% of the flux within 31.7 beams of 9.8 au). 2 ring(s) [46, 105 au] and 2 gap(s) [20, 66 au] detected at >= 3 sigma. Negative bowl present: missing short spacings bias sizes, contrasts and temperatures.

Target

Coordinates (J2000)	05:29:08.39 +11:52:12.7 (RA 82.28497, Dec +11.87019 deg, J2000)
Distance	402.0 pc
Class	Multiple system
Versus TW Hya	Triple star; broken, misaligned rings at ~46/188/338 au; 400 pc

Observation used

Project / band	2018.1.00813.S / Band 6 (225.0 GHz)
Beam	0.0244 x 0.0192 arcsec = 9.8 x 7.7 au
Archive resolution	0.0217 arcsec (8.7 au
Largest scale	0.42 arcsec (170 au)
Exposure / release	5552 s (1.54 h) / 2020-11-08
Package	2018.1.00813.S_uid___A001_X13b4_X5_001_of_001.tar (0.25 GB)
FITS image	member.uid___A001_X13b4_X5.GW_Ori_sci.spw25_27_29_31.cont.I.manual.pbcor.fits
Archive observations	18 total; finest 0.0217 arcsec; used #1
Stellar parameters	Stellar parameters: none stored for this target (pass lstar= / mstar=).

Geometry and radial profile

Inclination / PA	17.5 / 176.2 deg
Geometry source	FITTED FROM THE IMAGE (radial-symmetry criterion), accuracy about +-5 deg.
Peak of mean profile	0.648 mJy/beam at r = 46.6 au
R50 / R90	164 / 311 au (16.7 / 31.7 beams)
Outer edge (>= 3 sigma)	61 au
Integrated flux	85.0 mJy (n/a of catalogue expectation)
Image noise / dyn. range	0.00974 mJy/beam / 9
Shape	the profile is ring-like, not centrally peaked. The maximum is at 47 au and the innermost bin has only 20% of it, which points to a central cavity: the mean rises to half its peak at ~42 au.

Substructure (baseline-subtracted profile)

type	radius	residual	modulation	significance	
gap	r= 20.0 au	resid=-0.208 mJy/b	modul=-109.1%	signif=-107.0 sigma	SIGNIFICANT
ring	r= 45.8 au	resid=0.424 mJy/b	modul=+188.8%	signif=+173.1 sigma	SIGNIFICANT
gap	r= 66.2 au	resid=-0.134 mJy/b	modul=-99.2%	signif= -86.2 sigma	SIGNIFICANT
ring	r= 105.4 au	resid=0.0423 mJy/b	modul=n/a	signif= +26.2 sigma	SIGNIFICANT

Sequence	G 20 au -> R 46 au -> G 66 au -> R 105 au
Published features	none stored in the catalogue

Gaussian ring fit

Model / chi2_red tapered power law + 3 Gaussian ring(s); chi2_red = 25.50

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed
1	r=4.5+/-2.6	amp=0.137+/-0.039	S/N=3.5	FWHM_int=< 13.5 (unresolved)	core
2	r=45.9+/-0.3	amp=0.66+/-0.042	S/N=15.6	FWHM_int=12.7 +/- 0.7	det
3	r=174.6+/-1.1	amp=0.072+/-0.0042	S/N=17.2	FWHM_int=46.7 +/- 3.0	-

Brightness temperature and optical depth

T_d, T_b/T_d and tau not computed: no stellar luminosity stored for this target.

Azimuthal asymmetry

ring radius	m=1 amplitude	m=2 amplitude
r= 45.8 au	m=1: 10.4% (S/N 17.3)	m=2: 8.5% (S/N 14.0)
r= 105.4 au	m=1: 7.5% (S/N 1.8)	m=2: 12.1% (S/N 2.9)

Gap-opening planet masses

Not available (needs L* and M*, which are not stored for this target; pass lstar= and mstar=).

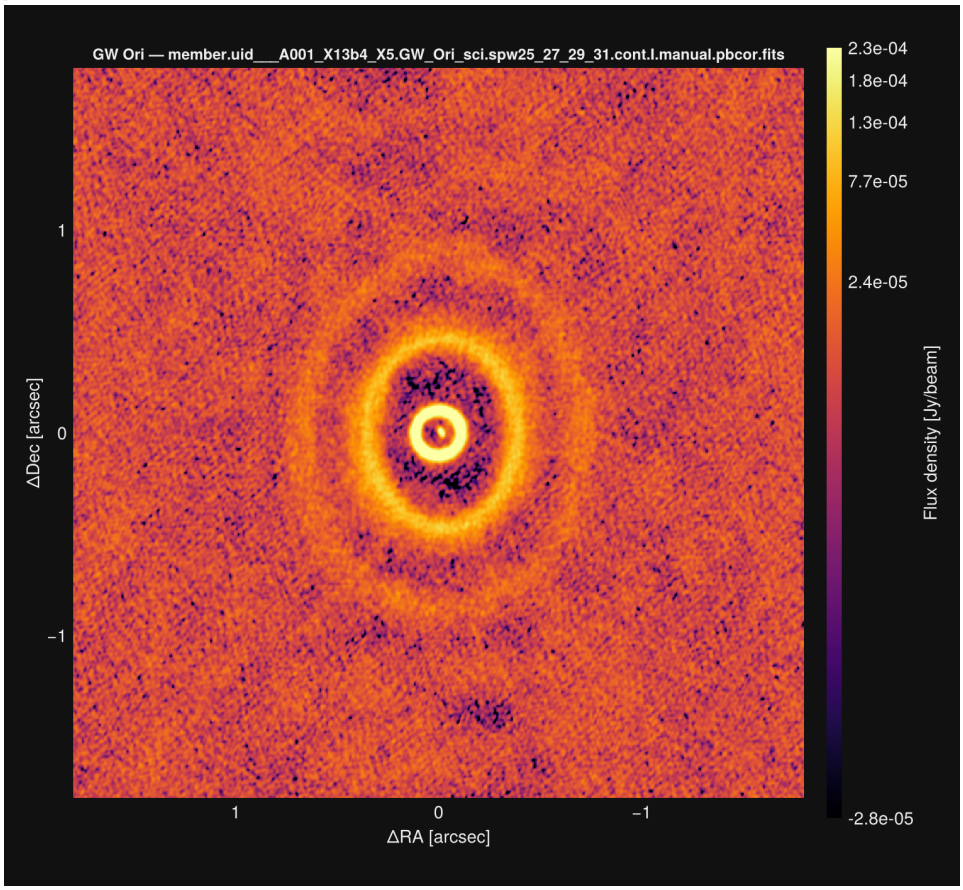
Pipeline warnings and notes

- THE FIT DISAGREES WITH THE PROFILE: 1 ring(s) detected in the profile at 3 sigma or more (105 au) have no fitted component within 1 beam. The fitted radii and widths, the asymmetry windows and the trapping limits do not describe the rings seen in the profile; the temperature table uses the detected rings.

- the Gaussian fit disagrees with the profile (section 1), so the windows below use the detected rings instead: +/- 4.2 au around each radius (one sigma of the beam major axis), amplitudes from the baseline-subtracted profile.

- Negative bowl: the azimuthal mean is significantly negative over r = 424-453 au (7 bins, down to -4.5 sigma, -0.00357 mJy/beam = -0.6% of the peak, at 434 au), which real emission cannot produce. It is the signature of missing short spacings (spatial filtering): the extended emission is partly resolved out, so sizes, contrasts, asymmetries and brightness temperatures are biased and the smooth baseline is underestimat...

Continuum image



11. AS 205

Resolved

Well resolved (90% of the flux within 26.7 beams of 4.9 au). 2 ring(s) [23, 45 au] and 2 gap(s) [33, 64 au] detected at >= 3 sigma. Image holds 99-105% of the expected catalogue flux (consistent).

Target

Coordinates (J2000)	16:11:31.35 -18:38:26.0 (RA 242.88061, Dec -18.64054 deg, J2000)
Distance	128.0 pc
Class	Multiple system
Versus TW Hya	Close binary (N/S), separate disks; spirals; Bosschaart+26: ED, R68 35 au
Catalogue	Bosschaart+26 (AS205): class ED (extended disc); R68 35 au; F(1.33 mm) 358 mJy; M_dust 61 Earth masses; M* 0.87 Msun; distance 128 pc; this image: 68%-flux radius 43 au

Observation used

Project / band	2016.1.00484.L / Band 6 (239.0 GHz)
Beam	0.0381 x 0.0244 arcsec = 4.9 x 3.1 au
Archive resolution	0.0237 arcsec (3.0 au
Largest scale	0.54 arcsec (69 au)
Exposure / release	2032 s (0.56 h) / 2019-02-20
Package	2016.1.00484.L_uid___A001_X8c5_X4a_001_of_001.tar (1.83 GB)
FITS image	member.uid___A001_X8c5_X4a.AS_205_sci.spw0_1_2_3.mfs.I.manual.image.pbcor.fits
Archive observations	10 total; finest 0.0237 arcsec; used #1
Stellar parameters	Stellar parameters (Andrews+18 Table 1 (MIST mass); triple system; verify): L* = 2.14 Lsun, M* = 0.87 Msun

Geometry and radial profile

Inclination / PA	33.8 / 110.6 deg
Geometry source	FITTED FROM THE IMAGE (radial-symmetry criterion), accuracy about +-5 deg.
Peak of mean profile	4.15 mJy/beam at r = 1.2 au
R50 / R90	33 / 130 au (6.7 / 26.7 beams)
Outer edge (>= 3 sigma)	69 au
Integrated flux	424.6 mJy (99-105% of catalogue expectation)
Image noise / dyn. range	0.0783 mJy/beam / 71
Shape	the emission is centrally peaked (the maximum lies at 1 au, within ~1.5 beams of the star), like TW Hya: a smooth, steeply falling profile on which any rings are shoulders and dips.

Substructure (baseline-subtracted profile)

type	radius	residual		modulation	significance	
ring	r= 23.2 au	resid=0.142	mJy/b	modul=+14.2%	signif= +7.5 sigma	SIGNIFICANT
gap	r= 32.9 au	resid=0.0112	mJy/b	modul=+1.4%	signif= -4.0 sigma	SIGNIFICANT
ring	r= 45.1 au	resid=0.0754	mJy/b	modul=+16.5%	signif= +5.6 sigma	SIGNIFICANT
gap	r= 64.4 au	resid=-0.0803	mJy/b	modul=-152.3%	signif= -9.9 sigma	SIGNIFICANT
ring	r= 67.7 au	resid=0.0324	mJy/b	modul=+120.3%	signif= +2.9 sigma	noise-level

Sequence	R 23 au -> G 33 au -> R 45 au -> G 64 au -> R 68 au
Published features	none stored in the catalogue

Gaussian ring fit

Model / chi2_red tapered power law + 3 Gaussian ring(s); chi2_red = 1.64

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed	
1	r=23.7+/-14.7	amp=0.815+/-1.5	S/N=0.6	FWHM_int=< 58.2 (unresolved)	det	<3sig
2	r=44.1+/-3.5	amp=0.393+/-0.41	S/N=1.0	FWHM_int=17.3 +/- 4.2	det	<3sig
3	r=70.0+/-1.6	amp=0.0341+/-0.0091	S/N=3.7	FWHM_int=10.8 +/- 4.2	det	

Brightness temperature and optical depth

type	radius	brightness T	dust T	ratio	optical depth	scale height
gap	r= 64.4 au	T_b= 4.0 K	T_d= 18.7 K	T_b/T_d= 0.22	tau= 0.24	H/r=0.075
ring	r= 70.0 au	T_b= 4.2 K	T_d= 18.1 K	T_b/T_d= 0.23	tau= 0.26	H/r=0.076

Azimuthal asymmetry

ring	radius	m=1 amplitude	m=2 amplitude
r=	23.2 au	m=1: 13.0% (S/N 0.9)	m=2: 12.0% (S/N 0.8)
r=	45.1 au	m=1: 30.4% (S/N 1.6)	m=2: 176.5% (S/N 9.0)
r=	70.0 au	m=1: 232.8% (S/N 10.3)	m=2: 39.2% (S/N 1.7)

Gap-opening planet masses

Not evaluated: 64 au: gap contrast only 0.3 sigma (< 3.0)

Pipeline warnings and notes

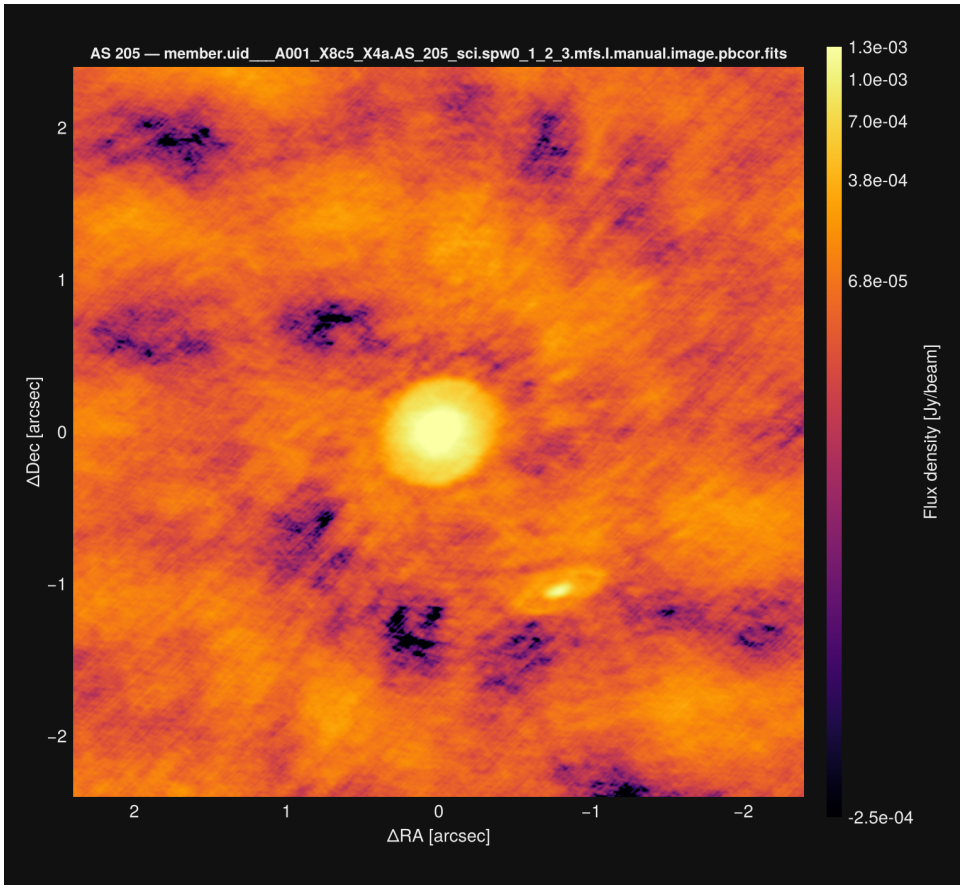
- 2 ring(s) detected in the profile at 3 sigma or more (23, 45 au) are covered only by a fitted component with amplitude S/N < 3; the fit is otherwise consistent with the profile, so it is used, and for the asymmetry and temperature steps those rings are taken from the detection (position and residual amplitude) instead of the weak component.

- at 70 au the local azimuthal-mean flux is below 2x the image rms per beam (0.0783 mJy/beam): the ring is detected only by averaging over many beams, and the S/N and false-alarm values above assume white noise at the beam scale. Correlated residuals (sidelobes, large-scale noise, weak background sources) are not in that noise and can make a dipole or m=2 pattern of this size, so these asymmetries are unconfirmed; an '...

- the feature(s) at 23, 45, 70 au are shoulders (the raw-profile peak finder finds no peak or dip within 1 beam): an amplitude measured against a shoulder baseline can be large without a real azimuthal structure; the percentages of the local flux are the safer numbers.

- gap at 64 au and ring at 68 au are 3.4 au apart (< 1 beam): probably one blended feature or noise.

Continuum image



12. AA Tau

Resolved

Well resolved (90% of the flux within 7.8 beams of 13.4 au). 2 ring(s) [44, 90 au] and 2 gap(s) [77, 105 au] detected at >= 3 sigma. Recovered 2 of 3 testable published features. Negative bowl present: missing short spacings bias sizes, contrasts and temperatures. A finer observation (0.0189 arcsec) exists but the package exceeded the download limit; finer structure may exist.

Target

Coordinates (J2000)	04:34:55.42 +24:28:53.0 (RA 68.73092, Dec +24.48140 deg, J2000)
Distance	136.0 pc
Class	T Tauri analogue
Versus TW Hya	Moderately inclined; warped inner disk

Observation used

Project / band	2016.1.01205.S / Band 6 (221.4 GHz)
Beam	0.0989 x 0.0607 arcsec = 13.4 x 8.3 au
Archive resolution	0.0641 arcsec (8.7 au
Largest scale	1.27 arcsec (173 au)
Exposure / release	2903 s (0.81 h) / 2018-11-03
Package	2016.1.01205.S_uid___A001_X87a_X5_001_of_001.tar (28.15 GB)
FITS image	member.uid___A001_X87a_X5.ari_l.AA_Tau_sci.spw0_1_2_3_221365MHz.12m.cont.l.pbcor.fits
Archive observations	13 total; finest 0.0189 arcsec; used #2 (fallback: finer packages too large)
Stellar parameters	Stellar parameters: none stored for this target (pass lstar= / mstar=).

Geometry and radial profile

Inclination / PA	59.1 / 93.0 deg
Geometry source	catalogue (literature) values.
Peak of mean profile	1.08 mJy/beam at r = 44.2 au
R50 / R90	58 / 105 au (4.3 / 7.8 beams)
Outer edge (>= 3 sigma)	139 au
Integrated flux	50.8 mJy (n/a of catalogue expectation)
Image noise / dyn. range	0.022 mJy/beam / 22
Shape	the peak lies off-centre (44 au, 3.3 beams) but the inner profile is not strongly depleted (86% of the peak); this may be a bright inner ring on a filled disk.

Substructure (baseline-subtracted profile)

type	radius	residual	modulation	significance	
ring	r= 43.5 au	resid=0.205 mJy/b	modul=+23.5%	signif= +26.1 sigma	SIGNIFICANT
gap	r= 76.8 au	resid=-0.113 mJy/b	modul=-53.9%	signif= -30.0 sigma	SIGNIFICANT
ring	r= 90.4 au	resid=0.0676 mJy/b	modul=+32.4%	signif= +12.4 sigma	SIGNIFICANT
gap	r= 105.4 au	resid=-0.0367 mJy/b	modul=-27.3%	signif= -8.6 sigma	SIGNIFICANT

Sequence	R 44 au -> G 77 au -> R 90 au -> G 105 au
Published features	recovered 2 of 3 testable

Gaussian ring fit

Model / chi2_red tapered power law + 3 Gaussian ring(s); chi2_red = 4.88

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed
1	r=43.8+/-0.3	amp=1.05+/-0.032	S/N=32.5	FWHM_int=25.6 +/- 1.1	det
2	r=87.5+/-0.8	amp=0.261+/-0.012	S/N=22.3	FWHM_int=29.8 +/- 3.0	det
3	r=126.8+/-3.8	amp=0.0374+/-0.0081	S/N=4.6	FWHM_int=21.6 +/- 9.1	lit

Brightness temperature and optical depth

T_d, T_b/T_d and tau not computed: no stellar luminosity stored for this target.

Azimuthal asymmetry

ring radius	m=1 amplitude	m=2 amplitude
r= 43.8 au	m=1: 0.8% (S/N 1.3)	m=2: 16.2% (S/N 28.1)
r= 87.5 au	m=1: 4.5% (S/N 2.9)	m=2: 2.8% (S/N 1.8)
r= 126.8 au	m=1: 38.5% (S/N 3.8)	m=2: 29.5% (S/N 2.9)

Gap-opening planet masses

Not available (needs L* and M*, which are not stored for this target; pass lstar= and mstar=).

Pipeline warnings and notes

- the smooth disk collapsed to the unresolved core (characteristic radius 10 au < 3 beams): the extended emission is shared among the ring components, so overlapping neighbours (shoulders) have degenerate widths and amplitudes; trust the isolated rings, not the blended ones.

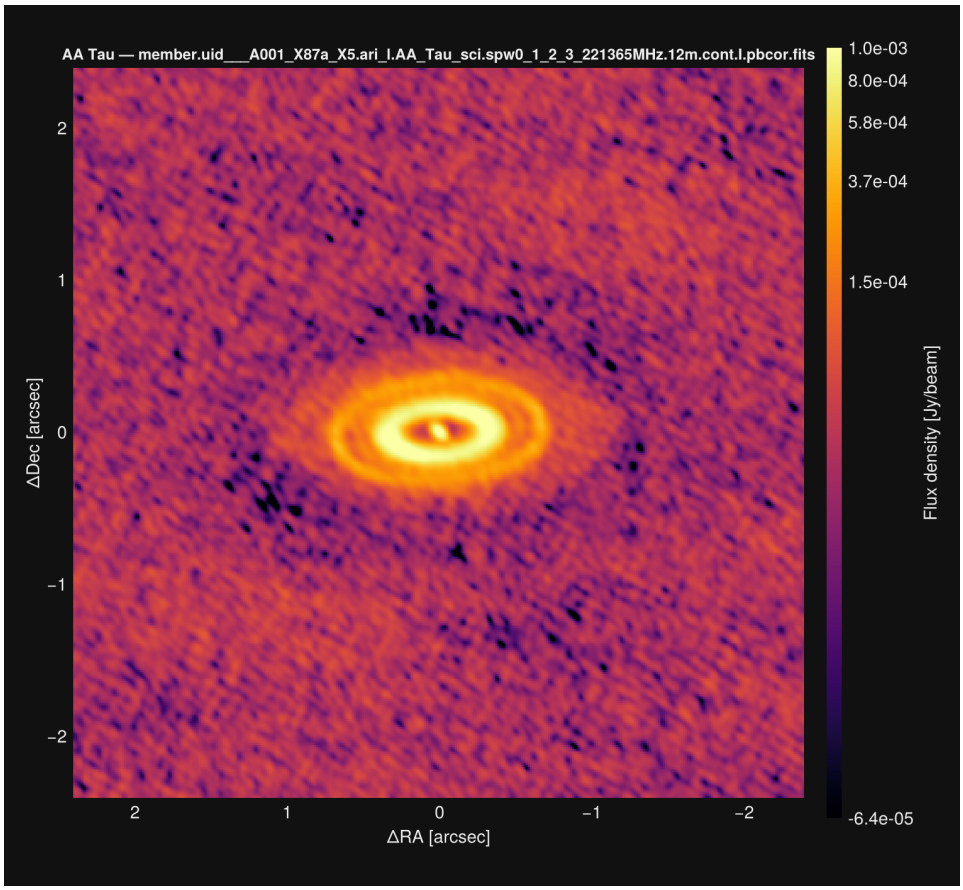
- at 127 au the local azimuthal-mean flux is below 2x the image rms per beam (0.022 mJy/beam): the ring is detected only by averaging over many beams, and the S/N and false-alarm values above assume white noise at the beam scale. Correlated residuals (sidelobes, large-scale noise, weak background sources) are not in that noise and can make a dipole or m=2 pattern of this size, so these asymmetries are unconfirmed; an '...

- the feature(s) at 127 au are shoulders (the raw-profile peak finder finds no peak or dip within 1 beam): an amplitude measured against a shoulder baseline can be large without a real azimuthal structure; the percentages of the local flux are the safer numbers.

- Negative bowl: the azimuthal mean is significantly negative over r = 167-228 au (10 bins, down to -6.4 sigma, -0.0246 mJy/beam = -2.3% of the peak, at 187 au), which real emission cannot produce. It is the signature of missing short spacings (spatial filtering): the extended emission is partly resolved out, so sizes, contrasts, asymmetries and brightness temperatures are biased and the smooth baseline is underestimat...

- the notebook fell back from observation #1 to #2 because the finer packages exceeded the size limit.

Continuum image



13. Sz 111

Resolved

Well resolved (90% of the flux within 52.6 beams of 4.8 au). 1 ring(s) [60 au] and 2 gap(s) [50, 78 au] detected at >= 3 sigma. Image holds 87% of the expected catalogue flux (consistent). Negative bowl present: missing short spacings bias sizes, contrasts and temperatures.

Target

Coordinates (J2000)	16:08:54.68 -39:37:43.1 (RA 242.22785, Dec -39.62865 deg, J2000)
Distance	158.0 pc
Class	T Tauri analogue
Versus TW Hya	Closest analogue: axisymmetric rings/gaps; Bosschaart+26: TD, R68 77 au
Catalogue	Bosschaart+26 (Sz111): class TD (transition disc); R68 77 au; F(1.33 mm) 60 mJy; M_dust 16 Earth masses; M* 0.51 Msun; distance 158 pc; this image: 68%-flux radius 74 au

Observation used

Project / band	2018.1.00689.S / Band 6 (225.0 GHz)
Beam	0.0306 x 0.0254 arcsec = 4.8 x 4.0 au
Archive resolution	0.0214 arcsec (3.4 au
Largest scale	0.41 arcsec (64 au)
Exposure / release	2976 s (0.83 h) / 2020-11-11
Package	2018.1.00689.S_uid___A001_X133d_X13fb_001_of_001.tar (0.18 GB)
FITS image	member.uid___A001_X133d_X13fb.Sz111_sci.spw19_23_25_27_29.cont.I.pbcor.fits
Archive observations	10 total; finest 0.0214 arcsec; used #1
Stellar parameters	Stellar parameters (Bosschaart+26 catalogue stellar mass; verify): L* = ? Lsun, M* = 0.51 Msun

Geometry and radial profile

Inclination / PA	55.0 / 43.1 deg
Geometry source	FITTED FROM THE IMAGE (radial-symmetry criterion), accuracy about +-5 deg.
Peak of mean profile	0.188 mJy/beam at r = 61.5 au
R50 / R90	66 / 254 au (13.7 / 52.6 beams)
Outer edge (>= 3 sigma)	93 au
Integrated flux	52.1 mJy (87% of catalogue expectation)
Image noise / dyn. range	0.0143 mJy/beam / 17
Shape	the profile is ring-like, not centrally peaked. The maximum is at 62 au and the innermost bin has only -20% of it, which points to a central cavity: the mean rises to half its peak at ~52 au. Negative central values are an artifact of missing short baselines (spatial filtering), not a real flux deficit.

Substructure (baseline-subtracted profile)

type	radius	residual	modulation	significance	
ring	r= 24.1 au	resid=0.00893 mJy/b	modul=n/a	signif= +1.9 sigma	noise-level
gap	r= 49.5 au	resid=-0.0257 mJy/b	modul=-23.2%	signif= -10.7 sigma	SIGNIFICANT
ring	r= 60.5 au	resid=0.0517 mJy/b	modul=+38.0%	signif= +17.8 sigma	SIGNIFICANT
gap	r= 78.4 au	resid=-0.0278 mJy/b	modul=-33.3%	signif= -11.9 sigma	SIGNIFICANT

Sequence	R 24 au -> G 49 au -> R 60 au -> G 78 au
Published features	none stored in the catalogue

Gaussian ring fit

Model / chi2_red tapered power law + 1 Gaussian ring(s); chi2_red = 16.93

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed
1	r=61.4+/-0.5	amp=0.173+/-0.0076	S/N=22.9	FWHM_int=27.2 +/- 1.5	det

Brightness temperature and optical depth

T_d, T_b/T_d and tau not computed: no stellar luminosity stored for this target.

Azimuthal asymmetry

ring radius	m=1 amplitude	m=2 amplitude
r= 61.4 au	m=1: 5.0% (S/N 6.5)	m=2: 3.9% (S/N 5.1)

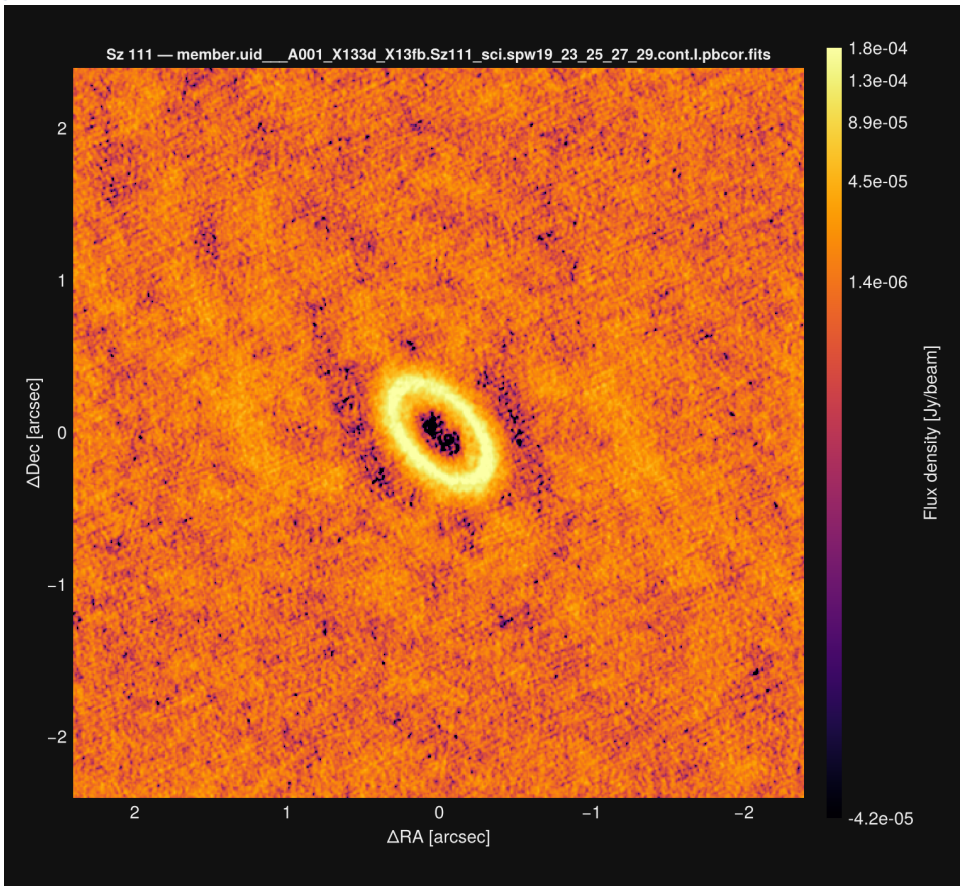
Gap-opening planet masses

Not available (needs L* and M*, which are not stored for this target; pass lstar= and mstar=).

Pipeline warnings and notes

- Negative bowl: the azimuthal mean is significantly negative over r = 103-131 au (13 bins, down to -5.2 sigma, -0.0109 mJy/beam = -5.8% of the peak, at 119 au), which real emission cannot produce. It is the signature of missing short spacings (spatial filtering): the extended emission is partly resolved out, so sizes, contrasts, asymmetries and brightness temperatures are biased and the smooth baseline is underestimat...

Continuum image



14. TMC1A

Resolved

Well resolved (90% of the flux within 43.7 beams of 5.1 au). 2 ring(s) [22, 50 au] and 1 gap(s) [37 au] detected at >= 3 sigma. Negative bowl present: missing short spacings bias sizes, contrasts and temperatures. A finer observation (0.0225 arcsec) exists but the package exceeded the download limit; finer structure may exist.

Target

Coordinates (J2000)	04:39:35.19 +25:41:44.7 (RA 69.89664, Dec +25.69576 deg, J2000)
Distance	140.0 pc
Class	Young/embedded
Versus TW Hya	Class I; embedded

Observation used

Project / band	2015.1.01415.S / Band 6 (224.6 GHz)
Beam	0.0364 x 0.0250 arcsec = 5.1 x 3.5 au
Archive resolution	0.0271 arcsec (3.8 au
Largest scale	0.71 arcsec (100 au)
Exposure / release	4246 s (1.18 h) / 2016-12-10
Package	2015.1.01415.S_uid___A001_X2d8_X4a_001_of_001.tar (1.40 GB)
FITS image	member.uid___A001_X2d8_X4a.ari_l.TMC1A_sci.spw0_1_2_3_4_224607MHz.12m.cont.I.pbcor.fits
Archive observations	25 total; finest 0.0225 arcsec; used #2 (fallback: finer packages too large)
Stellar parameters	Stellar parameters (Aso+15 (dynamical mass, inclination-corrected; Lbol is the total, accretion included); verify): L* = 2.70 Lsun, M* = 0.68 Msun

Geometry and radial profile

Inclination / PA	50.0 / 76.9 deg
Geometry source	FITTED FROM THE IMAGE (radial-symmetry criterion), accuracy about +-5 deg.
Peak of mean profile	3.49 mJy/beam at r = 1.3 au
R50 / R90	22 / 223 au (4.2 / 43.7 beams)
Outer edge (>= 3 sigma)	55 au
Integrated flux	107.0 mJy (n/a of catalogue expectation)
Image noise / dyn. range	0.0228 mJy/beam / 81
Shape	the emission is centrally peaked (the maximum lies at 1 au, within ~1.5 beams of the star), like TW Hya: a smooth, steeply falling profile on which any rings are shoulders and dips.

Substructure (baseline-subtracted profile)

type	radius	residual	modulation	significance
ring	r= 21.6 au	resid=0.133 mJy/b	modul=+18.4%	signif= +19.6 sigma SIGNIFICANT
gap	r= 36.9 au	resid=-0.0467 mJy/b	modul=-31.3%	signif= -17.7 sigma SIGNIFICANT
ring	r= 49.6 au	resid=0.0443 mJy/b	modul=n/a	signif= +9.7 sigma SIGNIFICANT

Sequence	R 22 au -> G 37 au -> R 50 au
Published features	none stored in the catalogue

Gaussian ring fit

Model / chi2_red tapered power law + 2 Gaussian ring(s); chi2_red = 0.78

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed
1	r=20.7+/-0.5	amp=0.279+/-0.024	S/N=11.5	FWHM_int=12.3 +/- 1.1	det
2	r=51.2+/-1.1	amp=0.0184+/-0.0031	S/N=5.9	FWHM_int=10.7 +/- 2.8	det

Brightness temperature and optical depth

type	radius	brightness T	dust T	ratio	optical depth	scale height
ring	r= 20.7 au	T_b= 27.7 K	T_d= 34.3 K	T_b/T_d= 0.81	tau= 1.65	H/r=0.066
gap	r= 36.9 au	T_b= 6.7 K	T_d= 26.3 K	T_b/T_d= 0.26	tau= 0.30	H/r=0.076
ring	r= 51.2 au	T_b= 3.9 K	T_d= 22.1 K	T_b/T_d= 0.18	tau= 0.19	H/r=0.083

Azimuthal asymmetry

ring radius	m=1 amplitude	m=2 amplitude
r= 20.7 au	m=1: 23.5% (S/N 13.7)	m=2: 16.2% (S/N 9.4)
r= 51.2 au	m=1: 89.4% (S/N 5.1)	m=2: 154.8% (S/N 8.8)

Gap-opening planet masses

Not evaluated: 37 au: no local maximum outward of the gap minimum at 39.5 au (the profile does not rise there)

Pipeline warnings and notes

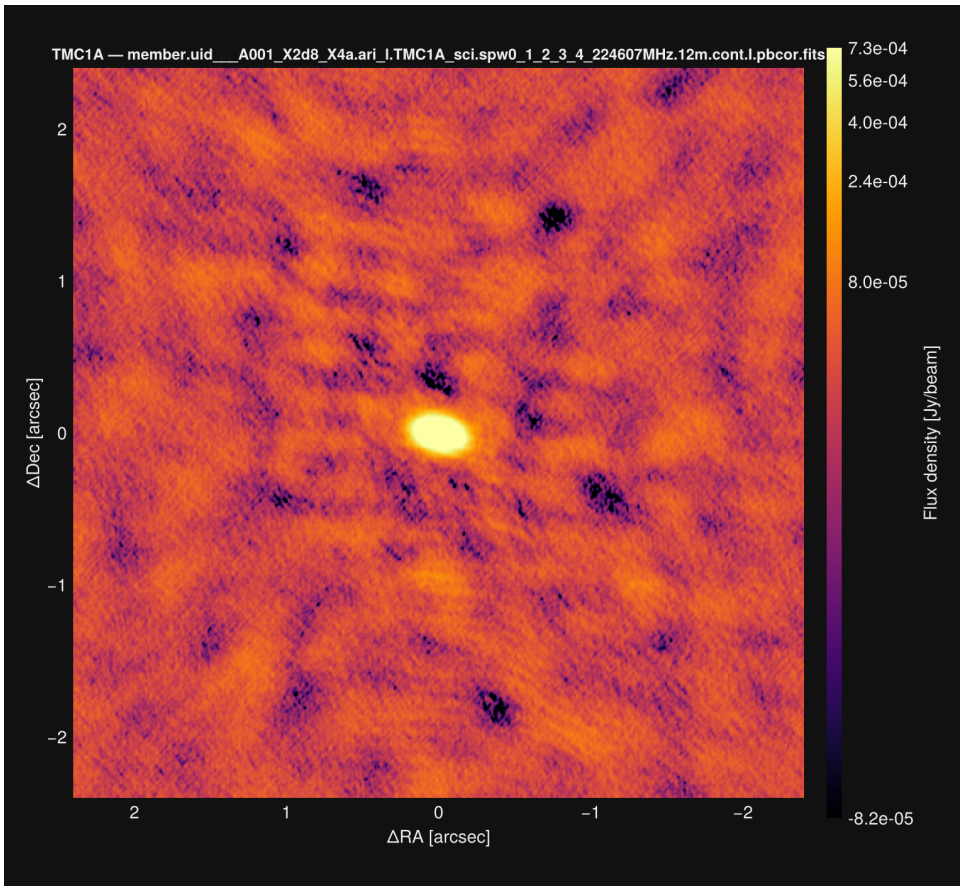
- at 51 au the local azimuthal-mean flux is below 2x the image rms per beam (0.0228 mJy/beam): the ring is detected only by averaging over many beams, and the S/N and false-alarm values above assume white noise at the beam scale. Correlated residuals (sidelobes, large-scale noise, weak background sources) are not in that noise and can make a dipole or m=2 pattern of this size, so these asymmetries are unconfirmed; an '...

- the feature(s) at 21, 51 au are shoulders (the raw-profile peak finder finds no peak or dip within 1 beam): an amplitude measured against a shoulder baseline can be large without a real azimuthal structure; the percentages of the local flux are the safer numbers.

- Negative bowl: the azimuthal mean is significantly negative over r = 75-101 au (11 bins, down to -5.2 sigma, -0.0175 mJy/beam = -0.5% of the peak, at 88 au), which real emission cannot produce. It is the signature of missing short spacings (spatial filtering): the extended emission is partly resolved out, so sizes, contrasts, asymmetries and brightness temperatures are biased and the smooth baseline is underestimated...

- the notebook fell back from observation #1 to #2 because the finer packages exceeded the size limit.

Continuum image



15. DM Tau

Resolved

Well resolved (90% of the flux within 20.7 beams of 3.9 au). 1 ring(s) [24 au] and 2 gap(s) [14, 42 au] detected at >= 3 sigma. Image holds 48-49% of the expected catalogue flux.

Target

Coordinates (J2000)	04:33:48.74 +18:10:09.8 (RA 68.45310, Dec +18.16940 deg, J2000)
Distance	145.0 pc
Class	T Tauri analogue
Versus TW Hya	Closest analogue: axisymmetric rings/gaps; Bosschaart+26: RD, R68 118 au
Catalogue	Bosschaart+26 (DMTau): class RD (ring disc); R68 118 au; F(1.33 mm) 89 mJy; M_dust 20 Earth masses; M* 0.29 Msun; distance 144 pc; this image: 68%-flux radius 38 au

Observation used

Project / band	2018.1.01755.S / Band 6 (221.7 GHz)
Beam	0.0269 x 0.0178 arcsec = 3.9 x 2.6 au
Archive resolution	0.0238 arcsec (3.4 au
Largest scale	0.48 arcsec (70 au)
Exposure / release	8056 s (2.24 h) / 2020-10-18
Package	2018.1.01755.S_uid___A001_X133d_X4079_001_of_001.tar (0.02 GB)
FITS image	member.uid___A001_X133d_X4079.dm_tau_sci.spw19-21-23-25.mfs.I.manual.image.pbcor.fits
Archive observations	34 total; finest 0.0238 arcsec; used #1
Stellar parameters	Stellar parameters (Bosschaart+26 catalogue stellar mass; verify): L* = ? Lsun, M* = 0.29 Msun

Geometry and radial profile

Inclination / PA	35.0 / 157.0 deg
Geometry source	catalogue (literature) values.
Peak of mean profile	0.197 mJy/beam at r = 24.3 au
R50 / R90	32 / 81 au (8.2 / 20.7 beams)
Outer edge (>= 3 sigma)	55 au
Integrated flux	41.7 mJy (48-49% of catalogue expectation)
Image noise / dyn. range	0.0114 mJy/beam / 9
Shape	the profile is ring-like, not centrally peaked. The maximum is at 24 au and the innermost bin has only 23% of it, which points to a central cavity: the mean rises to half its peak at ~22 au.

Substructure (baseline-subtracted profile)

type	radius	residual		modulation	significance	
gap	r= 13.6 au	resid=-0.0776	mJy/b	modul=-111.9%	signif= -39.3 sigma	SIGNIFICANT
ring	r= 24.0 au	resid=0.113	mJy/b	modul=+135.3%	signif= +46.0 sigma	SIGNIFICANT
gap	r= 41.8 au	resid=-0.0271	mJy/b	modul=-43.4%	signif= -16.9 sigma	SIGNIFICANT

Sequence	G 14 au -> R 24 au -> G 42 au
Published features	none stored in the catalogue

Gaussian ring fit

Model / chi2_red tapered power law + 3 Gaussian ring(s); chi2_red = 2.21

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed
1	r=23.3+/-0.1	amp=0.137+/-0.019	S/N=7.1	FWHM_int=3.6 +/- 0.6	-
2	r=27.9+/-0.7	amp=0.104+/-0.0094	S/N=11.1	FWHM_int=9.5 +/- 1.0	det
3	r=37.9+/-0.1	amp=0.0404+/-0.0025	S/N=16.4	FWHM_int=16.5 +/- 0.8	-

Brightness temperature and optical depth

T_d, T_b/T_d and tau not computed: no stellar luminosity stored for this target.

Azimuthal asymmetry

ring	radius	m=1 amplitude		m=2 amplitude	
r=	23.3 au	m=1:	10.4% (S/N 5.8)	m=2:	5.8% (S/N 3.2)
r=	27.9 au	m=1:	3.9% (S/N 2.6)	m=2:	3.2% (S/N 2.1)
r=	37.9 au	m=1:	8.3% (S/N 3.2)	m=2:	4.6% (S/N 1.8)

Gap-opening planet masses

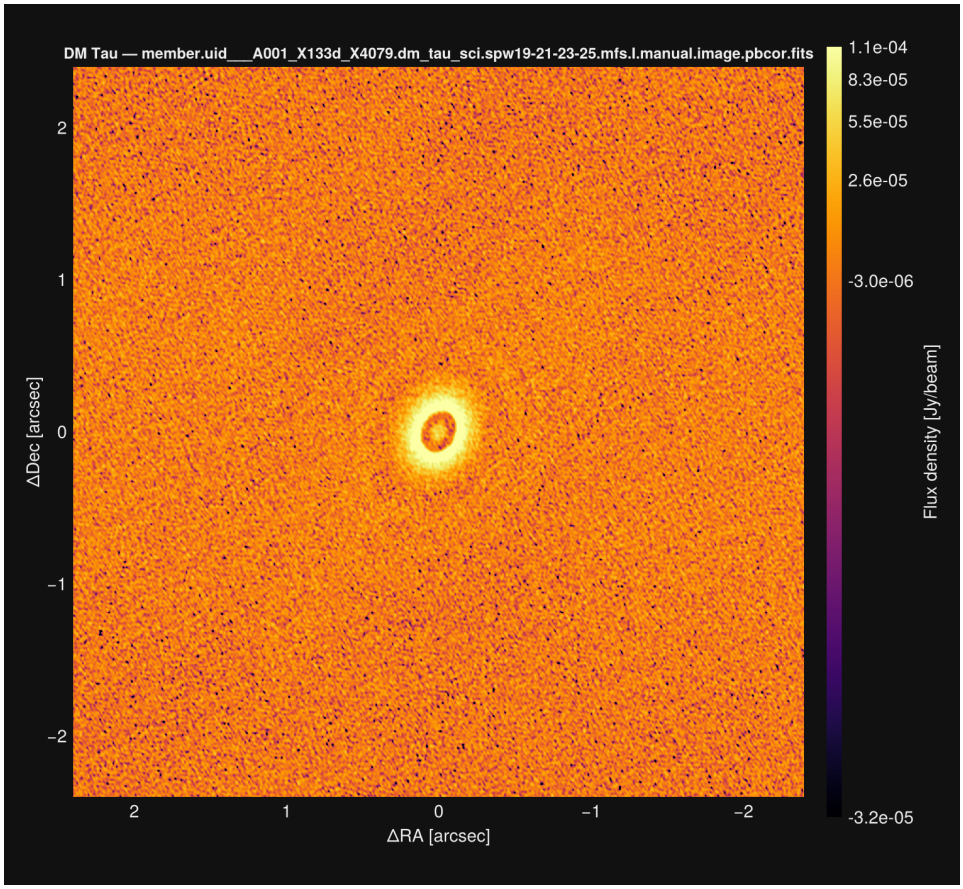
Not available (needs L* and M*, which are not stored for this target; pass lstar= and mstar=).

Pipeline warnings and notes

- the feature(s) at 38 au are shoulders (the raw-profile peak finder finds no peak or dip within 1 beam): an amplitude measured against a shoulder baseline can be large without a real azimuthal structure; the percentages of the local flux are the safer numbers.

- the image is spatially filtered (flux check above): arc contrasts and asymmetry amplitudes describe only the emission that survived the missing short spacings.

Continuum image



16. WaOph 6

Resolved

Well resolved (90% of the flux within 9.8 beams of 8.9 au). 2 ring(s) [29, 56 au] and 1 gap(s) [42 au] detected at >= 3 sigma. Recovered 0 of 2 testable published features. Image holds 68-72% of the expected catalogue flux. Negative bowl present: missing short spacings bias sizes, contrasts and temperatures.

Target

Coordinates (J2000)	16:48:45.63 -14:16:35.8 (RA 252.19014, Dec -14.27662 deg, J2000)
Distance	123.0 pc
Class	T Tauri analogue
Versus TW Hya	Closest analogue: axisymmetric rings/gaps; Bosschaart+26: RD, R68 103 au
Catalogue	Bosschaart+26 (WaOph6): class RD (ring disc); R68 103 au; F(1.33 mm) 161 mJy; M_dust 25 Earth masses; M* 0.68 Msun; distance 123 pc; this image: 68%-flux radius 60 au

Observation used

Project / band	2016.1.00484.L / Band 6 (238.9 GHz)
Beam	0.0725 x 0.0356 arcsec = 8.9 x 4.4 au
Archive resolution	0.0373 arcsec (4.6 au
Largest scale	1.34 arcsec (164 au)
Exposure / release	4137 s (1.15 h) / 2018-10-16
Package	2016.1.00484.L_uid___A001_X8c5_X6a_001_of_001.tar (1.45 GB)
FITS image	member.uid___A001_X8c5_X6a.ari_l.Wa_Oph_6_sci.spw0_1_2_3_238897MHz.12m.cont.l.pbcor.fits
Archive observations	10 total; finest 0.0373 arcsec; used #1
Stellar parameters	Stellar parameters (Bosschaart+26 catalogue stellar mass; verify): L* = ? Lsun, M* = 0.68 Msun

Geometry and radial profile

Inclination / PA	47.3 / 174.2 deg
Geometry source	catalogue (literature) values.
Peak of mean profile	3.59 mJy/beam at r = 2.2 au
R50 / R90	42 / 87 au (4.8 / 9.8 beams)
Outer edge (>= 3 sigma)	114 au
Integrated flux	129.7 mJy (68-72% of catalogue expectation)
Image noise / dyn. range	0.0274 mJy/beam / 109
Shape	the emission is centrally peaked (the maximum lies at 2 au, within ~1.5 beams of the star), like TW Hya: a smooth, steeply falling profile on which any rings are shoulders and dips.

Substructure (baseline-subtracted profile)

type	radius	residual		modulation	significance	
ring	r= 29.1 au	resid=0.107	mJy/b	modul=+18.3%	signif= +13.8 sigma	SIGNIFICANT
gap	r= 42.5 au	resid=-0.0328	mJy/b	modul=-8.9%	signif= -8.1 sigma	SIGNIFICANT
ring	r= 55.9 au	resid=0.0199	mJy/b	modul=+8.4%	signif= +3.5 sigma	SIGNIFICANT

Sequence	R 29 au -> G 42 au -> R 56 au
Published features	recovered 0 of 2 testable

Gaussian ring fit

Model / chi2_red tapered power law + 3 Gaussian ring(s); chi2_red = 0.97

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed	
1	r=30.9+/-1.0	amp=0.203+/-0.068	S/N=3.0	FWHM_int=13.4 +/- 3.3	det	<3sig
2	r=51.7+/-2.1	amp=0.219+/-0.037	S/N=5.9	FWHM_int=32.8 +/- 8.6	det	
3	r=83.0+/-5.1	amp=0.111+/-0.028	S/N=4.0	FWHM_int=33.3 +/- 4.9	lit	

Brightness temperature and optical depth

T_d, T_b/T_d and tau not computed: no stellar luminosity stored for this target.

Azimuthal asymmetry

ring	radius	m=1 amplitude		m=2 amplitude	
r=	29.1 au	m=1: 36.9%	(S/N 4.7)	m=2: 112.0%	(S/N 14.2)
r=	51.7 au	m=1: 8.9%	(S/N 6.0)	m=2: 21.2%	(S/N 14.2)
r=	83.0 au	m=1: 7.7%	(S/N 3.3)	m=2: 24.3%	(S/N 10.6)

Gap-opening planet masses

Not available (needs L* and M*, which are not stored for this target; pass lstar= and mstar=).

Pipeline warnings and notes

- the smooth disk collapsed to the unresolved core (characteristic radius 24 au < 3 beams): the extended emission is shared among the ring components, so overlapping neighbours (shoulders) have degenerate widths and amplitudes; trust the isolated rings, not the blended ones.

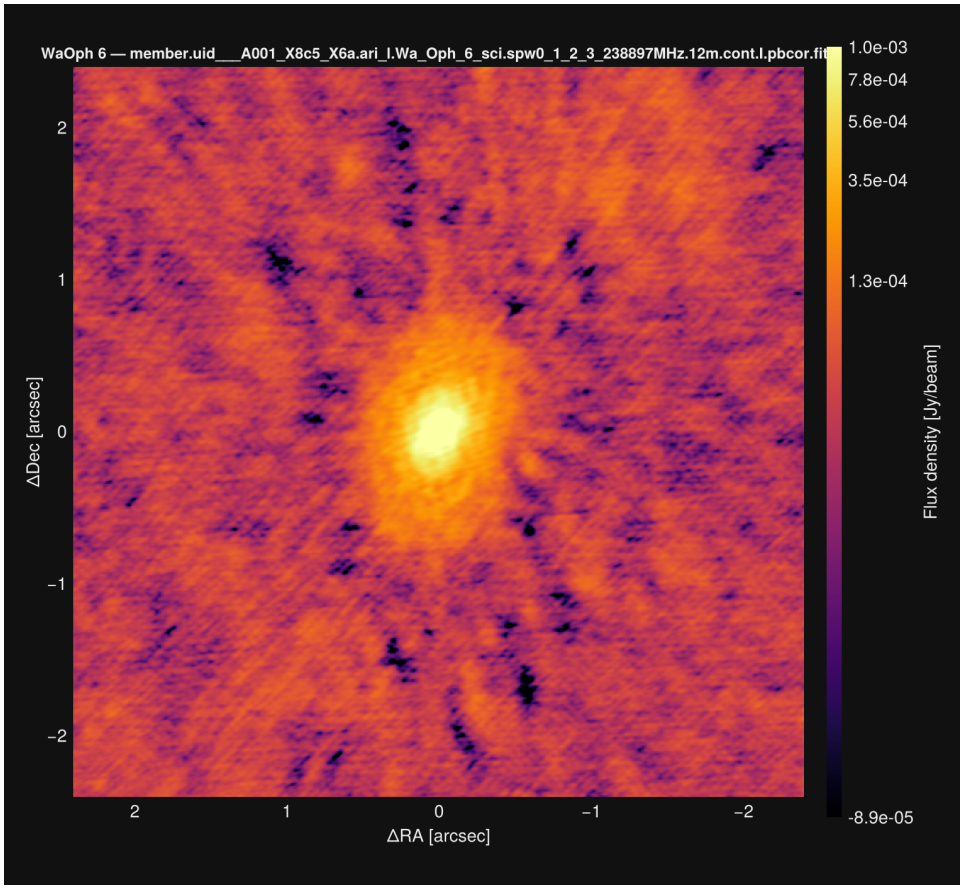
- 1 ring(s) detected in the profile at 3 sigma or more (29 au) are covered only by a fitted component with amplitude S/N < 3; the fit is otherwise consistent with the profile, so it is used, and for the asymmetry and temperature steps those rings are taken from the detection (position and residual amplitude) instead of the weak component.

- the feature(s) at 29, 52, 83 au are shoulders (the raw-profile peak finder finds no peak or dip within 1 beam): an amplitude measured against a shoulder baseline can be large without a real azimuthal structure; the percentages of the local flux are the safer numbers.

- the image is spatially filtered (flux check above): arc contrasts and asymmetry amplitudes describe only the emission that survived the missing short spacings.

- Negative bowl: the azimuthal mean is significantly negative over r = 127-163 au (9 bins, down to -4.9 sigma, -0.0173 mJy/beam = -0.5% of the peak, at 136 au), which real emission cannot produce. It is the signature of missing short spacings (spatial filtering): the extended emission is partly resolved out, so sizes, contrasts, asymmetries and brightness temperatures are biased and the smooth baseline is underestimate...

Continuum image



17. BP Tau

Resolved

Well resolved (90% of the flux within 7.7 beams of 4.5 au). 1 ring(s) [33 au] and 2 gap(s) [17, 44 au] detected at >= 3 sigma. Image holds 52% of the expected catalogue flux. Negative bowl present: missing short spacings bias sizes, contrasts and temperatures.

Target

Coordinates (J2000)	04:19:15.83 +29:06:26.9 (RA 64.81597, Dec +29.10748 deg, J2000)
Distance	128.0 pc
Class	T Tauri analogue
Versus TW Hya	Closest analogue: axisymmetric rings/gaps; Bosschaart+26: CD, R68 29 au
Catalogue	Bosschaart+26 (BPTau): class CD (compact disc); R68 29 au; F(1.33 mm) 45 mJy; M_dust 7.8 Earth masses; M* 0.51 Msun; distance 128 pc; this image: 68%-flux radius 28 au

Observation used

Project / band	2019.1.00607.S / Band 6 (224.7 GHz)
Beam	0.0351 x 0.0201 arcsec = 4.5 x 2.6 au
Archive resolution	0.0239 arcsec (3.1 au
Largest scale	0.46 arcsec (59 au)
Exposure / release	2268 s (0.63 h) / 2022-12-02
Package	2019.1.00607.S_uid___A001_X14ee_X21_001_of_001.tar (0.32 GB)
FITS image	member.uid___A001_X14ee_X21.BP_Tau_sci.spw25_27_29_31.mfs.I.manual.image.pbcor.fits
Archive observations	6 total; finest 0.0239 arcsec; used #1
Stellar parameters	Stellar parameters (Bosschaart+26 catalogue stellar mass; verify): L* = ? Lsun, M* = 0.51 Msun

Geometry and radial profile

Inclination / PA	41.2 / 151.9 deg
Geometry source	FITTED FROM THE IMAGE (radial-symmetry criterion), accuracy about +-5 deg.
Peak of mean profile	0.216 mJy/beam at r = 1.1 au
R50 / R90	24 / 35 au (5.2 / 7.7 beams)
Outer edge (>= 3 sigma)	42 au
Integrated flux	23.2 mJy (52% of catalogue expectation)
Image noise / dyn. range	0.0148 mJy/beam / 30
Shape	the emission is centrally peaked (the maximum lies at 1 au, within ~1.5 beams of the star), like TW Hya: a smooth, steeply falling profile on which any rings are shoulders and dips.

Substructure (baseline-subtracted profile)

type	radius	residual	modulation	significance
gap	r= 16.8 au	resid=-0.00908 mJy/b	modul=-6.2%	signif= -4.8 sigma SIGNIFICANT
ring	r= 32.6 au	resid=0.0101 mJy/b	modul=+15.0%	signif= +3.5 sigma SIGNIFICANT
gap	r= 43.8 au	resid=-0.0115 mJy/b	modul=-65.6%	signif= -6.2 sigma SIGNIFICANT
ring	r= 57.1 au	resid=0.00392 mJy/b	modul=n/a	signif= +1.8 sigma noise-level

Sequence	G 17 au -> R 33 au -> G 44 au -> R 57 au
Published features	none stored in the catalogue

Gaussian ring fit

Model / chi2_red tapered power law + 2 Gaussian ring(s); chi2_red = 2.05

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed
1	r=0.4+/-0.2	amp=0.15+/-0.047	S/N=3.2	FWHM_int=< 2.5 (unresolved)	core
2	r=30.0+/-1.5	amp=0.0731+/-0.017	S/N=4.3	FWHM_int=14.8 +/- 1.7	det

Brightness temperature and optical depth

T_d, T_b/T_d and tau not computed: no stellar luminosity stored for this target.

Azimuthal asymmetry

ring radius	m=1 amplitude	m=2 amplitude
r= 30.0 au	m=1: 5.7% (S/N 2.3)	m=2: 13.5% (S/N 5.5)

Gap-opening planet masses

Not available (needs L* and M*, which are not stored for this target; pass lstar= and mstar=).

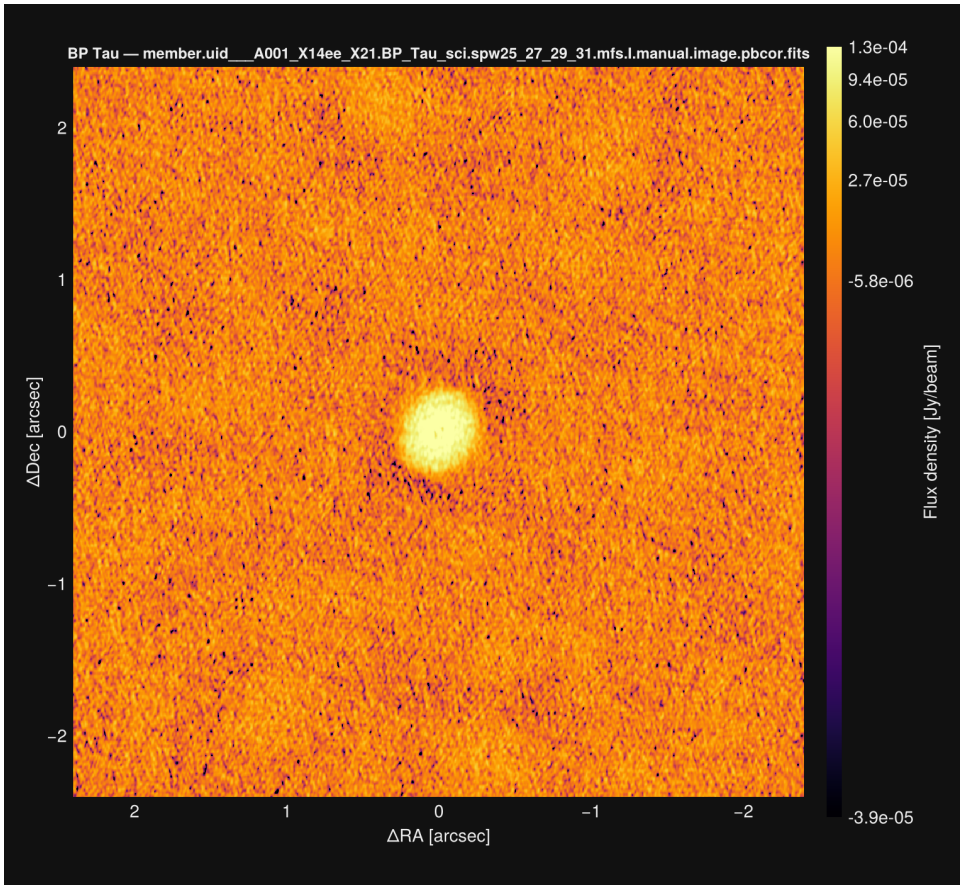
Pipeline warnings and notes

- the feature(s) at 30 au are shoulders (the raw-profile peak finder finds no peak or dip within 1 beam): an amplitude measured against a shoulder baseline can be large without a real azimuthal structure; the percentages of the local flux are the safer numbers.

- the image is spatially filtered (flux check above): arc contrasts and asymmetry amplitudes describe only the emission that survived the missing short spacings.

- Negative bowl: the azimuthal mean is significantly negative over r = 53-68 au (8 bins, down to -4.7 sigma, -0.00972 mJy/beam = -4.5% of the peak, at 64 au), which real emission cannot produce. It is the signature of missing short spacings (spatial filtering): the extended emission is partly resolved out, so sizes, contrasts, asymmetries and brightness temperatures are biased and the smooth baseline is underestimated.

Continuum image



18. GO Tau

Resolved

Well resolved (90% of the flux within 6.2 beams of 18.1 au). 1 ring(s) [73 au] and 2 gap(s) [57, 93 au] detected at >= 3 sigma. Recovered 4 of 4 testable published features. Image holds 71% of the expected catalogue flux. A finer observation (0.1041 arcsec) exists but the package exceeded the download limit; finer structure may exist.

Target

Coordinates (J2000)	04:43:03.08 +25:20:18.7 (RA 70.76282, Dec +25.33853 deg, J2000)
Distance	144.0 pc
Class	T Tauri analogue
Versus TW Hya	Closest analogue: axisymmetric rings/gaps; Bosschaart+26: RD, R68 100 au
Catalogue	Bosschaart+26 (GOTau): class RD (ring disc); R68 100 au; F(1.33 mm) 55 mJy; M_dust 12 Earth masses; M* 0.35 Msun; distance 143 pc; this image: 68%-flux radius 76 au

Observation used

Project / band	2016.1.01164.S / Band 6 (225.5 GHz)
Beam	0.1260 x 0.0923 arcsec = 18.1 x 13.3 au
Archive resolution	0.1153 arcsec (16.6 au
Largest scale	1.77 arcsec (255 au)
Exposure / release	254 s (0.07 h) / 2018-12-07
Package	2016.1.01164.S_uid___A001_X885_X1ec_001_of_001.tar (0.09 GB)
FITS image	member.uid___A001_X885_X1ec.GO_Tau_sci.spw67_73.mfs.I.manual.image.pbcor.fits
Archive observations	4 total; finest 0.1041 arcsec; used #2 (fallback: finer packages too large)
Stellar parameters	Stellar parameters (Long+18 Table 2; verify): L* = 0.21 Lsun, M* = 0.36 Msun

Geometry and radial profile

Inclination / PA	53.9 / 20.9 deg
Geometry source	catalogue (literature) values.
Peak of mean profile	5.52 mJy/beam at r = 3.6 au
R50 / R90	40 / 112 au (2.2 / 6.2 beams)
Outer edge (>= 3 sigma)	126 au
Integrated flux	39.2 mJy (71% of catalogue expectation)
Image noise / dyn. range	0.0754 mJy/beam / 164
Shape	the emission is centrally peaked (the maximum lies at 4 au, within ~1.5 beams of the star), like TW Hya: a smooth, steeply falling profile on which any rings are shoulders and dips.

Substructure (baseline-subtracted profile)

type	radius	residual	modulation	significance	
gap	r= 56.9 au	resid=-0.0659 mJy/b	modul=-23.5%	signif= -4.6 sigma	SIGNIFICANT
ring	r= 72.8 au	resid=0.0862 mJy/b	modul=+35.2%	signif= +3.2 sigma	SIGNIFICANT
gap	r= 93.0 au	resid=-0.0637 mJy/b	modul=-30.8%	signif= -4.0 sigma	SIGNIFICANT
ring	r= 109.1 au	resid=0.0358 mJy/b	modul=+22.6%	signif= +1.6 sigma	noise-level
gap	r= 126.0 au	resid=-0.0198 mJy/b	modul=-18.2%	signif= -1.2 sigma	noise-level

Sequence	G 57 au -> R 73 au -> G 93 au -> R 109 au -> G 126 au
Published features	recovered 4 of 4 testable

Gaussian ring fit

Model / chi2_red tapered power law + 2 Gaussian ring(s); chi2_red = 0.47

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed
1	r=73.1+/-1.1	amp=0.291+/-0.028	S/N=10.5	FWHM_int=12.1 +/- 4.3	det
2	r=108.2+/-2.3	amp=0.189+/-0.015	S/N=12.9	FWHM_int=34.9 +/- 5.7	det

Brightness temperature and optical depth

type	radius	brightness T	dust T	ratio	optical depth	scale height
ring	r= 73.1 au	T_b= 3.8 K	T_d= 10.0 K	T_b/T_d= 0.38	tau= 0.48	H/r=0.092
ring	r= 108.2 au	T_b= 3.2 K	T_d= 10.0 K	T_b/T_d= 0.32	tau= 0.39	H/r=0.112

Azimuthal asymmetry

ring radius	m=1 amplitude	m=2 amplitude
r= 73.1 au	m=1: 5.6% (S/N 0.8)	m=2: 14.1% (S/N 1.9)
r= 108.2 au	m=1: 0.6% (S/N 0.1)	m=2: 8.2% (S/N 1.0)

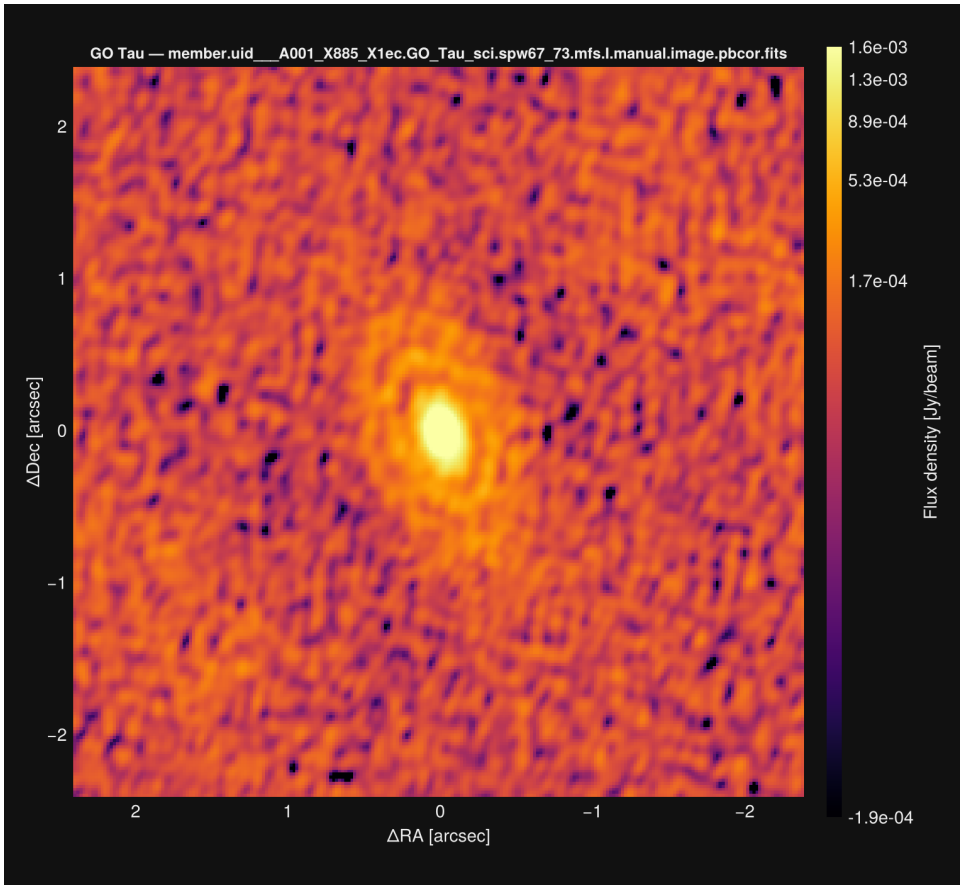
Gap-opening planet masses

Not available (no gaps were detected).

Pipeline warnings and notes

- the smooth disk collapsed to the unresolved core (characteristic radius 30 au < 3 beams): the extended emission is shared among the ring components, so overlapping neighbours (shoulders) have degenerate widths and amplitudes; trust the isolated rings, not the blended ones.
- the feature(s) at 73, 108 au are shoulders (the raw-profile peak finder finds no peak or dip within 1 beam): an amplitude measured against a shoulder baseline can be large without a real azimuthal structure; the percentages of the local flux are the safer numbers.
- the image is spatially filtered (flux check above): arc contrasts and asymmetry amplitudes describe only the emission that survived the missing short spacings.
- the notebook fell back from observation #1 to #2 because the finer packages exceeded the size limit.
- gap at 57 au and ring at 73 au are 16.0 au apart (< 1 beam): probably one blended feature or noise.
- gap at 93 au and ring at 109 au are 16.1 au apart (< 1 beam): probably one blended feature or noise.
- ring at 109 au and gap at 126 au are 16.9 au apart (< 1 beam): probably one blended feature or noise.

Continuum image



19. WSB 52

Resolved

Well resolved (90% of the flux within 5.3 beams of 6.0 au). 2 ring(s) [26, 53 au] and 1 gap(s) [34 au] detected at >= 3 sigma. Image holds 82-87% of the expected catalogue flux (consistent).

Target

Coordinates (J2000)	16:27:39.43 -24:39:15.5 (RA 246.91429, Dec -24.65431 deg, J2000)
Distance	135.0 pc
Class	T Tauri analogue
Versus TW Hya	Closest analogue: axisymmetric rings/gaps; Bosschaart+26: RD, R68 31 au
Catalogue	Bosschaart+26 (WSB52): class RD (ring disc); R68 31 au; F(1.33 mm) 67 mJy; M_dust 13 Earth masses; M* 0.48 Msun; distance 137 pc; this image: 68%-flux radius 23 au

Observation used

Project / band	2016.1.00484.L / Band 6 (239.0 GHz)
Beam	0.0444 x 0.0328 arcsec = 6.0 x 4.4 au
Archive resolution	0.0220 arcsec (3.0 au
Largest scale	1.00 arcsec (135 au)
Exposure / release	4064 s (1.13 h) / 2018-11-09
Package	2016.1.00484.L_uid___A001_X8c5_X66_001_of_001.tar (0.48 GB)
FITS image	member.uid___A001_X8c5_X66.WSB_52_sci.spw0_1_2_3_4_5_6_7.mfs.I.manual.image.pbcor.fits
Archive observations	2 total; finest 0.0220 arcsec; used #1
Stellar parameters	Stellar parameters (Andrews+18 Table 1 (MIST mass); verify): L* = 0.71 Lsun, M* = 0.48 Msun

Geometry and radial profile

Inclination / PA	54.4 / 138.4 deg
Geometry source	catalogue (literature) values.
Peak of mean profile	3.12 mJy/beam at r = 1.5 au
R50 / R90	19 / 32 au (3.3 / 5.3 beams)
Outer edge (>= 3 sigma)	38 au
Integrated flux	65.3 mJy (82-87% of catalogue expectation)
Image noise / dyn. range	0.0166 mJy/beam / 88
Shape	the emission is centrally peaked (the maximum lies at 2 au, within ~1.5 beams of the star), like TW Hya: a smooth, steeply falling profile on which any rings are shoulders and dips.

Substructure (baseline-subtracted profile)

type	radius	residual	modulation	significance	
ring	r= 25.5 au	resid=0.131 mJy/b	modul=+24.3%	signif= +24.2 sigma	SIGNIFICANT
gap	r= 34.5 au	resid=-0.0934 mJy/b	modul=-48.3%	signif= -31.1 sigma	SIGNIFICANT
ring	r= 53.0 au	resid=0.0523 mJy/b	modul=n/a	signif= +13.8 sigma	SIGNIFICANT

Sequence	R 26 au -> G 34 au -> R 53 au
Published features	none stored in the catalogue

Gaussian ring fit

Model / chi2_red tapered power law + 2 Gaussian ring(s); chi2_red = 0.93

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed
1	r=22.4+/-0.7	amp=0.56+/-0.12	S/N=4.6	FWHM_int=13.5 +/- 1.3	det
2	r=59.0+/-2.3	amp=0.0134+/-0.0017	S/N=8.0	FWHM_int=27.8 +/- 5.3	det

Brightness temperature and optical depth

type	radius	brightness T	dust T	ratio	optical depth	scale height
ring	r= 22.4 au	T_b= 17.2 K	T_d= 24.1 K	T_b/T_d= 0.72	tau= 1.26	H/r=0.068
gap	r= 34.5 au	T_b= 5.3 K	T_d= 19.4 K	T_b/T_d= 0.27	tau= 0.32	H/r=0.075
ring	r= 59.0 au	T_b= 2.7 K	T_d= 14.9 K	T_b/T_d= 0.18	tau= 0.20	H/r=0.086

Azimuthal asymmetry

ring radius	m=1 amplitude	m=2 amplitude
r= 22.4 au	m=1: 1.5% (S/N 2.0)	m=2: 6.2% (S/N 8.5)
r= 59.0 au	m=1: 64.3% (S/N 4.8)	m=2: 48.1% (S/N 3.6)

Gap-opening planet masses

Not evaluated: 34 au: no local maximum outward of the gap minimum at 34.5 au (the profile does not rise there)

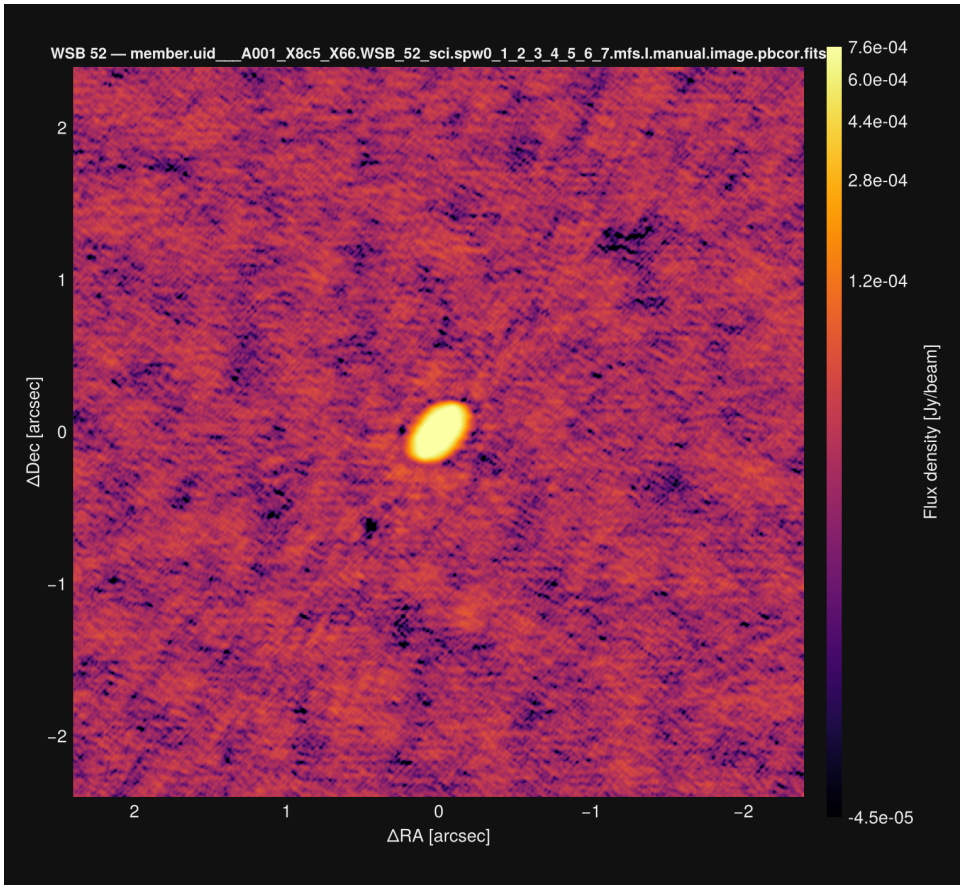
Pipeline warnings and notes

- the smooth disk collapsed to the unresolved core (characteristic radius 15 au < 3 beams): the extended emission is shared among the ring components, so overlapping neighbours (shoulders) have degenerate widths and amplitudes; trust the isolated rings, not the blended ones.

- at 59 au the local azimuthal-mean flux is below 2x the image rms per beam (0.0166 mJy/beam): the ring is detected only by averaging over many beams, and the S/N and false-alarm values above assume white noise at the beam scale. Correlated residuals (sidelobes, large-scale noise, weak background sources) are not in that noise and can make a dipole or m=2 pattern of this size, so these asymmetries are unconfirmed; an '...

- the feature(s) at 22, 59 au are shoulders (the raw-profile peak finder finds no peak or dip within 1 beam): an amplitude measured against a shoulder baseline can be large without a real azimuthal structure; the percentages of the local flux are the safer numbers.

Continuum image



20. Elias 27

Resolved

Well resolved (90% of the flux within 29.3 beams of 4.8 au). 1 ring(s) [35 au] and 1 gap(s) [66 au] detected at >= 3 sigma. Recovered 1 of 2 testable published features. Image holds 46-48% of the expected catalogue flux. Negative bowl present: missing short spacings bias sizes, contrasts and temperatures.

Target

Coordinates (J2000)	16:26:45.02 -24:23:07.8 (RA 246.68760, Dec -24.38550 deg, J2000)
Distance	116.0 pc
Class	T Tauri analogue
Versus TW Hya	Spirals; less suited to an azimuthal average; Bosschaart+26: RD, R68 254 au
Catalogue	Bosschaart+26 (Elias27): class RD (ring disc); R68 254 au; F(1.33 mm) 313 mJy; M_dust 44 Earth masses; M* 0.49 Msun; distance 116 pc; this image: 68%-flux radius 90 au

Observation used

Project / band	2016.1.00484.L / Band 6 (238.9 GHz)
Beam	0.0413 x 0.0246 arcsec = 4.8 x 2.9 au
Archive resolution	0.0201 arcsec (2.3 au
Largest scale	1.06 arcsec (123 au)
Exposure / release	4064 s (1.13 h) / 2018-10-18
Package	2016.1.00484.L_uid___A001_X8c5_X52_001_of_001.tar (1.02 GB)
FITS image	member.uid___A001_X8c5_X52.ari_l.Elias_27_sci.spw0_1_2_3_238892MHz.12m.cont.l.pbcor.fits
Archive observations	1 total; finest 0.0201 arcsec; used #1
Stellar parameters	Stellar parameters (Andrews+18 (approx.); verify): L* = 0.91 Lsun, M* = 0.49 Msun

Geometry and radial profile

Inclination / PA	56.2 / 118.8 deg
Geometry source	catalogue (literature) values.
Peak of mean profile	2.4 mJy/beam at r = 1.2 au
R50 / R90	52 / 140 au (10.8 / 29.3 beams)
Outer edge (>= 3 sigma)	184 au
Integrated flux	170.3 mJy (46-48% of catalogue expectation)
Image noise / dyn. range	0.0225 mJy/beam / 38
Shape	the emission is centrally peaked (the maximum lies at 1 au, within ~1.5 beams of the star), like TW Hya: a smooth, steeply falling profile on which any rings are shoulders and dips.

Substructure (baseline-subtracted profile)

type	radius	residual	modulation	significance	
ring	r= 34.8 au	resid=0.119 mJy/b	modul=+75.1%	signif= +22.6 sigma	SIGNIFICANT
gap	r= 66.0 au	resid=-0.0308 mJy/b	modul=-41.5%	signif= -13.0 sigma	SIGNIFICANT

Sequence	R 35 au -> G 66 au
Published features	recovered 1 of 2 testable

Gaussian ring fit

Model / chi2_red tapered power law + 2 Gaussian ring(s); chi2_red = 9.27

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed
1	r=30.0+/-3.4	amp=0.119+/-0.028	S/N=4.2	FWHM_int=33.0 +/- 8.3	det
2	r=90.8+/-2.6	amp=0.0214+/-0.006	S/N=3.6	FWHM_int=20.7 +/- 7.7	lit

Brightness temperature and optical depth

type	radius	brightness T	dust T	ratio	optical depth	scale height
ring	r= 30.0 au	T_b= 11.2 K	T_d= 22.2 K	T_b/T_d= 0.51	tau= 0.71	H/r=0.074
gap	r= 66.0 au	T_b= 4.4 K	T_d= 15.0 K	T_b/T_d= 0.29	tau= 0.35	H/r=0.090
ring	r= 90.8 au	T_b= 4.9 K	T_d= 12.8 K	T_b/T_d= 0.38	tau= 0.48	H/r=0.098

Azimuthal asymmetry

ring radius	m=1 amplitude	m=2 amplitude
r= 30.0 au	m=1: 8.8% (S/N 4.6)	m=2: 2.8% (S/N 1.4)
r= 90.8 au	m=1: 22.4% (S/N 2.9)	m=2: 99.7% (S/N 12.9)

Gap-opening planet masses

Not evaluated: 66 au: no local maximum outward of the gap minimum at 66.0 au (the profile does not rise there)

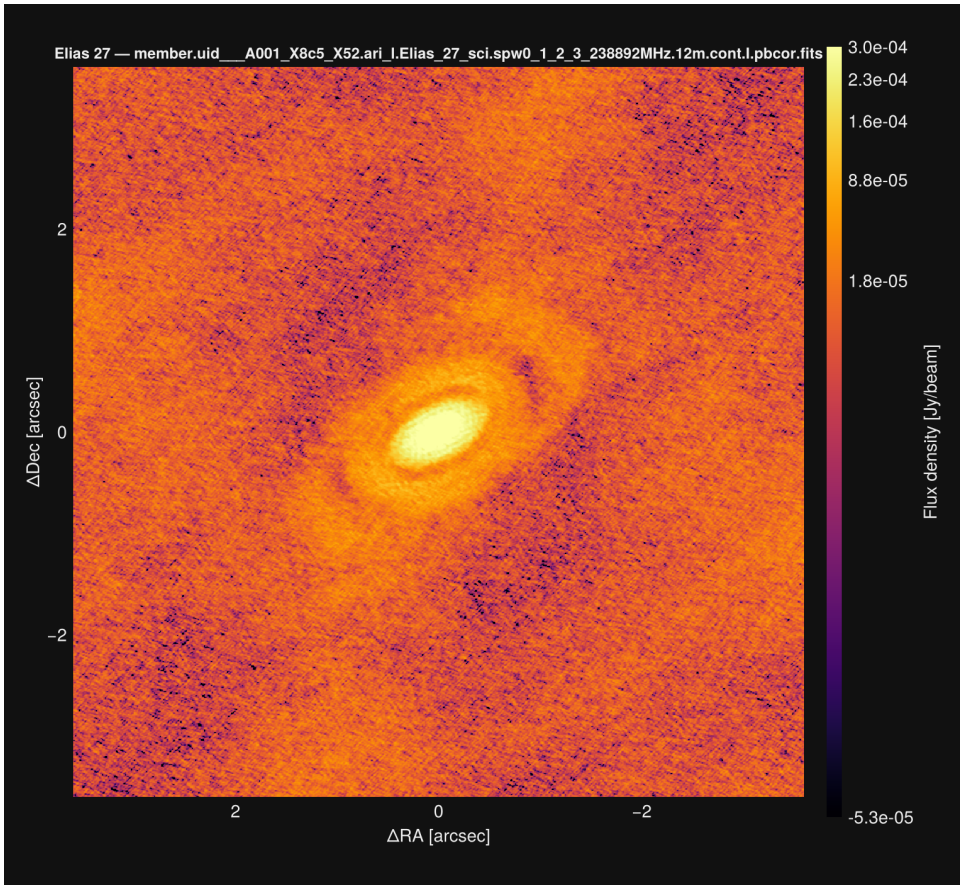
Pipeline warnings and notes

- the feature(s) at 30, 91 au are shoulders (the raw-profile peak finder finds no peak or dip within 1 beam): an amplitude measured against a shoulder baseline can be large without a real azimuthal structure; the percentages of the local flux are the safer numbers.

- the image is spatially filtered (flux check above): arc contrasts and asymmetry amplitudes describe only the emission that survived the missing short spacings.

- Negative bowl: the azimuthal mean is significantly negative over r = 296-347 au (22 bins, down to -4.4 sigma, -0.00731 mJy/beam = -0.3% of the peak, at 340 au), which real emission cannot produce. It is the signature of missing short spacings (spatial filtering): the extended emission is partly resolved out, so sizes, contrasts, asymmetries and brightness temperatures are biased and the smooth baseline is underestima...

Continuum image



21. IM Lup

Resolved

DATA CHECK: Catalogue entry is for 'Sz82', not 'IM Lup': published values (R68, flux, M_dust, ring radii) may belong to another source.

Well resolved (90% of the flux within 21.2 beams of 4.8 au). 1 ring(s) [40 au] and 1 gap(s) [13 au] detected at >= 3 sigma. Recovered 0 of 1 testable published features. Image holds 33-35% of the expected catalogue flux.

Target

Coordinates (J2000)	15:56:09.19 -37:56:06.0 (RA 239.03830, Dec -37.93500 deg, J2000)
Distance	158.0 pc
Class	T Tauri analogue
Versus TW Hya	Closest analogue: axisymmetric rings/gaps; Bosschaart+26: RD, R68 265 au
Catalogue	Bosschaart+26 (Sz82): class RD (ring disc); R68 265 au; F(1.33 mm) 205 mJy; M_dust 54 Earth masses; M* 0.95 Msun; distance 159 pc; this image: 68%-flux radius 54 au

Observation used

Project / band	2016.1.00484.L / Band 6 (239.0 GHz)
Beam	0.0304 x 0.0240 arcsec = 4.8 x 3.8 au
Archive resolution	0.0230 arcsec (3.6 au
Largest scale	0.58 arcsec (91 au)
Exposure / release	4170 s (1.16 h) / 2019-02-01
Package	2016.1.00484.L_uid___A001_X8c5_X72_001_of_001.tar (0.84 GB)
FITS image	member.uid___A001_X8c5_X72.IM_Lupi_sci.spw0_1_2_3_4_5_6_7.mfs.I.manual.image.pbcor.fits
Archive observations	57 total; finest 0.0230 arcsec; used #1
Stellar parameters	Stellar parameters (Andrews+18 (approx.); verify): L* = 2.57 Lsun, M* = 1.10 Msun

Geometry and radial profile

Inclination / PA	47.5 / 144.5 deg
Geometry source	catalogue (literature) values.
Peak of mean profile	2.45 mJy/beam at r = 1.2 au
R50 / R90	32 / 102 au (6.7 / 21.2 beams)
Outer edge (>= 3 sigma)	107 au
Integrated flux	81.8 mJy (33-35% of catalogue expectation)
Image noise / dyn. range	0.0158 mJy/beam / 46
Shape	the emission is centrally peaked (the maximum lies at 1 au, within ~1.5 beams of the star), like TW Hya: a smooth, steeply falling profile on which any rings are shoulders and dips.

Substructure (baseline-subtracted profile)

type	radius	residual		modulation	significance	
gap	r= 13.2 au	resid=-0.191	mJy/b	modul=-22.6%	signif= -48.3 sigma	SIGNIFICANT
ring	r= 39.5 au	resid=0.109	mJy/b	modul=n/a	signif= +30.4 sigma	SIGNIFICANT

Sequence	G 13 au -> R 40 au
Published features	recovered 0 of 1 testable

Gaussian ring fit

Model / chi2_red tapered power law + 2 Gaussian ring(s); chi2_red = 3.41

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed
1	r=6.0+/-2.6	amp=0.381+/-0.086	S/N=4.4	FWHM_int=16.9 +/- 3.3	core
2	r=137.3+/-1.7	amp=0.00845+/-0.0021	S/N=4.0	FWHM_int=12.9 +/- 4.2	lit

Brightness temperature and optical depth

type	radius	brightness T	dust T	ratio	optical depth	scale height
gap	r= 13.2 au	T_b= 24.5 K	T_d= 43.4 K	T_b/T_d= 0.56	tau= 0.83	H/r=0.046
ring	r= 137.3 au	T_b= 3.2 K	T_d= 13.4 K	T_b/T_d= 0.24	tau= 0.27	H/r=0.082

Azimuthal asymmetry

ring radius	m=1 amplitude	m=2 amplitude
r= 39.5 au	m=1: 4.8% (S/N 1.4)	m=2: 29.6% (S/N 8.7)
r= 137.3 au	m=1: 42.1% (S/N 2.9)	m=2: 25.5% (S/N 1.7)

Gap-opening planet masses

Not evaluated: 13 au: no local maximum outward of the gap minimum at 15.6 au (the profile does not rise there)

Pipeline warnings and notes

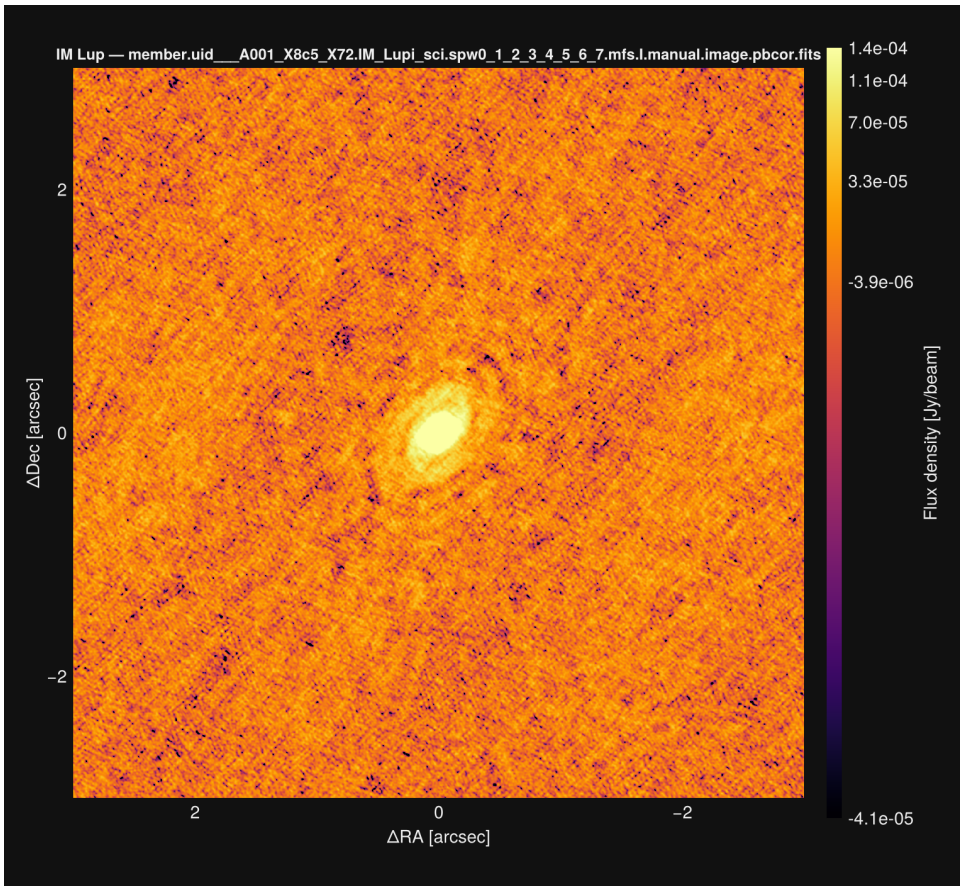
- 1 ring(s) detected in the profile at 3 sigma or more (40 au) have no fitted component, but each is a shoulder (the raw-profile peak finder finds no peak or dip within 1 beam) of the smooth disc, not an isolated ring; the fit is not counted as disagreeing with the profile. For the asymmetry and temperature steps those features are taken from the detection (position and residual amplitude).

- at 137 au the local azimuthal-mean flux is below 2x the image rms per beam (0.0158 mJy/beam): the ring is detected only by averaging over many beams, and the S/N and false-alarm values above assume white noise at the beam scale. Correlated residuals (sidelobes, large-scale noise, weak background sources) are not in that noise and can make a dipole or m=2 pattern of this size, so these asymmetries are unconfirmed; an ...

- the feature(s) at 40, 137 au are shoulders (the raw-profile peak finder finds no peak or dip within 1 beam): an amplitude measured against a shoulder baseline can be large without a real azimuthal structure; the percentages of the local flux are the safer numbers.

- the image is spatially filtered (flux check above): arc contrasts and asymmetry amplitudes describe only the emission that survived the missing short spacings.

Continuum image



22. HD 100546

Resolved

Well resolved (90% of the flux within 19.2 beams of 5.0 au). 1 ring(s) [24 au] and 1 gap(s) [46 au] detected at >= 3 sigma. Image holds 110-165% of the expected catalogue flux (consistent).

Target

Coordinates (J2000)	11:33:25.44 -70:11:41.3 (RA 173.35600, Dec -70.19480 deg, J2000)
Distance	108.0 pc
Class	Herbig
Versus TW Hya	Higher-mass star; warmer, larger disk; Bosschaart+26: TD, R68 59 au
Catalogue	Bosschaart+26 (HD100546): class TD (transition disc); R68 59 au; F(1.33 mm) 379 mJy; M_dust 48 Earth masses; M* 2.20 Msun; distance 110 pc; this image: 68%-flux radius 41 au

Observation used

Project / band	2015.1.00806.S / Band 7 (338.2 GHz)
Beam	0.0462 x 0.0314 arcsec = 5.0 x 3.4 au
Archive resolution	0.0217 arcsec (2.3 au
Largest scale	0.47 arcsec (51 au)
Exposure / release	1796 s (0.50 h) / 2017-03-06
Package	2015.1.00806.S_uid___A001_X2de_X155_001_of_001.tar (2.52 GB)
FITS image	member.uid___A001_X2de_X155.ari_l.hd_100546_sci.spw0_1_2_3_338172MHz.12m.cont.l.pbcor.fits
Archive observations	26 total; finest 0.0217 arcsec; used #1
Stellar parameters	Stellar parameters (A&A 2023 aa43601-22 Table 1 (d = 110 pc); VERIFY; verify): L* = 25.00 Lsun, M* = 2.13 Msun

Geometry and radial profile

Inclination / PA	38.8 / 146.2 deg
Geometry source	FITTED FROM THE IMAGE (radial-symmetry criterion), accuracy about +-5 deg.
Peak of mean profile	6.48 mJy/beam at r = 23.6 au
R50 / R90	34 / 96 au (6.7 / 19.2 beams)
Outer edge (>= 3 sigma)	83 au
Integrated flux	1409.6 mJy (110-165% of catalogue expectation)
Image noise / dyn. range	0.165 mJy/beam / 20
Shape	the profile is ring-like, not centrally peaked. The maximum is at 24 au and the innermost bin has only 34% of it, which points to a central cavity: the mean rises to half its peak at ~16 au.

Substructure (baseline-subtracted profile)

type	radius	residual	modulation	significance
ring	r= 24.2 au	resid=1.28 mJy/b	modul=+24.0%	signif= +29.7 sigma SIGNIFICANT
gap	r= 45.9 au	resid=-0.412 mJy/b	modul=-26.3%	signif= -17.9 sigma SIGNIFICANT

Sequence	R 24 au -> G 46 au
Published features	none stored in the catalogue

Gaussian ring fit

Model / chi2_red tapered power law + 2 Gaussian ring(s); chi2_red = 1.34

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed
1	r=21.7+/-0.3	amp=1.71+/-0.16	S/N=10.5	FWHM_int=8.1 +/- 1.1	-
2	r=28.0+/-0.2	amp=5.21+/-0.13	S/N=41.4	FWHM_int=23.2 +/- 0.3	det

Brightness temperature and optical depth

type	radius	brightness T	dust T	ratio	optical depth	scale height
ring	r= 21.7 au	T_b= 53.6 K	T_d= 60.6 K	T_b/T_d= 0.88	tau= 2.16	H/r=0.049
ring	r= 28.0 au	T_b= 51.9 K	T_d= 52.1 K	T_b/T_d= 1.00	tau= 5.53	H/r=0.053
gap	r= 45.9 au	T_b= 15.2 K	T_d= 41.1 K	T_b/T_d= 0.37	tau= 0.46	H/r=0.060

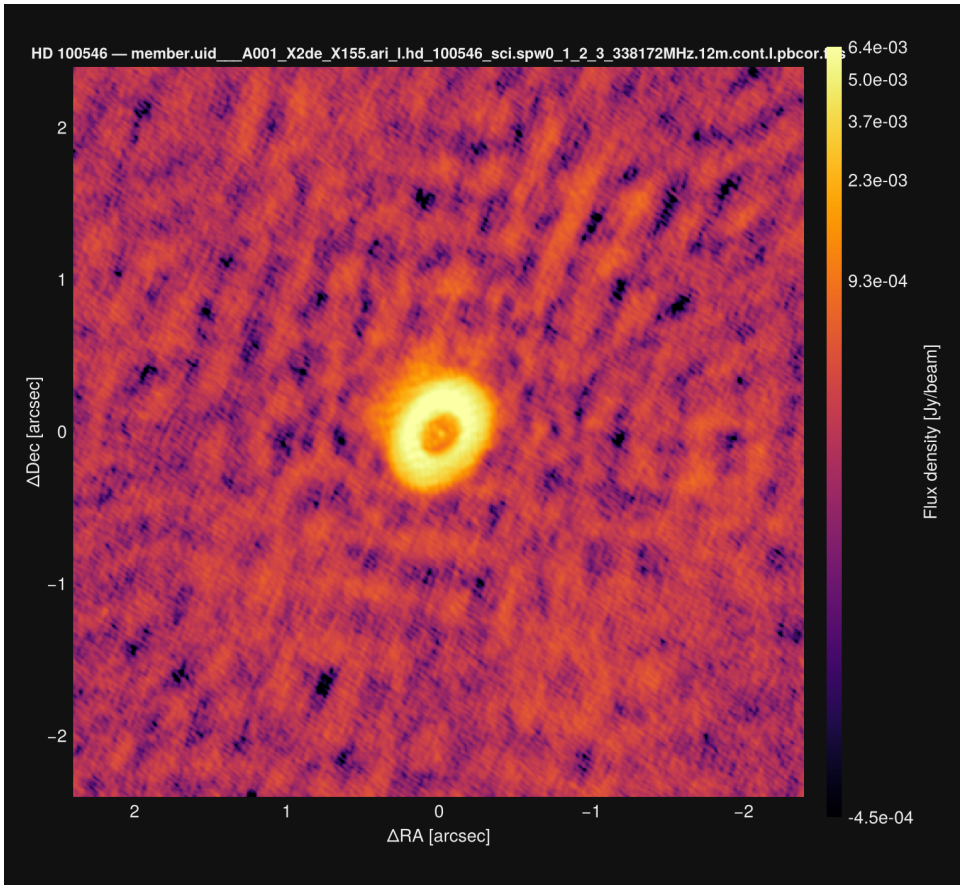
Azimuthal asymmetry

ring radius	m=1 amplitude	m=2 amplitude
r= 21.7 au	m=1: 85.8% (S/N 41.1)	m=2: 23.6% (S/N 11.3)
r= 28.0 au	m=1: 19.7% (S/N 52.5)	m=2: 5.0% (S/N 13.3)

Gap-opening planet masses

Not evaluated: 46 au: no local maximum outward of the gap minimum at 48.4 au (the profile does not rise there)

Continuum image



23. IRS 63

Resolved

DATA CHECK: Catalogue entry is for 'ODISEA_C4_130', not 'IRS 63': published values (R68, flux, M_dust, ring radii) may belong to another source.

Well resolved (90% of the flux within 13.2 beams of 6.0 au). 1 ring(s) [28 au] and 1 gap(s) [80 au] detected at >= 3 sigma. Image holds 68-70% of the expected catalogue flux. Negative bowl present: missing short spacings bias sizes, contrasts and temperatures.

Target

Coordinates (J2000)	16:31:35.65 -24:01:29.6 (RA 247.89854, Dec -24.02489 deg, J2000)
Distance	144.0 pc
Class	Young/embedded
Versus TW Hya	Class I; embedded, rings (eDisk); Bosschaart+26: RD, R68 32 au
Catalogue	Bosschaart+26 (ODISEA_C4_130): class RD (ring disc); R68 32 au; F(1.33 mm) 335 mJy; M_dust 64 Earth masses; distance 135 pc; this image: 68%-flux radius 50 au

Observation used

Project / band	2015.1.01512.S / Band 6 (233.0 GHz)
Beam	0.0417 x 0.0249 arcsec = 6.0 x 3.6 au
Archive resolution	0.0229 arcsec (3.3 au
Largest scale	0.76 arcsec (109 au)
Exposure / release	6641 s (1.84 h) / 2018-10-19
Package	2015.1.01512.S_uid___A001_X2d1_X96_001_of_001.tar (0.12 GB)
FITS image	member.uid___A001_X2d1_X96.ari_l.IRS_63_sci.spw17_19_21_23_233009MHz.12m.cont.I.pbcor.fits
Archive observations	36 total; finest 0.0229 arcsec; used #1
Stellar parameters	Stellar parameters: none stored for this target (pass lstar= / mstar=).

Geometry and radial profile

Inclination / PA	38.8 / 159.4 deg
Geometry source	FITTED FROM THE IMAGE (radial-symmetry criterion), accuracy about +-5 deg.
Peak of mean profile	3.33 mJy/beam at r = 1.5 au
R50 / R90	37 / 79 au (6.2 / 13.2 beams)
Outer edge (>= 3 sigma)	86 au
Integrated flux	250.5 mJy (68-70% of catalogue expectation)
Image noise / dyn. range	0.022 mJy/beam / 107
Shape	the emission is centrally peaked (the maximum lies at 2 au, within ~1.5 beams of the star), like TW Hya: a smooth, steeply falling profile on which any rings are shoulders and dips.

Substructure (baseline-subtracted profile)

type	radius	residual		modulation	significance	
ring	r= 28.5 au	resid=0.0867	mJy/b	modul=+13.0%	signif= +16.4 sigma	SIGNIFICANT
gap	r= 79.5 au	resid=-0.0132	mJy/b	modul=-28.9%	signif= -6.3 sigma	SIGNIFICANT

Sequence	R 29 au -> G 79 au
Published features	none stored in the catalogue

Gaussian ring fit

Model / chi2_red	tapered power law + 0 Gaussian ring(s); chi2_red = 40.68

No ring components fitted.

Brightness temperature and optical depth

T_d, T_b/T_d and tau not computed: no stellar luminosity stored for this target.

Azimuthal asymmetry

ring radius	m=1 amplitude	m=2 amplitude
r= 28.5 au	m=1: 26.2% (S/N 4.0)	m=2: 49.7% (S/N 7.5)

Gap-opening planet masses

Not available (needs L* and M*, which are not stored for this target; pass lstar= and mstar=).

Pipeline warnings and notes

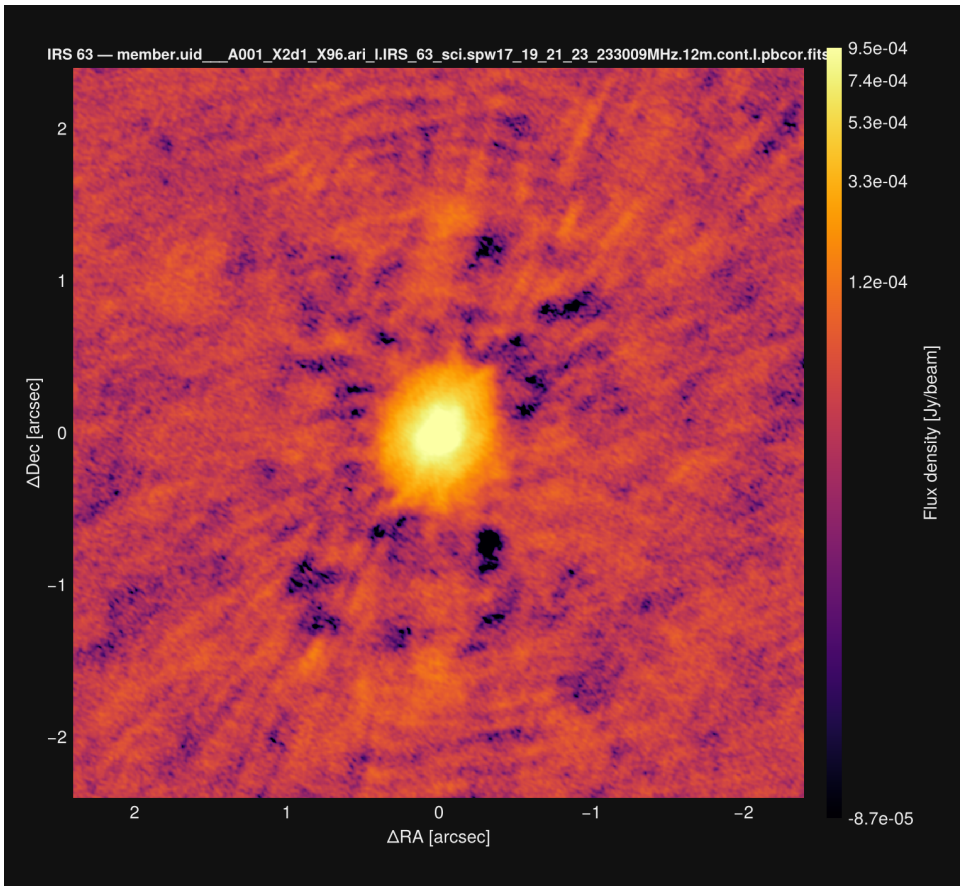
- 1 ring(s) detected in the profile at 3 sigma or more (29 au) have no fitted component, but each is a shoulder (the raw-profile peak finder finds no peak or dip within 1 beam) of the smooth disc, not an isolated ring; the fit is not counted as disagreeing with the profile. For the asymmetry and temperature steps those features are taken from the detection (position and residual amplitude).

- the feature(s) at 29 au are shoulders (the raw-profile peak finder finds no peak or dip within 1 beam): an amplitude measured against a shoulder baseline can be large without a real azimuthal structure; the percentages of the local flux are the safer numbers.

- the image is spatially filtered (flux check above): arc contrasts and asymmetry amplitudes describe only the emission that survived the missing short spacings.

- Negative bowl: the azimuthal mean is significantly negative over r = 97-134 au (13 bins, down to -9.4 sigma, -0.0247 mJy/beam = -0.7% of the peak, at 115 au), which real emission cannot produce. It is the signature of missing short spacings (spatial filtering): the extended emission is partly resolved out, so sizes, contrasts, asymmetries and brightness temperatures are biased and the smooth baseline is underestimate...

Continuum image



24. Sz 129

Resolved

Well resolved (90% of the flux within 13.2 beams of 3.7 au). 0 ring(s) [au] and 2 gap(s) [19, 61 au] detected at >= 3 sigma. Recovered 2 of 3 testable published features. Image holds 2.1% of the expected catalogue flux. Negative bowl present: missing short spacings bias sizes, contrasts and temperatures.

Target

Coordinates (J2000)	15:59:16.47 -41:57:10.3 (RA 239.81863, Dec -41.95286 deg, J2000)
Distance	160.0 pc
Class	T Tauri analogue
Versus TW Hya	Closest analogue: axisymmetric rings/gaps; Bosschaart+26: RD, R68 76 au
Catalogue	Bosschaart+26 (Sz129): class RD (ring disc); R68 76 au; F(1.33 mm) 76 mJy; M_dust 21 Earth masses; M* 0.78 Msun; distance 162 pc; this image: 68%-flux radius 36 au

Observation used

Project / band	2018.1.01054.S / Band 6 (224.5 GHz)
Beam	0.0228 x 0.0200 arcsec = 3.7 x 3.2 au
Archive resolution	0.0207 arcsec (3.3 au
Largest scale	0.35 arcsec (55 au)
Exposure / release	454 s (0.13 h) / 2020-10-17
Package	2018.1.01054.S_uid___A001_X133d_X292d_001_of_001.tar (0.01 GB)
FITS image	member.uid___A001_X133d_X292d.Sz129_sci.spw19_23_25_27.mfs.I.manual.image.pbcor.fits
Archive observations	12 total; finest 0.0207 arcsec; used #1
Stellar parameters	Stellar parameters (Andrews+18 Table 1 (MIST mass); verify): L* = 0.44 Lsun, M* = 0.83 Msun

Geometry and radial profile

Inclination / PA	34.1 / 151.0 deg
Geometry source	catalogue (literature) values.
Peak of mean profile	0.248 mJy/beam at r = 10.0 au
R50 / R90	26 / 48 au (7.2 / 13.2 beams)
Outer edge (>= 3 sigma)	52 au
Integrated flux	1.5 mJy (2.1% of catalogue expectation)
Image noise / dyn. range	0.0444 mJy/beam / 16
Shape	the profile is ring-like, not centrally peaked. The maximum is at 10 au and the innermost bin has only 5% of it, which points to a central cavity: the mean rises to half its peak at ~8 au.

Substructure (baseline-subtracted profile)

type	radius	residual	modulation	significance	
gap	r= 19.2 au	resid=-0.0285 mJy/b	modul=-16.1%	signif= -4.3 sigma	SIGNIFICANT
ring	r= 30.1 au	resid=0.00247 mJy/b	modul=+3.1%	signif= +0.3 sigma	noise-level
gap	r= 35.6 au	resid=-0.0119 mJy/b	modul=-19.3%	signif= -1.6 sigma	noise-level
ring	r= 47.4 au	resid=0.0225 mJy/b	modul=+62.2%	signif= +2.9 sigma	noise-level
gap	r= 61.3 au	resid=-0.0162 mJy/b	modul=n/a	signif= -3.4 sigma	SIGNIFICANT
ring	r= 68.3 au	resid=0.00706 mJy/b	modul=n/a	signif= +1.1 sigma	noise-level
gap	r= 116.4 au	resid=-0.00462 mJy/b	modul=n/a	signif= -2.0 sigma	noise-level

Sequence	G 19 au -> R 30 au -> G 36 au -> R 47 au -> G 61 au -> R 68 au -> G 116 au
Published features	recovered 2 of 3 testable

Gaussian ring fit

Model / chi2_red tapered power law (inner hole at 7.0 +/- 1.2 au) + 2 Gaussian ring(s); chi2_red = 2.21

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed
1	r=9.0+/-0.8	amp=0.0927+/-0.058	S/N=1.6	FWHM_int=< 6.1 (unresolved)	core <3sig
2	r=47.4+/-0.9	amp=0.0445+/-0.009	S/N=4.9	FWHM_int=9.2 +/- 2.6	det

Brightness temperature and optical depth

type	radius	brightness T	dust T	ratio	optical depth	scale height
ring	r= 47.4 au	T_b= 6.9 K	T_d= 14.8 K	T_b/T_d= 0.47	tau= 0.63	H/r=0.058

Azimuthal asymmetry

ring radius	m=1 amplitude	m=2 amplitude
r= 47.4 au	m=1: 12.3% (S/N 1.1)	m=2: 23.2% (S/N 2.0)

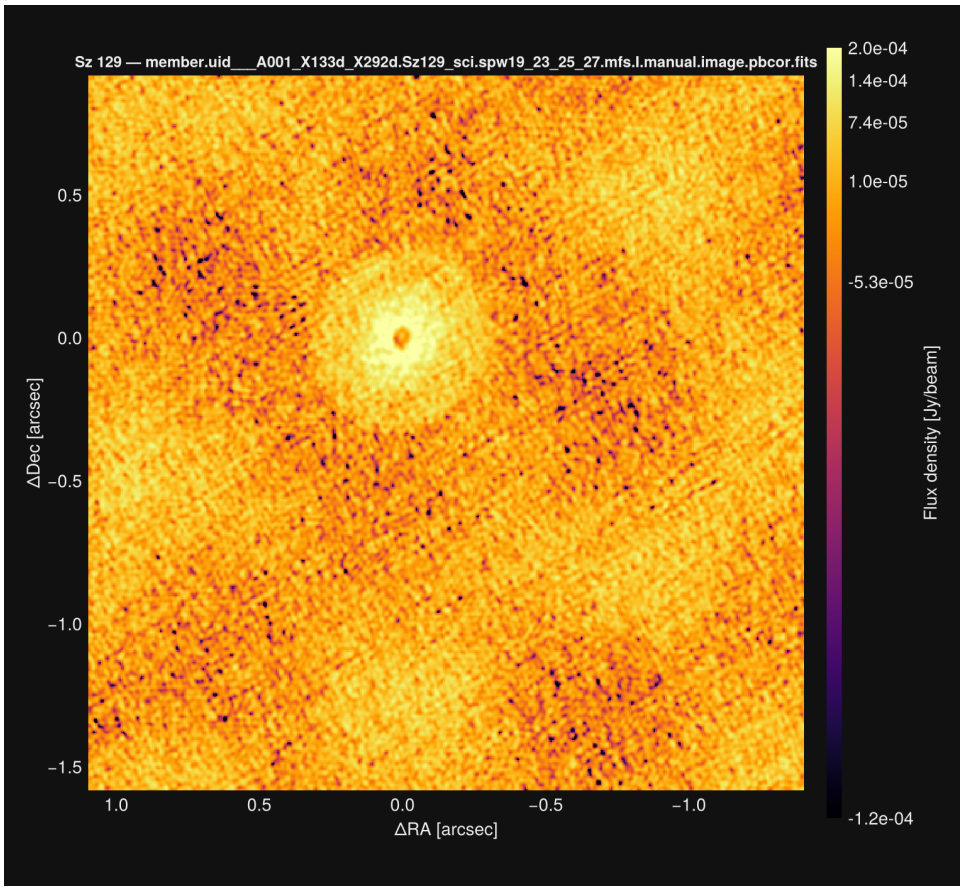
Gap-opening planet masses

Not available (no gaps were detected).

Pipeline warnings and notes

- at 47 au the local azimuthal-mean flux is below 2x the image rms per beam (0.0444 mJy/beam): the ring is detected only by averaging over many beams, and the S/N and false-alarm values above assume white noise at the beam scale. Correlated residuals (sidelobes, large-scale noise, weak background sources) are not in that noise and can make a dipole or m=2 pattern of this size, so these asymmetries are unconfirmed; an '...
- the feature(s) at 47 au are shoulders (the raw-profile peak finder finds no peak or dip within 1 beam): an amplitude measured against a shoulder baseline can be large without a real azimuthal structure; the percentages of the local flux are the safer numbers.
- the image is spatially filtered (flux check above): arc contrasts and asymmetry amplitudes describe only the emission that survived the missing short spacings.
- Negative bowl: the azimuthal mean is significantly negative over r = 72-107 au (20 bins, down to -4.7 sigma, -0.0269 mJy/beam = -10.9% of the peak, at 85 au), which real emission cannot produce. It is the signature of missing short spacings (spatial filtering): the extended emission is partly resolved out, so sizes, contrasts, asymmetries and brightness temperatures are biased and the smooth baseline is underestimate...

Continuum image



25. L1551 IRS 5

Resolved

Well resolved (90% of the flux within 6.2 beams of 7.8 au). 1 ring(s) [28 au] and 1 gap(s) [49 au] detected at >= 3 sigma. A finer observation (0.0372 arcsec) exists but the package exceeded the download limit; finer structure may exist.

Target

Coordinates (J2000)	04:31:34.08 +18:08:04.9 (RA 67.89199, Dec +18.13470 deg, J2000)
Distance	140.0 pc
Class	Young/embedded
Versus TW Hya	Class I binary; circumbinary disk

Observation used

Project / band	2017.1.00388.S / Band 4 (159.9 GHz)
Beam	0.0557 x 0.0404 arcsec = 7.8 x 5.7 au
Archive resolution	0.0446 arcsec (6.2 au
Largest scale	0.73 arcsec (102 au)
Exposure / release	345 s (0.10 h) / 2019-04-27
Package	2017.1.00388.S_uid___A001_X1288_Xb6c_001_of_001.tar (0.19 GB)
FITS image	member.uid___A001_X1288_Xb6c.L1551_IRS_5_sci.spw31.cont.I.manual.repr.pbcor.fits
Archive observations	37 total; finest 0.0372 arcsec; used #2 (fallback: finer packages too large)
Stellar parameters	Stellar parameters: none stored for this target (pass lstar= / mstar=).

Geometry and radial profile

Inclination / PA	75.0 / 170.6 deg
Geometry source	FITTED FROM THE IMAGE (radial-symmetry criterion), accuracy about +-5 deg.
Peak of mean profile	5.27 mJy/beam at r = 29.2 au
R50 / R90	33 / 49 au (4.2 / 6.2 beams)
Outer edge (>= 3 sigma)	64 au
Integrated flux	156.7 mJy (n/a of catalogue expectation)
Image noise / dyn. range	0.144 mJy/beam / 16
Shape	the profile is ring-like, not centrally peaked. The maximum is at 29 au and the innermost bin has only 6% of it, which points to a central cavity: the mean rises to half its peak at ~18 au.

Substructure (baseline-subtracted profile)

type	radius	residual	modulation	significance	
ring	r= 27.5 au	resid=1.09 mJy/b	modul=+26.0%	signif= +14.3 sigma	SIGNIFICANT
gap	r= 48.6 au	resid=-0.614 mJy/b	modul=-32.7%	signif= -11.9 sigma	SIGNIFICANT

Sequence	R 28 au -> G 49 au
Published features	none stored in the catalogue

Gaussian ring fit

Model / chi2_red tapered power law (inner hole at 21.9 +/- 0.9 au) + 1 Gaussian ring(s); chi2_red = 0.90

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed
1	r=28.9+/-0.5	amp=3.89+/-0.21	S/N=18.4	FWHM_int=26.1 +/- 0.5	det

Brightness temperature and optical depth

T_d, T_b/T_d and tau not computed: no stellar luminosity stored for this target.

Azimuthal asymmetry

ring	radius	m=1 amplitude	m=2 amplitude
r=	28.9 au	m=1: 41.8% (S/N 38.2)	m=2: 91.3% (S/N 83.4)

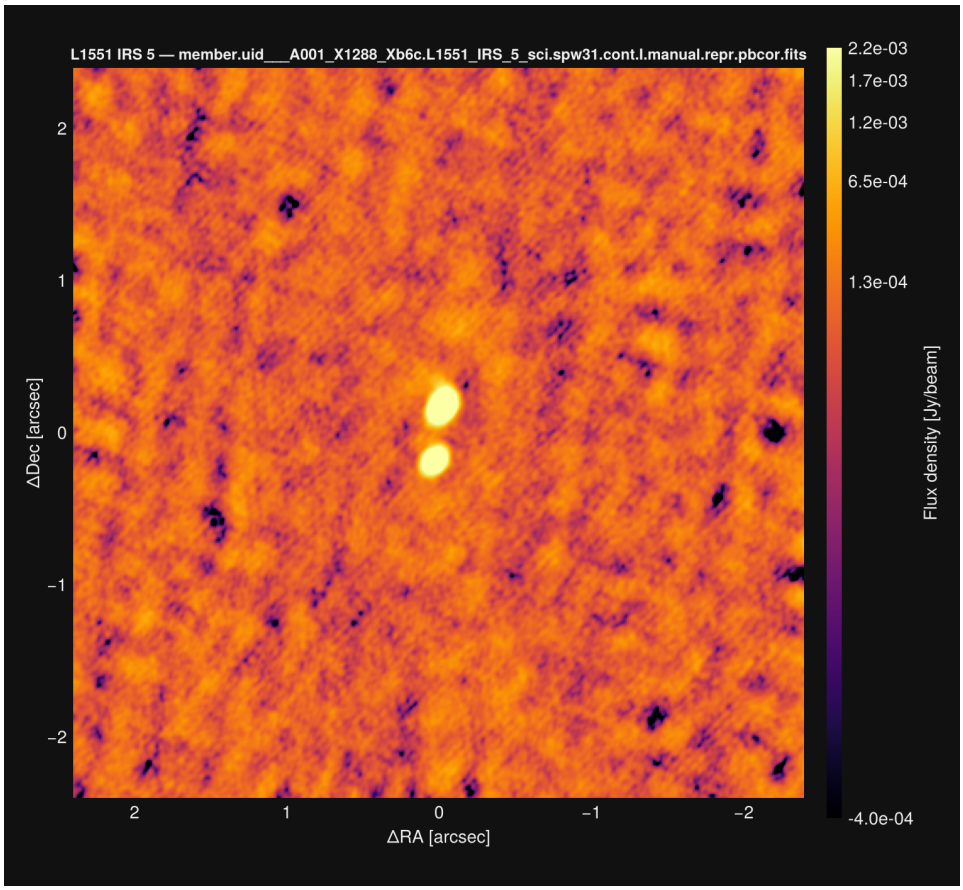
Gap-opening planet masses

Not available (needs L* and M*, which are not stored for this target; pass lstar= and mstar=).

Pipeline warnings and notes

- the smooth disk collapsed to the unresolved core (characteristic radius 9 au < 3 beams): the extended emission is shared among the ring components, so overlapping neighbours (shoulders) have degenerate widths and amplitudes; trust the isolated rings, not the blended ones.
- the notebook fell back from observation #1 to #2 because the finer packages exceeded the size limit.

Continuum image



26. V883 Ori

Resolved

Well resolved (90% of the flux within 55.9 beams of 11.7 au). 1 ring(s) [74 au] and 0 gap(s) [au] detected at >= 3 sigma. Negative bowl present: missing short spacings bias sizes, contrasts and temperatures.

Target

Coordinates (J2000)	05:38:18.10 -07:02:26.0 (RA 84.57542, Dec -7.04056 deg, J2000)
Distance	388.0 pc
Class	Outbursting
Versus TW Hya	FU Ori-type; water snowline resolved

Observation used

Project / band	2015.1.00350.S / Band 6 (225.3 GHz)
Beam	0.0302 x 0.0262 arcsec = 11.7 x 10.2 au
Archive resolution	0.0257 arcsec (10.0 au
Largest scale	0.43 arcsec (166 au)
Exposure / release	1397 s (0.39 h) / 2016-12-01
Package	2015.1.00350.S_uid___A001_X2d6_X2c8_001_of_001.tar (0.30 GB)
FITS image	member.uid___A001_X2d6_X2c8.ari_l.V883_Ori_sci.spw0_1_2_3_4_225286MHz.12m.cont.I.pbcor.fits
Archive observations	47 total; finest 0.0257 arcsec; used #1
Stellar parameters	Stellar parameters: none stored for this target (pass lstar= / mstar=).

Geometry and radial profile

Inclination / PA	40.0 / 33.8 deg
Geometry source	FITTED FROM THE IMAGE (radial-symmetry criterion), accuracy about +-5 deg.
Peak of mean profile	12.6 mJy/beam at r = 2.9 au
R50 / R90	68 / 655 au (5.8 / 55.9 beams)
Outer edge (>= 3 sigma)	144 au
Integrated flux	377.7 mJy (n/a of catalogue expectation)
Image noise / dyn. range	0.0589 mJy/beam / 55
Shape	the emission is centrally peaked (the maximum lies at 3 au, within ~1.5 beams of the star), like TW Hya: a smooth, steeply falling profile on which any rings are shoulders and dips.

Substructure (baseline-subtracted profile)

type	radius	residual		modulation	significance	
ring	r= 73.5 au	resid=0.698	mJy/b	modul=+188.8%	signif= +46.1 sigma	SIGNIFICANT

Sequence	R 73 au
Published features	none stored in the catalogue

Gaussian ring fit

Model / chi2_red tapered power law + 1 Gaussian ring(s); chi2_red = 2.36

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed
1	r=69.0+/-3.6	amp=0.948+/-0.061	S/N=15.6	FWHM_int=72.4 +/- 3.9	det

Brightness temperature and optical depth

T_d, T_b/T_d and tau not computed: no stellar luminosity stored for this target.

Azimuthal asymmetry

ring radius	m=1 amplitude	m=2 amplitude
r= 69.0 au	m=1: 5.5% (S/N 7.7)	m=2: 2.4% (S/N 3.4)

Gap-opening planet masses

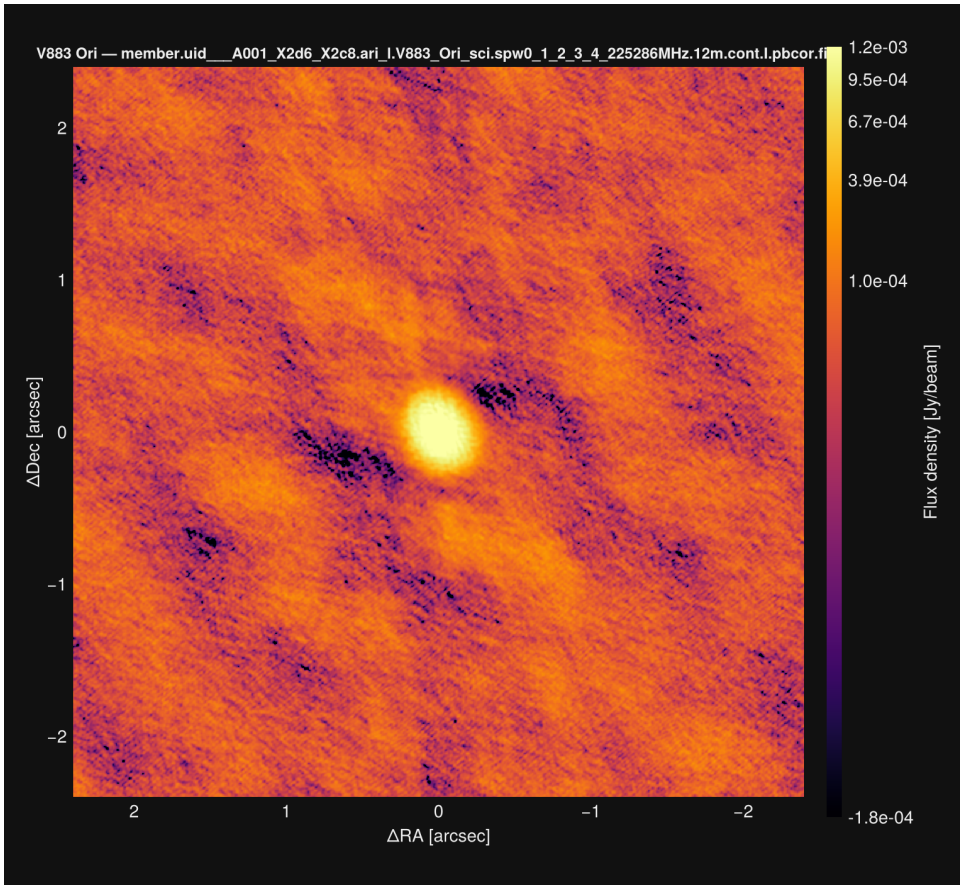
Not available (needs L* and M*, which are not stored for this target; pass lstar= and mstar=).

Pipeline warnings and notes

- the smooth disk collapsed to the unresolved core (characteristic radius 31 au < 3 beams): the extended emission is shared among the ring components, so overlapping neighbours (shoulders) have degenerate widths and amplitudes; trust the isolated rings, not the blended ones.
- the feature(s) at 69 au are shoulders (the raw-profile peak finder finds no peak or dip within 1 beam): an amplitude measured against a shoulder baseline can be large without a real azimuthal structure; the percentages of the local flux are the safer numbers.
- Negative bowl: the azimuthal mean is significantly negative over r = 385-461 au (14 bins, down to -4.3 sigma, -0.0262 mJy/beam = -0.2% of the

peak, at 438 au), which real emission cannot produce. It is the signature of missing short spacings (spatial filtering): the extended emission is partly resolved out, so sizes, contrasts, asymmetries and brightness temperatures are biased and the smooth baseline is underestimat...

Continuum image



27. Elias 20

Resolved

Well resolved (90% of the flux within 9.8 beams of 5.2 au). 1 ring(s) [32 au] and 0 gap(s) [au] detected at >= 3 sigma. Recovered 2 of 4 testable published features. Image holds 91-96% of the expected catalogue flux (consistent).

Target

Coordinates (J2000)	16:26:18.88 -24:28:19.7 (RA 246.57866, Dec -24.47214 deg, J2000)
Distance	138.0 pc
Class	T Tauri analogue
Versus TW Hya	Closest analogue: axisymmetric rings/gaps; Bosschaart+26: RD, R68 63 au
Catalogue	Bosschaart+26 (Elias20): class RD (ring disc); R68 63 au; F(1.33 mm) 97 mJy; M_dust 19 Earth masses; M* 0.48 Msun; distance 138 pc; this image: 68%-flux radius 37 au

Observation used

Project / band	2016.1.00484.L / Band 6 (239.0 GHz)
Beam	0.0373 x 0.0269 arcsec = 5.2 x 3.7 au
Archive resolution	0.0230 arcsec (3.2 au
Largest scale	0.57 arcsec (78 au)
Exposure / release	4064 s (1.13 h) / 2018-11-15
Package	2016.1.00484.L_uid___A001_X8c5_X5a_001_of_001.tar (0.49 GB)
FITS image	member.uid___A001_X8c5_X5a.Elias_20_sci.spw0_1_2_3_4_5_6_7.mfs.I.manual.image.pbcor.fits
Archive observations	4 total; finest 0.0230 arcsec; used #1
Stellar parameters	Stellar parameters (Andrews+18 (approx.); verify): L* = 2.24 Lsun, M* = 0.48 Msun

Geometry and radial profile

Inclination / PA	49.0 / 153.2 deg
Geometry source	catalogue (literature) values.
Peak of mean profile	2.76 mJy/beam at r = 1.3 au
R50 / R90	27 / 50 au (5.3 / 9.8 beams)
Outer edge (>= 3 sigma)	74 au
Integrated flux	105.2 mJy (91-96% of catalogue expectation)
Image noise / dyn. range	0.0164 mJy/beam / 60
Shape	the emission is centrally peaked (the maximum lies at 1 au, within ~1.5 beams of the star), like TW Hya: a smooth, steeply falling profile on which any rings are shoulders and dips.

Substructure (baseline-subtracted profile)

type	radius	residual		modulation	significance	
ring	r= 32.2 au	resid=0.111	mJy/b	modul=+46.6%	signif= +27.0 sigma	SIGNIFICANT

Sequence	R 32 au
Published features	recovered 2 of 4 testable

Gaussian ring fit

Model / chi2_red tapered power law + 1 Gaussian ring(s); chi2_red = 2.67

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed
1	r=37.4+/-0.9	amp=0.104+/-0.0094	S/N=11.1	FWHM_int=22.0 +/- 2.1	det

Brightness temperature and optical depth

type	radius	brightness T	dust T	ratio	optical depth	scale height
ring	r= 37.4 au	T_b= 10.9 K	T_d= 24.9 K	T_b/T_d= 0.44	tau= 0.58	H/r=0.089

Azimuthal asymmetry

ring radius	m=1 amplitude	m=2 amplitude
r= 32.2 au	m=1: 10.6% (S/N 3.1)	m=2: 24.4% (S/N 7.0)
r= 37.4 au	m=1: 5.1% (S/N 2.7)	m=2: 23.0% (S/N 12.0)

Gap-opening planet masses

Not available (no gaps were detected).

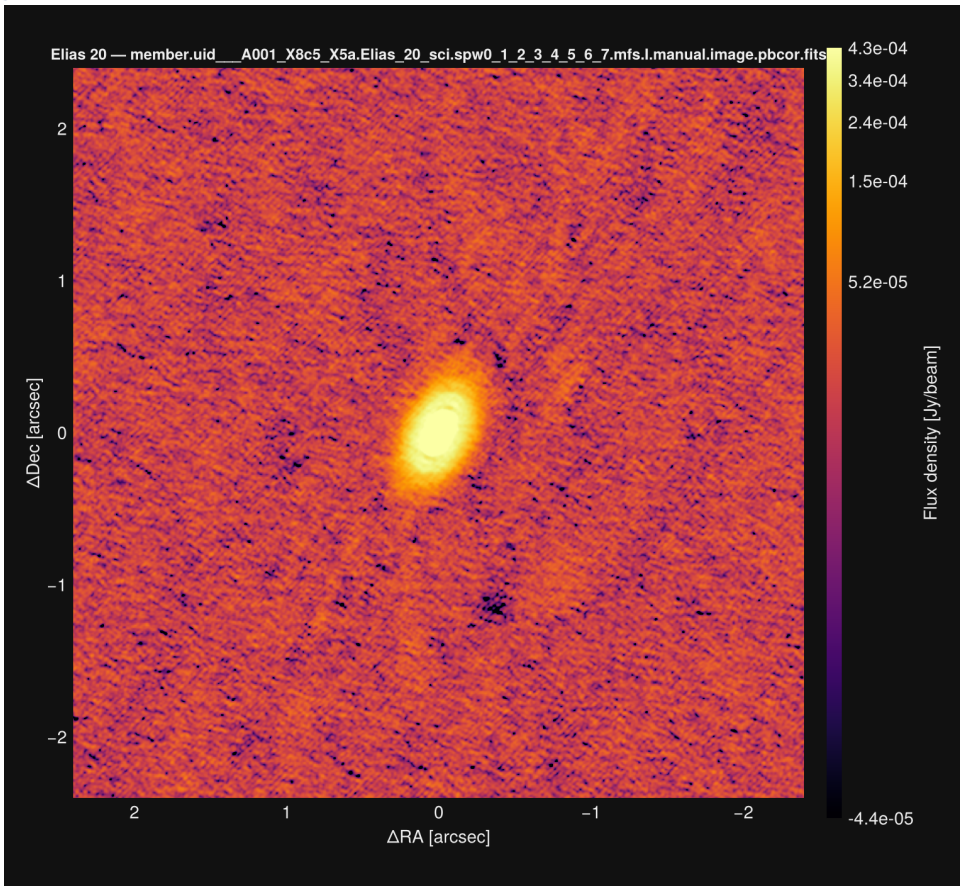
Pipeline warnings and notes

- 1 ring(s) detected in the profile at 3 sigma or more (32 au) have no fitted component, but each is a shoulder (the raw-profile peak finder finds no

peak or dip within 1 beam) of the smooth disc, not an isolated ring; the fit is not counted as disagreeing with the profile. For the asymmetry and temperature steps those features are taken from the detection (position and residual amplitude).

- the feature(s) at 32, 37 au are shoulders (the raw-profile peak finder finds no peak or dip within 1 beam): an amplitude measured against a shoulder baseline can be large without a real azimuthal structure; the percentages of the local flux are the safer numbers.

Continuum image



28. UZ Tau

Resolved

Well resolved (90% of the flux within 9.3 beams of 8.1 au). 0 ring(s) [au] and 1 gap(s) [22 au] detected at >= 3 sigma. Recovered 1 of 2 testable published features. Negative bowl present: missing short spacings bias sizes, contrasts and temperatures.

Target

Coordinates (J2000)	04:32:43.02 +25:52:30.9 (RA 68.17926, Dec +25.87525 deg, J2000)
Distance	131.0 pc
Class	Multiple system
Versus TW Hya	Multiple star; circumbinary disk

Observation used

Project / band	2017.1.00388.S / Band 4 (159.9 GHz)
Beam	0.0622 x 0.0387 arcsec = 8.1 x 5.1 au
Archive resolution	0.0485 arcsec (6.4 au
Largest scale	0.80 arcsec (105 au)
Exposure / release	381 s (0.11 h) / 2019-04-27
Package	2017.1.00388.S_uid___A001_X1288_Xb6c_001_of_001.tar (0.19 GB)
FITS image	member.uid___A001_X1288_Xb6c.UZ_Tau_E_sci.spw31.cont.I.manual.repr.pbcor.fits
Archive observations	7 total; finest 0.0485 arcsec; used #1
Stellar parameters	Stellar parameters (Long+18 Table 2 (disk E component); verify): L* = 0.35 Lsun, M* = 0.39 Msun

Geometry and radial profile

Inclination / PA	56.1 / 90.4 deg
Geometry source	catalogue (literature) values.
Peak of mean profile	0.848 mJy/beam at r = 10.2 au
R50 / R90	39 / 76 au (4.8 / 9.3 beams)
Outer edge (>= 3 sigma)	88 au
Integrated flux	25.2 mJy (n/a of catalogue expectation)
Image noise / dyn. range	0.0652 mJy/beam / 25
Shape	the emission is centrally peaked (the maximum lies at 10 au, within ~1.5 beams of the star), like TW Hya: a smooth, steeply falling profile on which any rings are shoulders and dips.

Substructure (baseline-subtracted profile)

type	radius	residual		modulation	significance		
gap	r= 22.5 au	resid=-0.0809	mJy/b	modul=-15.8%	signif= -4.3 sigma	SIGNIFICANT	
ring	r= 38.9 au	resid=0.0231	mJy/b	modul=+9.8%	signif= +1.2 sigma	noise-level	
gap	r= 74.5 au	resid=-0.0107	mJy/b	modul=-15.7%	signif= -2.5 sigma	noise-level	
ring	r= 79.8 au	resid=0.0267	mJy/b	modul=+46.7%	signif= +2.0 sigma	noise-level	
gap	r= 101.2 au	resid=-0.0218	mJy/b	modul=n/a	signif= -2.7 sigma	noise-level	
ring	r= 182.6 au	resid=0.0103	mJy/b	modul=n/a	signif= +1.2 sigma	noise-level	
gap	r= 191.0 au	resid=-0.0119	mJy/b	modul=n/a	signif= -2.5 sigma	noise-level	

Sequence	G 23 au -> R 39 au -> G 75 au -> R 80 au -> G 101 au -> R 183 au -> G 191 au
Published features	recovered 1 of 2 testable

Gaussian ring fit

Model / chi2_red tapered power law + 2 Gaussian ring(s); chi2_red = 1.34

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed
1	r=8.7+/-1.3	amp=0.516+/-0.16	S/N=3.3	FWHM_int=15.1 +/- 3.9	core
2	r=80.1+/-1.5	amp=0.0601+/-0.015	S/N=3.9	FWHM_int=13.2 +/- 5.0	det

Brightness temperature and optical depth

type	radius	brightness T	dust T	ratio	optical depth	scale height
gap	r= 22.5 au	T_b= 12.0 K	T_d= 20.1 K	T_b/T_d= 0.59	tau= 0.90	H/r=0.069
ring	r= 80.1 au	T_b= 4.4 K	T_d= 10.7 K	T_b/T_d= 0.41	tau= 0.52	H/r=0.094

Azimuthal asymmetry

ring radius	m=1 amplitude	m=2 amplitude
r= 80.1 au	m=1: 26.8% (S/N 1.5)	m=2: 22.5% (S/N 1.3)

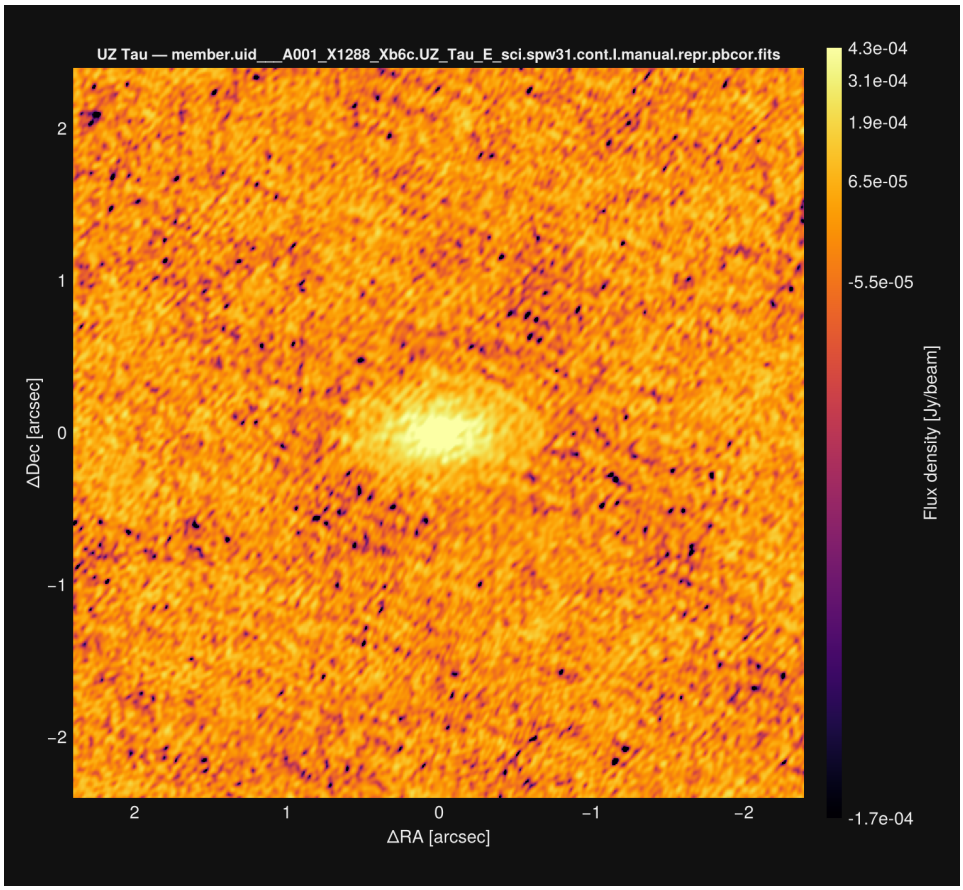
Gap-opening planet masses

Not evaluated: 23 au: no local maximum outward of the gap minimum at 22.5 au (the profile does not rise there)

Pipeline warnings and notes

- at 80 au the local azimuthal-mean flux is below 2x the image rms per beam (0.0652 mJy/beam): the ring is detected only by averaging over many beams, and the S/N and false-alarm values above assume white noise at the beam scale. Correlated residuals (sidelobes, large-scale noise, weak background sources) are not in that noise and can make a dipole or m=2 pattern of this size, so these asymmetries are unconfirmed; an '...
- the feature(s) at 80 au are shoulders (the raw-profile peak finder finds no peak or dip within 1 beam): an amplitude measured against a shoulder baseline can be large without a real azimuthal structure; the percentages of the local flux are the safer numbers.
- Negative bowl: the azimuthal mean is significantly negative over r = 121-125 au (2 bins, down to -3.3 sigma, -0.035 mJy/beam = -4.1% of the peak, at 121 au), which real emission cannot produce. It is the signature of missing short spacings (spatial filtering): the extended emission is partly resolved out, so sizes, contrasts, asymmetries and brightness temperatures are biased and the smooth baseline is underestimated...
- gap at 75 au and ring at 80 au are 5.3 au apart (< 1 beam): probably one blended feature or noise.

Continuum image



29. Sz 91

Resolved

Well resolved (90% of the flux within 7.8 beams of 13.7 au). 1 ring(s) [88 au] and 0 gap(s) [au] detected at >= 3 sigma. Image holds 15-25% of the expected catalogue flux.

Target

Coordinates (J2000)	16:07:11.59 -39:03:47.5 (RA 241.79828, Dec -39.06319 deg, J2000)
Distance	159.0 pc
Class	T Tauri + cavity
Versus TW Hya	Ring, not centrally peaked; gaps = cavity physics; Bosschaart+26: TD, R68 100 au
Catalogue	Bosschaart+26 (Sz91): class TD (transition disc); R68 100 au; F(1.33 mm) 13 mJy; M_dust 3.4 Earth masses; M* 0.51 Msun; distance 159 pc; this image: 68%-flux radius 93 au

Observation used

Project / band	2018.1.01020.S / Band 4 (139.4 GHz)
Beam	0.0861 x 0.0690 arcsec = 13.7 x 11.0 au
Archive resolution	0.0699 arcsec (11.1 au
Largest scale	1.61 arcsec (256 au)
Exposure / release	6568 s (1.82 h) / 2020-11-22
Package	2018.1.01020.S_uid___A001_X133d_X28b0_001_of_001.tar (0.02 GB)
FITS image	member.uid___A001_X133d_X28b0.Sz_91_sci.spw23_25_27_29.mfs.I.manual.image.pbcor.fits
Archive observations	14 total; finest 0.0699 arcsec; used #1
Stellar parameters	Stellar parameters (Bosschaart+26 catalogue stellar mass; verify): L* = ? Lsun, M* = 0.51 Msun

Geometry and radial profile

Inclination / PA	46.2 / 18.8 deg
Geometry source	FITTED FROM THE IMAGE (radial-symmetry criterion), accuracy about +-5 deg.
Peak of mean profile	0.0329 mJy/beam at r = 86.4 au
R50 / R90	86 / 107 au (6.3 / 7.8 beams)
Outer edge (>= 3 sigma)	114 au
Integrated flux	0.8 mJy (15-25% of catalogue expectation)
Image noise / dyn. range	0.00704 mJy/beam / 19
Shape	the profile is ring-like, not centrally peaked. The maximum is at 86 au and the innermost bin has only 15% of it, which points to a central cavity: the mean rises to half its peak at ~80 au.

Substructure (baseline-subtracted profile)

type	radius	residual	modulation	significance
gap	r= 39.8 au	resid=-0.000744 mJy/b	modul=n/a	signif= -0.5 sigma noise-level
ring	r= 51.8 au	resid=0.000568 mJy/b	modul=+38.9%	signif= +0.2 sigma noise-level
gap	r= 65.7 au	resid=-0.00363 mJy/b	modul=-28.9%	signif= -2.1 sigma noise-level
ring	r= 88.3 au	resid=0.00665 mJy/b	modul=+25.4%	signif= +3.7 sigma SIGNIFICANT
gap	r= 107.2 au	resid=-0.00367 mJy/b	modul=-22.9%	signif= -2.8 sigma noise-level
ring	r= 152.2 au	resid=0.000821 mJy/b	modul=n/a	signif= +0.6 sigma noise-level
gap	r= 175.0 au	resid=-0.000782 mJy/b	modul=n/a	signif= -1.0 sigma noise-level

Sequence	G 40 au -> R 52 au -> G 66 au -> R 88 au -> G 107 au -> R 152 au -> G 175 au
Published features	none stored in the catalogue

Gaussian ring fit

Model / chi2_red tapered power law + 1 Gaussian ring(s); chi2_red = 1.82

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed
1	r=88.2+/-0.8	amp=0.0326+/-0.0017	S/N=18.9	FWHM_int=29.9 +/- 2.2	det

Brightness temperature and optical depth

T_d, T_b/T_d and tau not computed: no stellar luminosity stored for this target.

Azimuthal asymmetry

ring radius	m=1 amplitude	m=2 amplitude
r= 88.2 au	m=1: 2.6% (S/N 0.7)	m=2: 17.5% (S/N 4.5)

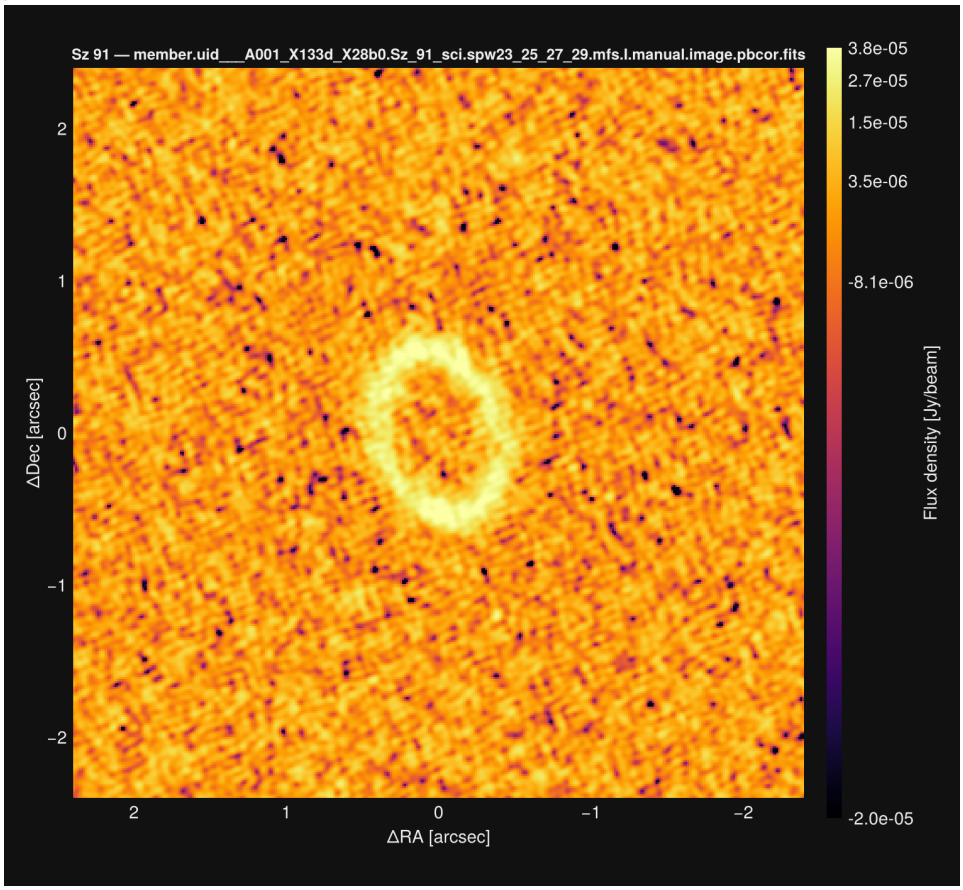
Gap-opening planet masses

Not available (needs L* and M*, which are not stored for this target; pass lstar= and mstar=).

Pipeline warnings and notes

- the image is spatially filtered (flux check above): arc contrasts and asymmetry amplitudes describe only the emission that survived the missing short spacings.
- gap at 40 au and ring at 52 au are 12.0 au apart (< 1 beam): probably one blended feature or noise.

Continuum image



30. AB Aur

Resolved

Well resolved (90% of the flux within 6.8 beams of 16.1 au). 1 ring(s) [84 au] and 0 gap(s) [au] detected at >= 3 sigma. Recovered 0 of 1 testable published features. Image holds 1.7-1.8% of the expected catalogue flux. Negative bowl present: missing short spacings bias sizes, contrasts and temperatures. A finer observation (0.0188 arcsec) exists but the package exceeded the download limit; finer structure may exist.

Target

Coordinates (J2000)	04:55:45.84 +30:33:04.3 (RA 73.94100, Dec +30.55120 deg, J2000)
Distance	156.0 pc
Class	Herbig
Versus TW Hya	Higher-mass star; warmer, larger disk; Bosschaart+26: TD, R68 193 au
Catalogue	Bosschaart+26 (ABAur): class TD (transition disc); R68 193 au; F(1.33 mm) 80 mJy; M_dust 29 Earth masses; M* 2.20 Msun; distance 163 pc; this image: 68%-flux radius 93 au

Observation used

Project / band	2015.1.00889.S / Band 6 (232.7 GHz)
Beam	0.1034 x 0.0779 arcsec = 16.1 x 12.1 au
Archive resolution	0.0285 arcsec (4.5 au
Largest scale	0.52 arcsec (80 au)
Exposure / release	5262 s (1.46 h) / 2017-02-25
Package	2015.1.00889.S_uid___A001_X2de_X10c_001_of_001.tar (2.48 GB)
FITS image	member.uid___A001_X2de_X10c.AB_Aur.spw0.232.cont.pbcor.fits
Archive observations	24 total; finest 0.0188 arcsec; used #2 (fallback: finer packages too large)
Stellar parameters	Stellar parameters (A&A 2023 aa43601-22 Table 1 (d = 163 pc); other tables give L* = 46; VERIFY; verify): L* = 65.10 Lsun, M* = 2.56 Msun

Geometry and radial profile

Inclination / PA	23.2 / 54.0 deg
Geometry source	catalogue (literature) values.
Peak of mean profile	0.14 mJy/beam at r = 84.7 au
R50 / R90	85 / 109 au (5.3 / 6.8 beams)
Outer edge (>= 3 sigma)	117 au
Integrated flux	1.5 mJy (1.7-1.8% of catalogue expectation)
Image noise / dyn. range	0.0387 mJy/beam / 17
Shape	the profile is ring-like, not centrally peaked. The maximum is at 85 au and the innermost bin has only -55% of it, which points to a central cavity: the mean rises to half its peak at ~69 au. Negative central values are an artifact of missing short baselines (spatial filtering), not a real flux deficit.

Substructure (baseline-subtracted profile)

type	radius	residual	modulation	significance
gap	r= 44.4 au	resid=-0.011 mJy/b	modul=n/a	signif= -2.0 sigma noise-level
ring	r= 83.7 au	resid=0.0422 mJy/b	modul=+43.4%	signif= +4.6 sigma SIGNIFICANT
gap	r= 108.9 au	resid=-0.014 mJy/b	modul=-22.5%	signif= -2.6 sigma noise-level
ring	r= 133.1 au	resid=0.00431 mJy/b	modul=+127.7%	signif= +0.6 sigma noise-level
gap	r= 150.3 au	resid=-0.00309 mJy/b	modul=n/a	signif= -1.1 sigma noise-level
ring	r= 156.1 au	resid=0.00691 mJy/b	modul=n/a	signif= +1.0 sigma noise-level
gap	r= 176.4 au	resid=-0.0041 mJy/b	modul=n/a	signif= -1.2 sigma noise-level
ring	r= 233.8 au	resid=0.00371 mJy/b	modul=+31.4%	signif= +0.7 sigma noise-level
gap	r= 284.0 au	resid=-0.00347 mJy/b	modul=n/a	signif= -1.2 sigma noise-level
ring	r= 293.9 au	resid=0.00255 mJy/b	modul=n/a	signif= +0.5 sigma noise-level
gap	r= 397.2 au	resid=-0.00292 mJy/b	modul=n/a	signif= -1.2 sigma noise-level

Sequence	G 44 au -> R 84 au -> G 109 au -> R 133 au -> G 150 au -> R 156 au -> G 176 au -> R 234 au -> G 284 au -> R 294 au -> G 397 au
Published features	recovered 0 of 1 testable

Gaussian ring fit

Model / chi2_red tapered power law + 1 Gaussian ring(s); chi2_red = 6.75

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed
1	r=85.0+/-2.4	amp=0.125+/-0.019	S/N=6.5	FWHM_int=35.9 +/- 8.2	det

Brightness temperature and optical depth

type	radius	brightness T	dust T	ratio	optical depth	scale height
ring	r= 85.0 au	T_b= 3.3 K	T_d= 38.4 K	T_b/T_d= 0.09	tau= 0.09	H/r=0.072

Azimuthal asymmetry

ring radius	m=1 amplitude	m=2 amplitude
r= 85.0 au	m=1: 26.0% (S/N 5.0)	m=2: 75.1% (S/N 14.6)

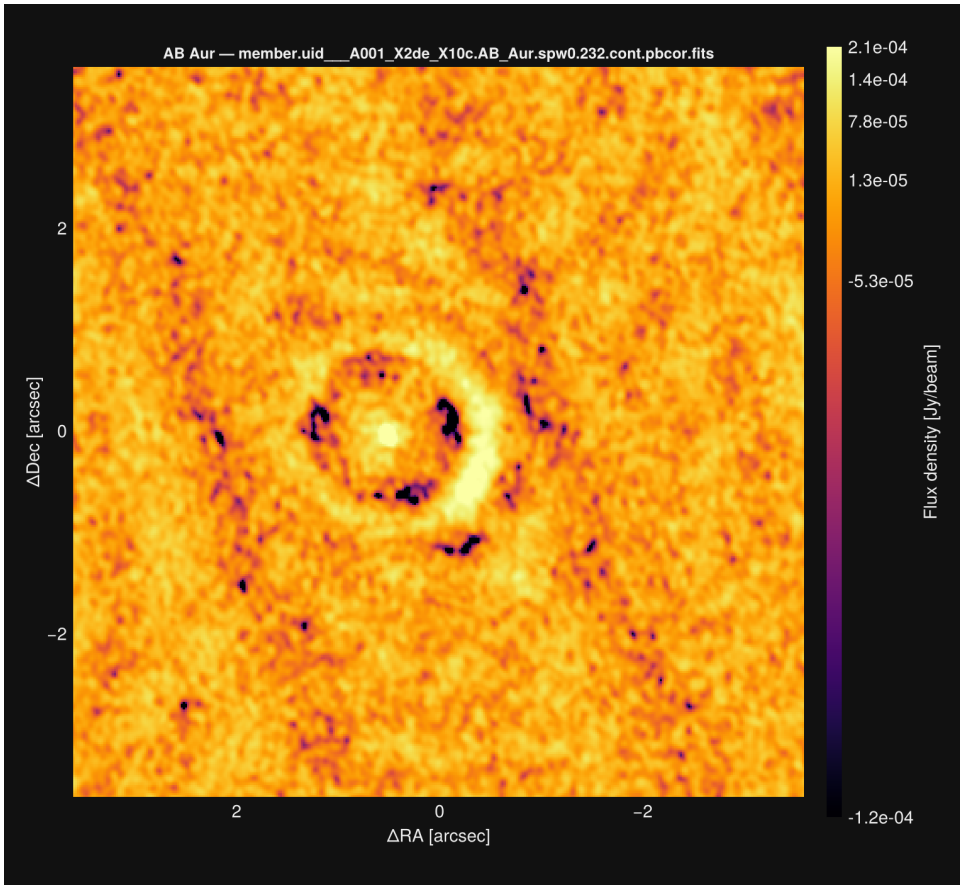
Gap-opening planet masses

Not available (no gaps were detected).

Pipeline warnings and notes

- the image is spatially filtered (flux check above): arc contrasts and asymmetry amplitudes describe only the emission that survived the missing short spacings.
- Negative bowl: the azimuthal mean is significantly negative over r = 173-190 au (3 bins, down to -3.7 sigma, -0.0236 mJy/beam = -16.9% of the peak, at 173 au), which real emission cannot produce. It is the signature of missing short spacings (spatial filtering): the extended emission is partly resolved out, so sizes, contrasts, asymmetries and brightness temperatures are biased and the smooth baseline is underestimat...
- the notebook fell back from observation #1 to #2 because the finer packages exceeded the size limit.
- gap at 150 au and ring at 156 au are 5.8 au apart (< 1 beam): probably one blended feature or noise.
- gap at 284 au and ring at 294 au are 9.9 au apart (< 1 beam): probably one blended feature or noise.

Continuum image



31. HD 135344B

Resolved

Well resolved (90% of the flux within 8.9 beams of 10.5 au). 10 candidate features found but none above 3 sigma (strongest 1.7 sigma): consistent with noise. Image holds 11-15% of the expected catalogue flux. A finer observation (0.0267 arcsec) exists but the package exceeded the download limit; finer structure may exist.

Target

Coordinates (J2000)	15:15:48.46 -37:09:15.8 (RA 228.95190, Dec -37.15440 deg, J2000)
Distance	135.0 pc
Class	Herbig
Versus TW Hya	Higher-mass star; warmer, larger disk; Bosschaart+26: TD, R68 85 au
Catalogue	Bosschaart+26 (HD135344B): class TD (transition disc); R68 85 au; F(1.33 mm) 636 mJy; M_dust 49 Earth masses; M* 1.60 Msun; distance 136 pc; this image: 68%-flux radius 68 au

Observation used

Project / band	2016.1.00340.S / Band 4 (154.7 GHz)
Beam	0.0777 x 0.0469 arcsec = 10.5 x 6.3 au
Archive resolution	0.0441 arcsec (6.0 au
Largest scale	1.43 arcsec (193 au)
Exposure / release	2685 s (0.75 h) / 2018-12-14
Package	2016.1.00340.S_uid___A001_X87c_X47b_001_of_001.tar (0.01 GB)
FITS image	member.uid___A001_X87c_X47b.ari_l.hd135344b_sci.spw17_19_21_23_154722MHz.12m.cont.I.tt0.pbcor.fits
Archive observations	23 total; finest 0.0267 arcsec; used #4 (fallback: finer packages too large)
Stellar parameters	Stellar parameters (A&A 2023 aa43601-22 Table 1 (d = 136 pc); VERIFY; verify): L* = 6.70 Lsun, M* = 1.51 Msun

Geometry and radial profile

Inclination / PA	61.2 / 84.4 deg
Geometry source	FITTED FROM THE IMAGE (radial-symmetry criterion), accuracy about +-5 deg.
Peak of mean profile	0.276 mJy/beam at r = 43.0 au
R50 / R90	53 / 94 au (5.1 / 8.9 beams)
Outer edge (>= 3 sigma)	114 au
Integrated flux	31.7 mJy (11-15% of catalogue expectation)
Image noise / dyn. range	0.0253 mJy/beam / 71
Shape	the profile is ring-like, not centrally peaked. The maximum is at 43 au and the innermost bin has only 59% of it, which points to a central cavity: the mean rises to half its peak at ~3 au.

Substructure (baseline-subtracted profile)

10 candidate features (strongest 1.7 sigma) none reaching 3 sigma. Candidate radii: 11 au, 41 au, 63 au, 73 au, 94 au, 109 au, 126 au, 148 au, 160 au, 174 au.

Published features	none stored in the catalogue

Gaussian ring fit

Model / chi2_red tapered power law + 1 Gaussian ring(s); chi2_red = 1.05

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed
1	r=42.1+/-0.8	amp=0.184+/-0.0098	S/N=18.8	FWHM_int=37.8 +/- 2.3	det

Brightness temperature and optical depth

type	radius	brightness T	dust T	ratio	optical depth	scale height
ring	r= 42.1 au	T_b= 6.9 K	T_d= 30.5 K	T_b/T_d= 0.23	tau= 0.26	H/r=0.059

Azimuthal asymmetry

ring radius	m=1 amplitude	m=2 amplitude
r= 42.1 au	m=1: 60.3% (S/N 23.1)	m=2: 14.2% (S/N 5.4)

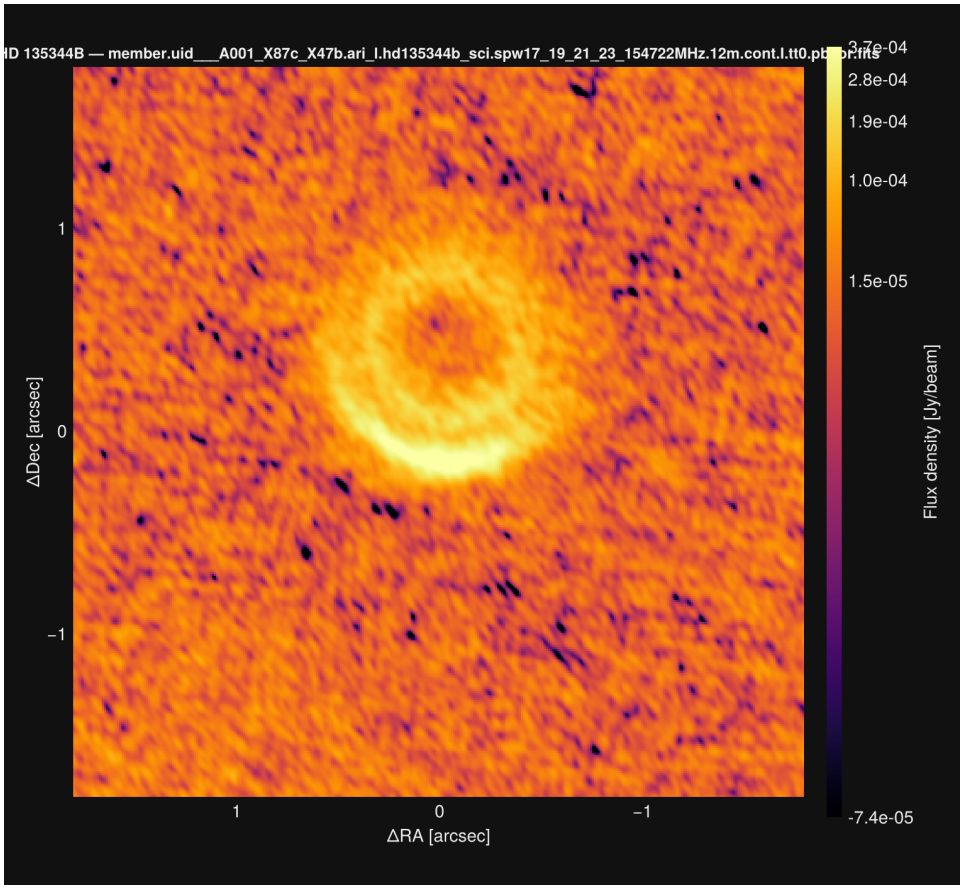
Gap-opening planet masses

Not available (no gaps were detected).

Pipeline warnings and notes

- the feature(s) at 42 au are shoulders (the raw-profile peak finder finds no peak or dip within 1 beam): an amplitude measured against a shoulder baseline can be large without a real azimuthal structure; the percentages of the local flux are the safer numbers.
- the image is spatially filtered (flux check above): arc contrasts and asymmetry amplitudes describe only the emission that survived the missing short spacings.
- the notebook fell back from observation #1 to #4 because the finer packages exceeded the size limit.

Continuum image



32. MWC 480

Resolved

Well resolved (90% of the flux within 6.9 beams of 24.2 au). 3 candidate features found but none above 3 sigma (strongest 2.0 sigma): consistent with noise. Recovered 2 of 2 testable published features. Image holds 111-245% of the expected catalogue flux (consistent).

Target

Coordinates (J2000)	04:58:46.27 +29:50:37.0 (RA 74.69280, Dec +29.84360 deg, J2000)
Distance	162.0 pc
Class	Herbig
Versus TW Hya	Higher-mass star; warmer, larger disk; Bosschaart+26: RD, R68 55 au
Catalogue	Bosschaart+26 (MWC480): class RD (ring disc); R68 55 au; F(1.33 mm) 268 mJy; M_dust 73 Earth masses; M* 1.80 Msun; distance 161 pc; this image: 68%-flux radius 101 au

Observation used

Project / band	2016.1.01042.S / Band 3 (102.4 GHz)
Beam	0.1492 x 0.0942 arcsec = 24.2 x 15.3 au
Archive resolution	0.0824 arcsec (13.3 au
Largest scale	2.71 arcsec (439 au)
Exposure / release	399 s (0.11 h) / 2019-09-17
Package	2016.1.01042.S_uid___A001_X892_X38_001_of_001.tar (9.44 GB)
FITS image	member.uid___A001_X892_X38.MWC480_sci.spw16_17_18_19_24_25_26.mfs.I.manual.image.tt0.pbcor.fits
Archive observations	33 total; finest 0.0824 arcsec; used #1
Stellar parameters	Stellar parameters (Oberg+21 MAPS I (dynamical mass); verify): L* = 21.90 Lsun, M* = 2.10 Msun

Geometry and radial profile

Inclination / PA	37.0 / 148.0 deg
Geometry source	catalogue (literature) values.
Peak of mean profile	5.67 mJy/beam at r = 4.0 au
R50 / R90	61 / 166 au (2.5 / 6.9 beams)
Outer edge (>= 3 sigma)	198 au
Integrated flux	61.6 mJy (111-245% of catalogue expectation)
Image noise / dyn. range	0.0716 mJy/beam / 189
Shape	the emission is centrally peaked (the maximum lies at 4 au, within ~1.5 beams of the star), like TW Hya: a smooth, steeply falling profile on which any rings are shoulders and dips.

Substructure (baseline-subtracted profile)

3 candidate features (strongest 2.0 sigma) none reaching 3 sigma. Candidate radii: 75 au, 96 au, 142 au.

Published features	recovered 2 of 2 testable

Gaussian ring fit

Model / chi2_red	tapered power law + 1 Gaussian ring(s); chi2_red = 3.62

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed
1	r=85.0+/-16.7	amp=0.246+/-0.029	S/N=8.4	FWHM_int=115.8 +/- 24.7	det

Brightness temperature and optical depth

type	radius	brightness T	dust T	ratio	optical depth	scale height
ring	r= 85.0 au	T_b= 4.3 K	T_d= 29.2 K	T_b/T_d= 0.15	tau= 0.16	H/r=0.069

Azimuthal asymmetry

ring radius	m=1 amplitude	m=2 amplitude
r= 85.0 au	m=1: 8.4% (S/N 2.1)	m=2: 17.3% (S/N 4.2)

Gap-opening planet masses

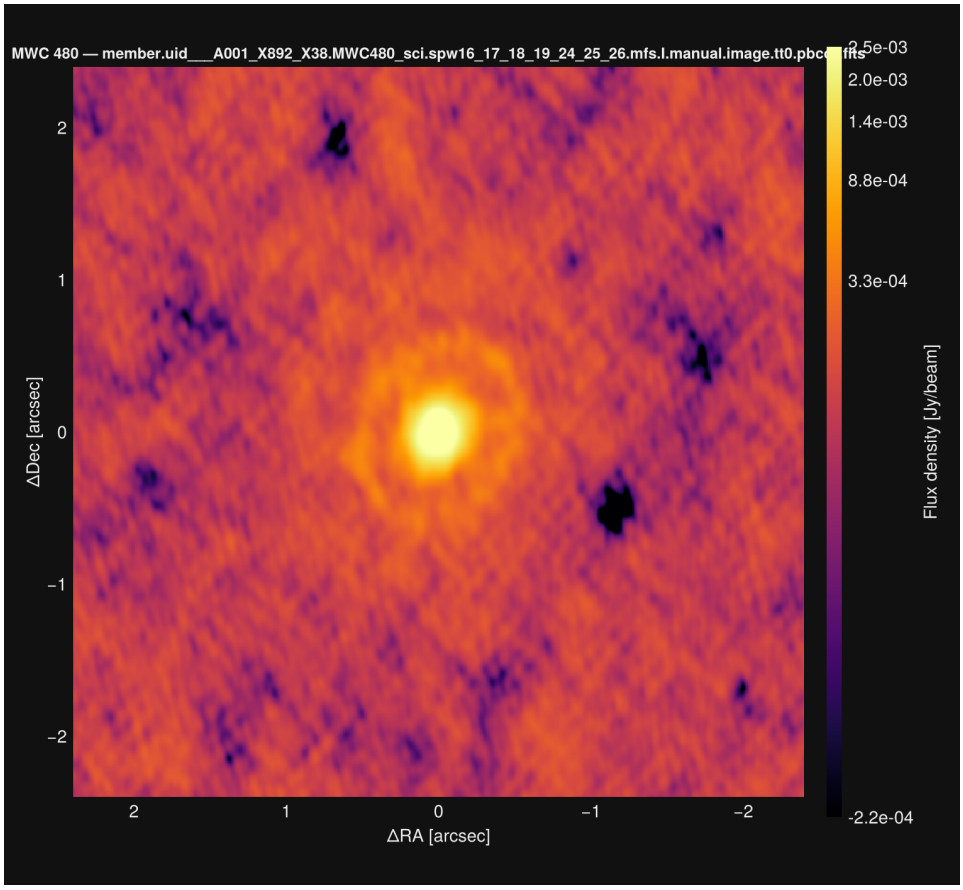
Not available (no gaps were detected).

Pipeline warnings and notes

- the smooth disk collapsed to the unresolved core (characteristic radius 32 au < 3 beams): the extended emission is shared among the ring components, so overlapping neighbours (shoulders) have degenerate widths and amplitudes; trust the isolated rings, not the blended ones.

- the feature(s) at 85 au are shoulders (the raw-profile peak finder finds no peak or dip within 1 beam): an amplitude measured against a shoulder baseline can be large without a real azimuthal structure; the percentages of the local flux are the safer numbers.

Continuum image



33. V1094 Sco

Resolved

Well resolved (90% of the flux within 6.4 beams of 40.1 au). No significant substructure detected. Image holds 21-44% of the expected catalogue flux. A finer observation (0.0208 arcsec) exists but the package exceeded the download limit; finer structure may exist.

Target

Coordinates (J2000)	16:08:36.18 -39:23:02.5 (RA 242.15074, Dec -39.38402 deg, J2000)
Distance	153.0 pc
Class	T Tauri analogue
Versus TW Hya	Very large disk with wide rings; Bosschaart+26: RD, R68 166 au
Catalogue	Bosschaart+26 (V1094Sco): class RD (ring disc); R68 166 au; F(1.33 mm) 180 mJy; M_dust 45 Earth masses; M* 0.83 Msun; distance 154 pc; this image: 68%-flux radius 187 au

Observation used

Project / band	2015.1.01137.S / Band 8 (484.5 GHz)
Beam	0.2618 x 0.2514 arcsec = 40.1 x 38.5 au
Archive resolution	0.2516 arcsec (38.5 au
Largest scale	2.69 arcsec (411 au)
Exposure / release	1693 s (0.47 h) / 2017-07-08
Package	2015.1.01137.S_uid___A001_X2fe_X487_001_of_001.tar (0.06 GB)
FITS image	member.uid___A001_X2fe_X487.ari_l.V1094_Sco_sci.spw0_1_2_3_484470MHz.12m.cont.l.pbcor.fits
Archive observations	15 total; finest 0.0208 arcsec; used #8 (fallback: finer packages too large)
Stellar parameters	Stellar parameters (Bosschaart+26 catalogue stellar mass; verify): L* = ? Lsun, M* = 0.83 Msun

Geometry and radial profile

Inclination / PA	48.8 / 108.8 deg
Geometry source	FITTED FROM THE IMAGE (radial-symmetry criterion), accuracy about +-5 deg.
Peak of mean profile	29.5 mJy/beam at r = 3.8 au
R50 / R90	134 / 256 au (3.3 / 6.4 beams)
Outer edge (>= 3 sigma)	302 au
Integrated flux	369.3 mJy (21-44% of catalogue expectation)
Image noise / dyn. range	0.275 mJy/beam / 429
Shape	the emission is centrally peaked (the maximum lies at 4 au, within ~1.5 beams of the star), like TW Hya: a smooth, steeply falling profile on which any rings are shoulders and dips.

Substructure (baseline-subtracted profile)

type	radius	residual	modulation	significance
ring	r= 133.9 au	resid=0.0291 mJy/b	modul=+0.8%	signif= +0.2 sigma noise-level
gap	r= 218.0 au	resid=-0.024 mJy/b	modul=-0.9%	signif= -0.4 sigma noise-level

Sequence	R 134 au -> G 218 au
Published features	none stored in the catalogue

Gaussian ring fit

Model / chi2_red	tapered power law + 1 Gaussian ring(s); chi2_red = 0.85

No ring components fitted.

Brightness temperature and optical depth

T_d, T_b/T_d and tau not computed: no stellar luminosity stored for this target.

Azimuthal asymmetry

No rings with a testable asymmetry.

Gap-opening planet masses

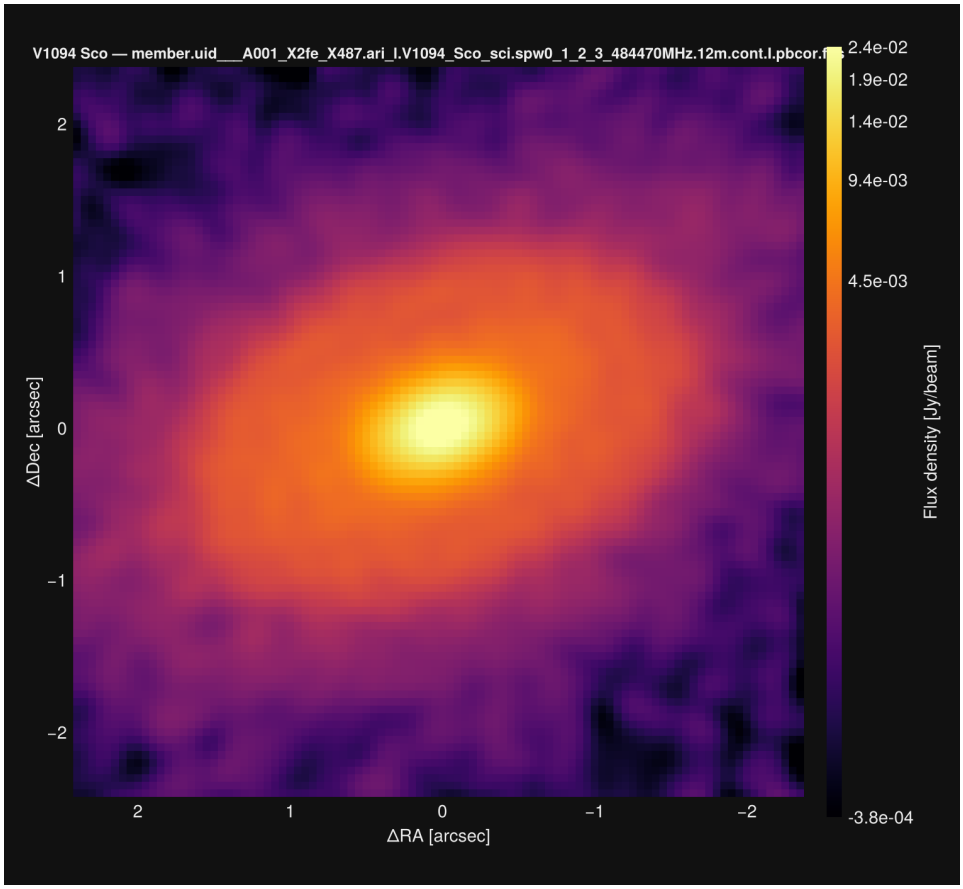
Not available (needs L* and M*, which are not stored for this target; pass lstar= and mstar=).

Pipeline warnings and notes

- the beam is 40 x 38 au (> 40 au). A typical disk (rings at tens of au) is not resolved, so the rings, asymmetries, temperatures and masses below are not meaningful. A finer observation is needed.

- the notebook fell back from observation #1 to #8 because the finer packages exceeded the size limit.

Continuum image



34. IQ Tau

Resolved

Well resolved (90% of the flux within 5.0 beams of 19.1 au). 9 candidate features found but none above 3 sigma (strongest 1.2 sigma): consistent with noise. Recovered 0 of 2 testable published features. Image holds 113% of the expected catalogue flux (consistent).

Target

Coordinates (J2000)	04:29:51.56 +26:06:44.9 (RA 67.46482, Dec +26.11246 deg, J2000)
Distance	131.0 pc
Class	T Tauri analogue
Versus TW Hya	Closest analogue: axisymmetric rings/gaps; Bosschaart+26: RD, R68 55 au
Catalogue	Bosschaart+26 (IQTau): class RD (ring disc); R68 55 au; F(1.33 mm) 64 mJy; M_dust 11 Earth masses; M* 0.44 Msun; distance 130 pc; this image: 68%-flux radius 62 au

Observation used

Project / band	2016.1.01164.S / Band 6 (225.5 GHz)
Beam	0.1455 x 0.0939 arcsec = 19.1 x 12.3 au
Archive resolution	0.1169 arcsec (15.3 au
Largest scale	1.86 arcsec (243 au)
Exposure / release	91 s (0.03 h) / 2018-12-07
Package	2016.1.01164.S_uid___A001_X885_X1ec_001_of_001.tar (0.09 GB)
FITS image	member.uid___A001_X885_X1ec.IQ_Tau_sci.spw47_55.mfs.I.manual.image.pbcor.fits
Archive observations	5 total; finest 0.1169 arcsec; used #1
Stellar parameters	Stellar parameters (Long+18 Table 2; verify): L* = 0.21 Lsun, M* = 0.50 Msun

Geometry and radial profile

Inclination / PA	62.1 / 42.4 deg
Geometry source	catalogue (literature) values.
Peak of mean profile	4.52 mJy/beam at r = 3.3 au
R50 / R90	49 / 95 au (2.6 / 5.0 beams)
Outer edge (>= 3 sigma)	121 au
Integrated flux	72.3 mJy (113% of catalogue expectation)
Image noise / dyn. range	0.118 mJy/beam / 173
Shape	the emission is centrally peaked (the maximum lies at 3 au, within ~1.5 beams of the star), like TW Hya: a smooth, steeply falling profile on which any rings are shoulders and dips.

Substructure (baseline-subtracted profile)

9 candidate features (strongest 1.2 sigma) none reaching 3 sigma. Candidate radii: 49 au, 69 au, 95 au, 115 au, 134 au, 160 au, 174 au, 174 au, 207 au.

Published features	recovered 0 of 2 testable

Gaussian ring fit

Model / chi2_red tapered power law + 2 Gaussian ring(s); chi2_red = 0.12

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed	
1	r=49.8+/-4.0	amp=0.473+/-0.4	S/N=1.2	FWHM_int=< 33.3 (unresolved)	det	<3sig
2	r=66.5+/-37.3	amp=0.409+/-0.28	S/N=1.4	FWHM_int=84.6 +/- 33.6	lit	<3sig

Brightness temperature and optical depth

No rings or gaps fitted/detected, so no radii to evaluate.

Azimuthal asymmetry

No rings with a testable asymmetry.

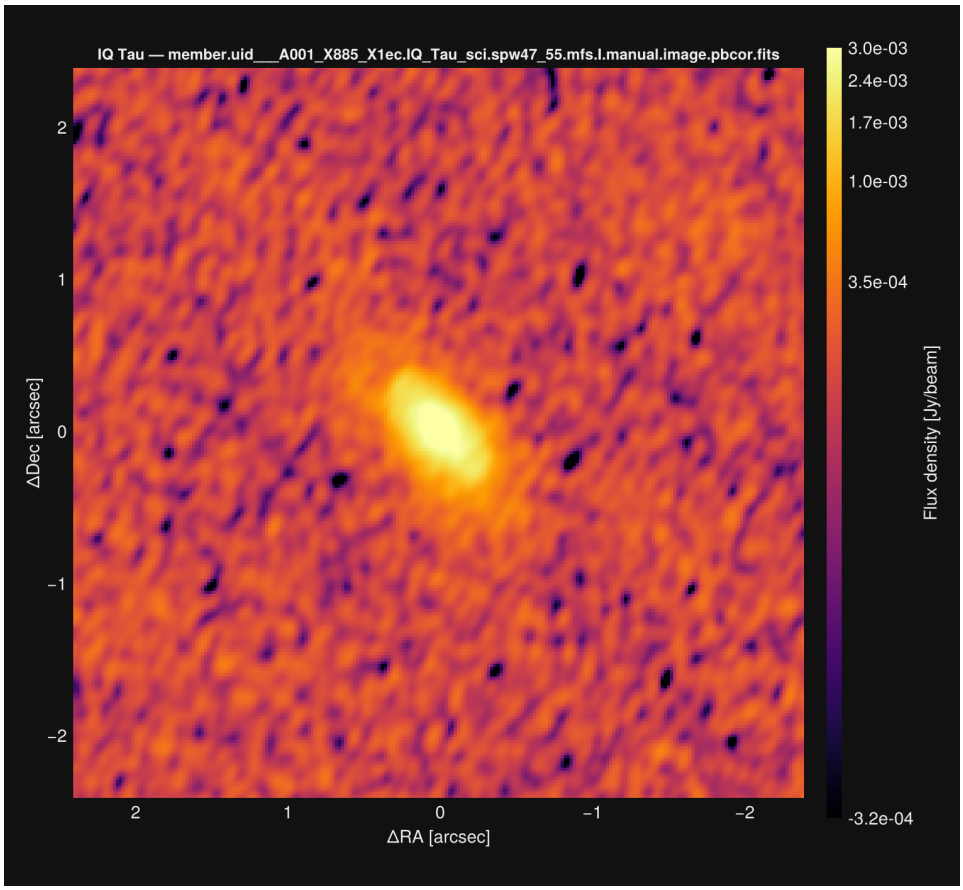
Gap-opening planet masses

Not available (no gaps were detected).

Pipeline warnings and notes

- the smooth disk collapsed to the unresolved core (characteristic radius 39 au < 3 beams): the extended emission is shared among the ring components, so overlapping neighbours (shoulders) have degenerate widths and amplitudes; trust the isolated rings, not the blended ones.

Continuum image



35. DoAr 33

Barely resolved

Only barely resolved (90% of the flux within 4.8 beams of 5 au); overall size and trend are trustworthy, narrow rings are not. 2 ring(s) [20, 44 au] and 1 gap(s) [37 au] detected at >= 3 sigma. Image holds 77-81% of the expected catalogue flux (consistent).

Target

Coordinates (J2000)	16:27:39.01 -23:58:18.7 (RA 246.91255, Dec -23.97186 deg, J2000)
Distance	140.0 pc
Class	T Tauri analogue
Versus TW Hya	Closest analogue: axisymmetric rings/gaps; Bosschaart+26: RD, R68 26 au
Catalogue	Bosschaart+26 (DoAr33): class RD (ring disc); R68 26 au; F(1.33 mm) 35 mJy; M_dust 7.1 Earth masses; M* 1.10 Msun; distance 139 pc; this image: 68%-flux radius 20 au

Observation used

Project / band	2016.1.00484.L / Band 6 (239.0 GHz)
Beam	0.0377 x 0.0209 arcsec = 5.3 x 2.9 au
Archive resolution	0.0212 arcsec (3.0 au
Largest scale	0.67 arcsec (94 au)
Exposure / release	4064 s (1.13 h) / 2018-10-31
Package	2016.1.00484.L_uid___A001_X8c5_X6e_001_of_001.tar (1.01 GB)
FITS image	member.uid___A001_X8c5_X6e.DoAr_33_sci.spw0_1_2_3_4_5_6_7.mfs.I.manual.image.pbcor.fits
Archive observations	2 total; finest 0.0212 arcsec; used #1
Stellar parameters	Stellar parameters (Andrews+18 Table 1 (MIST mass); verify): L* = 1.51 Lsun, M* = 1.10 Msun

Geometry and radial profile

Inclination / PA	41.8 / 81.1 deg
Geometry source	catalogue (literature) values.
Peak of mean profile	1.19 mJy/beam at r = 1.3 au
R50 / R90	17 / 25 au (3.3 / 4.8 beams)
Outer edge (>= 3 sigma)	30 au
Integrated flux	32.0 mJy (77-81% of catalogue expectation)
Image noise / dyn. range	0.0168 mJy/beam / 35
Shape	the emission is centrally peaked (the maximum lies at 1 au, within ~1.5 beams of the star), like TW Hya: a smooth, steeply falling profile on which any rings are shoulders and dips.

Substructure (baseline-subtracted profile)

type	radius	residual	modulation	significance	
ring	r= 19.8 au	resid=0.091 mJy/b	modul=+43.8%	signif= +20.3 sigma	SIGNIFICANT
gap	r= 36.6 au	resid=-0.0443 mJy/b	modul=-113.4%	signif= -19.6 sigma	SIGNIFICANT
ring	r= 44.5 au	resid=0.0213 mJy/b	modul=n/a	signif= +7.0 sigma	SIGNIFICANT

Sequence	R 20 au -> G 37 au -> R 44 au
Published features	none stored in the catalogue

Gaussian ring fit

Model / chi2_red tapered power law + 2 Gaussian ring(s); chi2_red = 0.72

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed	
1	r=17.3+/-1.0	amp=0.29+/-0.066	S/N=4.4	FWHM_int=14.7 +/- 1.4	det	
2	r=45.8+/-1.7	amp=0.00587+/-0.0023	S/N=2.6	FWHM_int=7.8 +/- 4.5	det	<3sig

Brightness temperature and optical depth

type	radius	brightness T	dust T	ratio	optical depth	scale height
ring	r= 17.3 au	T_b= 14.2 K	T_d= 33.3 K	T_b/T_d= 0.43	tau= 0.55	H/r=0.046
gap	r= 36.6 au	T_b= 2.0 K	T_d= 23.1 K	T_b/T_d= 0.09	tau= 0.09	H/r=0.055

Azimuthal asymmetry

ring radius	m=1 amplitude	m=2 amplitude
r= 17.3 au	m=1: 1.5% (S/N 1.4)	m=2: 3.6% (S/N 3.4)

Gap-opening planet masses

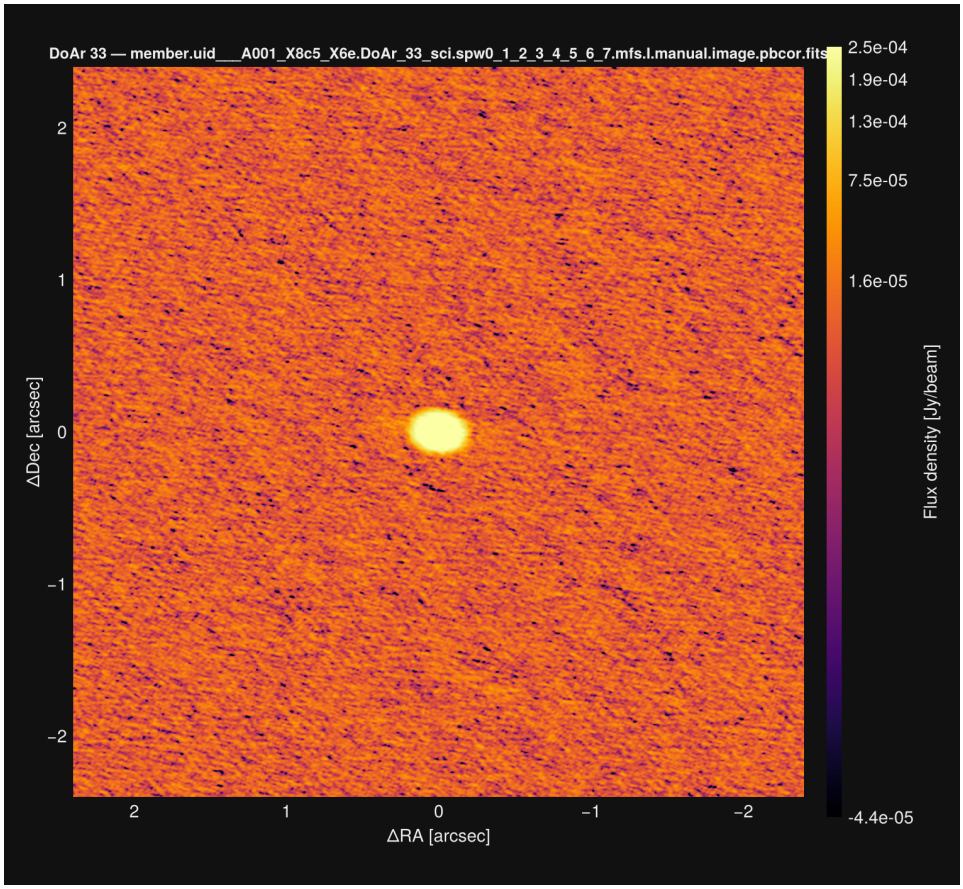
Not evaluated: 37 au: gap contrast only 0.8 sigma (< 3.0)

Pipeline warnings and notes

- the smooth disk collapsed to the unresolved core (characteristic radius 11 au < 3 beams): the extended emission is shared among the ring components, so overlapping neighbours (shoulders) have degenerate widths and amplitudes; trust the isolated rings, not the blended ones.

- the feature(s) at 17 au are shoulders (the raw-profile peak finder finds no peak or dip within 1 beam): an amplitude measured against a shoulder baseline can be large without a real azimuthal structure; the percentages of the local flux are the safer numbers.

Continuum image



36. T Cha

Barely resolved

Only barely resolved (90% of the flux within 4.4 beams of 13 au); overall size and trend are trustworthy, narrow rings are not. 1 ring(s) [34 au] and 1 gap(s) [51 au] detected at >= 3 sigma.

Target

Coordinates (J2000)	11:57:13.52 -79:21:31.5 (RA 179.30635, Dec -79.35876 deg, J2000)
Distance	107.0 pc
Class	T Tauri + cavity
Versus TW Hya	Transition disk; ~7 Myr; small ring

Observation used

Project / band	2015.1.00979.S / Band 3 (97.5 GHz)
Beam	0.1187 x 0.0620 arcsec = 12.7 x 6.6 au
Archive resolution	0.0683 arcsec (7.3 au
Largest scale	1.23 arcsec (132 au)
Exposure / release	7348 s (2.04 h) / 2017-01-26
Package	2015.1.00979.S_uid___A001_X2d1_X83_001_of_001.tar (2.84 GB)
FITS image	member.uid___A001_X2d1_X83.ari_l.TCha_sci.spw0_1_2_3_97494MHz.12m.cont.l.tt0.pbcor.fits
Archive observations	5 total; finest 0.0683 arcsec; used #1
Stellar parameters	Stellar parameters: none stored for this target (pass lstar= / mstar=).

Geometry and radial profile

Inclination / PA	71.2 / 114.4 deg
Geometry source	FITTED FROM THE IMAGE (radial-symmetry criterion), accuracy about +-5 deg.
Peak of mean profile	0.695 mJy/beam at r = 34.8 au
R50 / R90	35 / 56 au (2.7 / 4.4 beams)
Outer edge (>= 3 sigma)	72 au
Integrated flux	14.6 mJy (n/a of catalogue expectation)
Image noise / dyn. range	0.00898 mJy/beam / 40
Shape	the profile is ring-like, not centrally peaked. The maximum is at 35 au and the innermost bin has only 60% of it, which points to a central cavity: the mean rises to half its peak at ~3 au.

Substructure (baseline-subtracted profile)

type	radius	residual		modulation	significance	
ring	r= 33.7 au	resid=0.0734	mJy/b	modul=+11.8%	signif= +16.0 sigma	SIGNIFICANT
gap	r= 50.8 au	resid=-0.0482	mJy/b	modul=-17.5%	signif= -14.6 sigma	SIGNIFICANT

Sequence	R 34 au -> G 51 au
Published features	none stored in the catalogue

Gaussian ring fit

Model / chi2_red tapered power law + 1 Gaussian ring(s); chi2_red = 3.35

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed
1	r=33.7+/-0.5	amp=0.607+/-0.023	S/N=26.8	FWHM_int=27.0 +/- 1.1	det

Brightness temperature and optical depth

T_d, T_b/T_d and tau not computed: no stellar luminosity stored for this target.

Azimuthal asymmetry

ring	radius	m=1 amplitude		m=2 amplitude	
r=	33.7 au	m=1:	12.7% (S/N 25.1)	m=2:	21.3% (S/N 42.2)

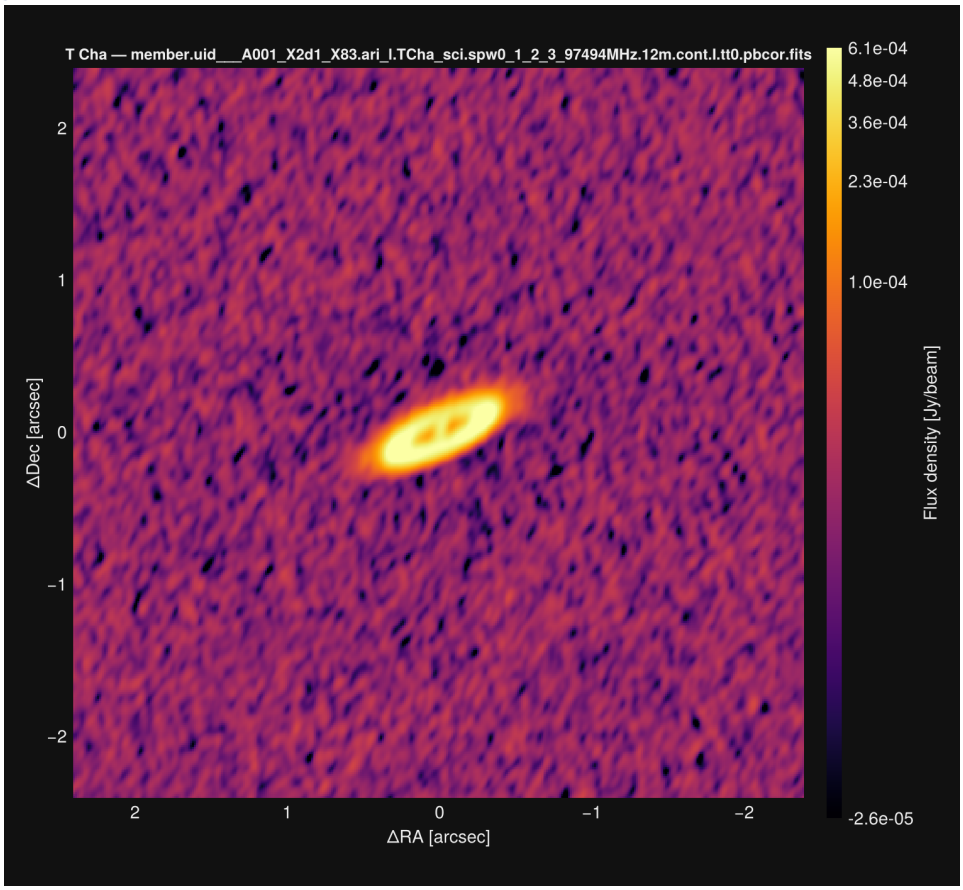
Gap-opening planet masses

Not available (needs L* and M*, which are not stored for this target; pass lstar= and mstar=).

Pipeline warnings and notes

- the smooth disk collapsed to the unresolved core (characteristic radius 37 au < 3 beams): the extended emission is shared among the ring components, so overlapping neighbours (shoulders) have degenerate widths and amplitudes; trust the isolated rings, not the blended ones.

Continuum image



37. LkCa 15

Barely resolved

DATA CHECK: Catalogue entry is for 'J04391779+2221034', not 'LkCa 15': published values (R68, flux, M_dust, ring radii) may belong to another source.

Only barely resolved (90% of the flux within 3.6 beams of 45 au); overall size and trend are trustworthy, narrow rings are not. 0 ring(s) [au] and 1 gap(s) [124 au] detected at >= 3 sigma. Image holds 97-101% of the expected catalogue flux (consistent). A finer observation (0.0281 arcsec) exists but the package exceeded the download limit; finer structure may exist.

Target

Coordinates (J2000)	04:39:17.81 +22:21:03.6 (RA 69.82420, Dec +22.35100 deg, J2000)
Distance	157.0 pc
Class	T Tauri + cavity
Versus TW Hya	Ring, not centrally peaked; gaps = cavity physics; Bosschaart+26: TD, R68 124 au
Catalogue	Bosschaart+26 (J04391779+2221034): class TD (transition disc); R68 124 au; F(1.33 mm) 127 mJy; M_dust 33 Earth masses; M* 0.78 Msun; distance 158 pc; this image: 68%-flux radius 117 au

Observation used

Project / band	2015.1.00118.S / Band 6 (233.0 GHz)
Beam	0.2847 x 0.1914 arcsec = 44.7 x 30.0 au
Archive resolution	0.1920 arcsec (30.1 au
Largest scale	5.33 arcsec (836 au)
Exposure / release	4351 s (1.21 h) / 2018-10-09
Package	2015.1.00118.S_uid___A001_X2fb_X4d1_001_of_001.tar (0.09 GB)
FITS image	member.uid___A001_X2fb_X4d1.LkCA_15_sci.spw17_19_21_23.mfs.IQUV.manual.pbcor.fits
Archive observations	30 total; finest 0.0281 arcsec; used #9 (fallback: finer packages too large)
Stellar parameters	Stellar parameters (Bosschaart+26 catalogue stellar mass; verify): L* = ? Lsun, M* = 0.78 Msun

Geometry and radial profile

Inclination / PA	50.0 / 60.0 deg
Geometry source	catalogue (literature) values.
Peak of mean profile	6.48 mJy/beam at r = 74.2 au
R50 / R90	95 / 159 au (2.1 / 3.6 beams)
Outer edge (>= 3 sigma)	230 au
Integrated flux	136.5 mJy (97-101% of catalogue expectation)
Image noise / dyn. range	0.0412 mJy/beam / 125
Shape	the azimuthal mean peaks at 74 au (1.7 beams) and the innermost bin holds 27% of the peak. A peak this close to the centre, with a dip of this size, is at the resolution limit: it is consistent with a centrally peaked disk smoothed by the 45 au beam, and a cavity smaller than ~89 au cannot be established.

Substructure (baseline-subtracted profile)

type	radius	residual	modulation	significance
ring	r= 72.3 au	resid=0.0166 mJy/b	modul=+0.3%	signif= +0.5 sigma noise-level
gap	r= 123.6 au	resid=-0.0501 mJy/b	modul=-1.7%	signif= -3.7 sigma SIGNIFICANT
ring	r= 144.8 au	resid=0.0408 mJy/b	modul=+2.8%	signif= +1.5 sigma noise-level
gap	r= 151.9 au	resid=-0.041 mJy/b	modul=-3.5%	signif= -2.6 sigma noise-level

Sequence	R 72 au -> G 124 au -> R 145 au -> G 152 au
Published features	none stored in the catalogue

Gaussian ring fit

Model / chi2_red tapered power law + 2 Gaussian ring(s); chi2_red = 0.54

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed
1	r=74.3+/-0.9	amp=6.36+/-0.092	S/N=68.8	FWHM_int=76.3 +/- 2.3	core
2	r=152.9+/-11.9	amp=0.555+/-0.11	S/N=5.2	FWHM_int=87.7 +/- 12.1	det

Brightness temperature and optical depth

T_d, T_b/T_d and tau not computed: no stellar luminosity stored for this target.

Azimuthal asymmetry

ring radius	m=1 amplitude	m=2 amplitude
r= 152.9 au	m=1: 62.1% (S/N 33.8)	m=2: 24.0% (S/N 13.1)

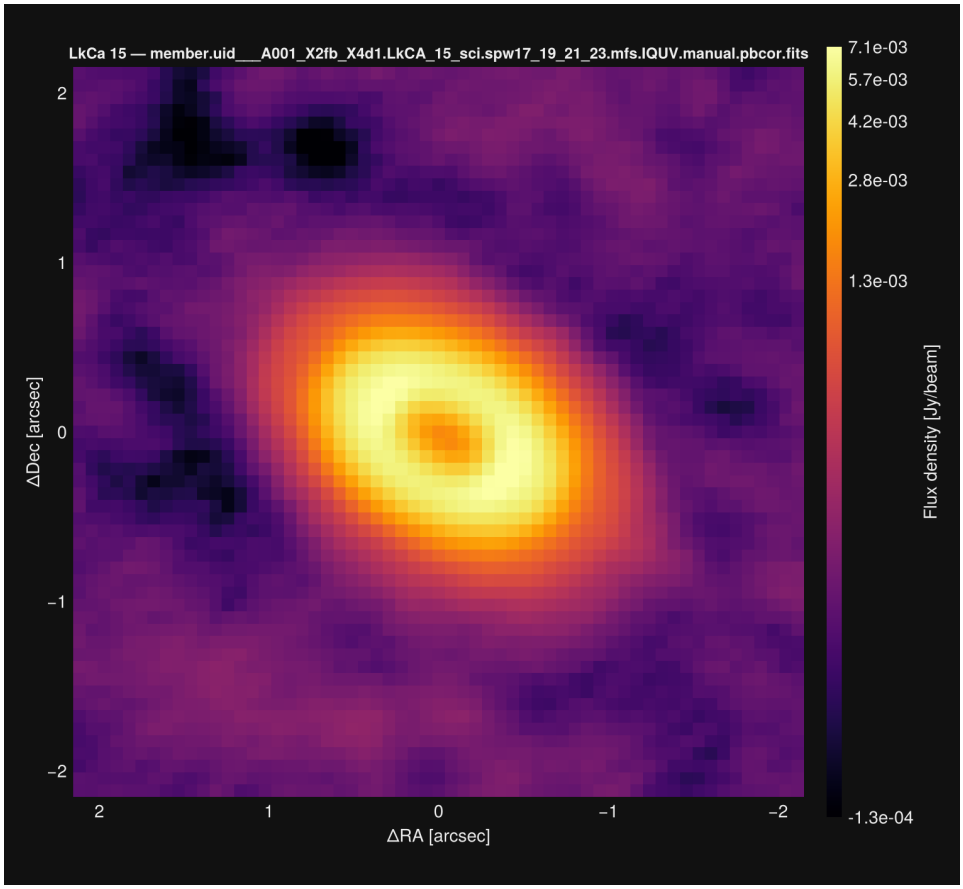
Gap-opening planet masses

Not available (needs L* and M*, which are not stored for this target; pass lstar= and mstar=).

Pipeline warnings and notes

- the beam is 45 x 30 au (> 40 au). A typical disk (rings at tens of au) is not resolved, so the rings, asymmetries, temperatures and masses below are not meaningful. A finer observation is needed.
- the smooth disk collapsed to the unresolved core (characteristic radius 19 au < 3 beams): the extended emission is shared among the ring components, so overlapping neighbours (shoulders) have degenerate widths and amplitudes; trust the isolated rings, not the blended ones.
- the feature(s) at 153 au are shoulders (the raw-profile peak finder finds no peak or dip within 1 beam): an amplitude measured against a shoulder baseline can be large without a real azimuthal structure; the percentages of the local flux are the safer numbers.
- the notebook fell back from observation #1 to #9 because the finer packages exceeded the size limit.
- gap at 124 au and ring at 145 au are 21.2 au apart (< 1 beam): probably one blended feature or noise.
- ring at 145 au and gap at 152 au are 7.1 au apart (< 1 beam): probably one blended feature or noise.

Continuum image



38. XZ Tau

Barely resolved

Only barely resolved (90% of the flux within 3.3 beams of 8 au); overall size and trend are trustworthy, narrow rings are not. 0 ring(s) [au] and 1 gap(s) [30 au] detected at >= 3 sigma. A finer observation (0.0290 arcsec) exists but the package exceeded the download limit; finer structure may exist.

Target

Coordinates (J2000)	04:31:40.09 +18:13:56.6 (RA 67.91703, Dec +18.23240 deg, J2000)
Distance	140.0 pc
Class	Multiple system
Versus TW Hya	Binary with interacting disks

Observation used

Project / band	2016.1.01488.S / Band 6 (240.0 GHz)
Beam	0.0580 x 0.0306 arcsec = 8.1 x 4.3 au
Archive resolution	0.0334 arcsec (4.7 au
Largest scale	0.65 arcsec (91 au)
Exposure / release	2939 s (0.82 h) / 2018-10-17
Package	2016.1.01488.S_uid___A001_X879_X525_001_of_001.tar (0.01 GB)
FITS image	member.uid___A001_X879_X525.XZ_Tau_B_sci.spw0_1_2.cont.I.manual.pbcor.fits
Archive observations	6 total; finest 0.0290 arcsec; used #2 (fallback: finer packages too large)
Stellar parameters	Stellar parameters (XZ Tau B only (A: 0.31 Lsun, 0.29 Msun); arXiv:2106.11924 Table 5 (Hartigan & Kenyon 2003, Herczeg & Hillenbrand 2014), mass also Osorio+16; verify): L* = 0.17 Lsun, M* = 0.37 Msun

Geometry and radial profile

Inclination / PA	71.2 / 127.5 deg (UNRELIABLE: disk barely resolved)
Geometry source	FITTED FROM THE IMAGE, but the disk is barely resolved (90% of its flux within 4 beams), so the fitted inclination and PA are UNRELIABLE (the +-5 deg accuracy holds only for resolved disks). Give incl_deg / pa_deg from the literature.
Peak of mean profile	1.34 mJy/beam at r = 18.3 au
R50 / R90	22 / 26 au (2.8 / 3.3 beams)
Outer edge (>= 3 sigma)	39 au
Integrated flux	20.6 mJy (n/a of catalogue expectation)
Image noise / dyn. range	0.0215 mJy/beam / 8
Shape	the azimuthal mean peaks at 18 au (2.3 beams) and the innermost bin holds 3% of the peak. A peak this close to the centre, with a dip of this size, is at the resolution limit: it is consistent with a centrally peaked disk smoothed by the 8 au beam, and a cavity smaller than ~16 au cannot be established.

Substructure (baseline-subtracted profile)

type	radius	residual	modulation	significance
gap	r= 30.5 au	resid=-0.32 mJy/b	modul=-69.3%	signif= -45.2 sigma SIGNIFICANT

Sequence	G 31 au
Published features	none stored in the catalogue

Gaussian ring fit

Model / chi2_red tapered power law (inner hole at 9.2 +/- 0.9 au) + 2 Gaussian ring(s); chi2_red = 0.53

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed	
1	r=19.8+/-0.7	amp=1.3+/-0.25	S/N=5.3	FWHM_int=4.7 +/- 1.5	core	
2	r=24.4+/-5.4	amp=0.274+/-1.1	S/N=0.3	FWHM_int=10.7 +/- 5.6	-	<3sig

Brightness temperature and optical depth

type	radius	brightness T	dust T	ratio	optical depth	scale height
gap	r= 30.5 au	T_b= 5.6 K	T_d= 14.5 K	T_b/T_d= 0.39	tau= 0.49	H/r=0.069

Azimuthal asymmetry

No rings with a testable asymmetry.

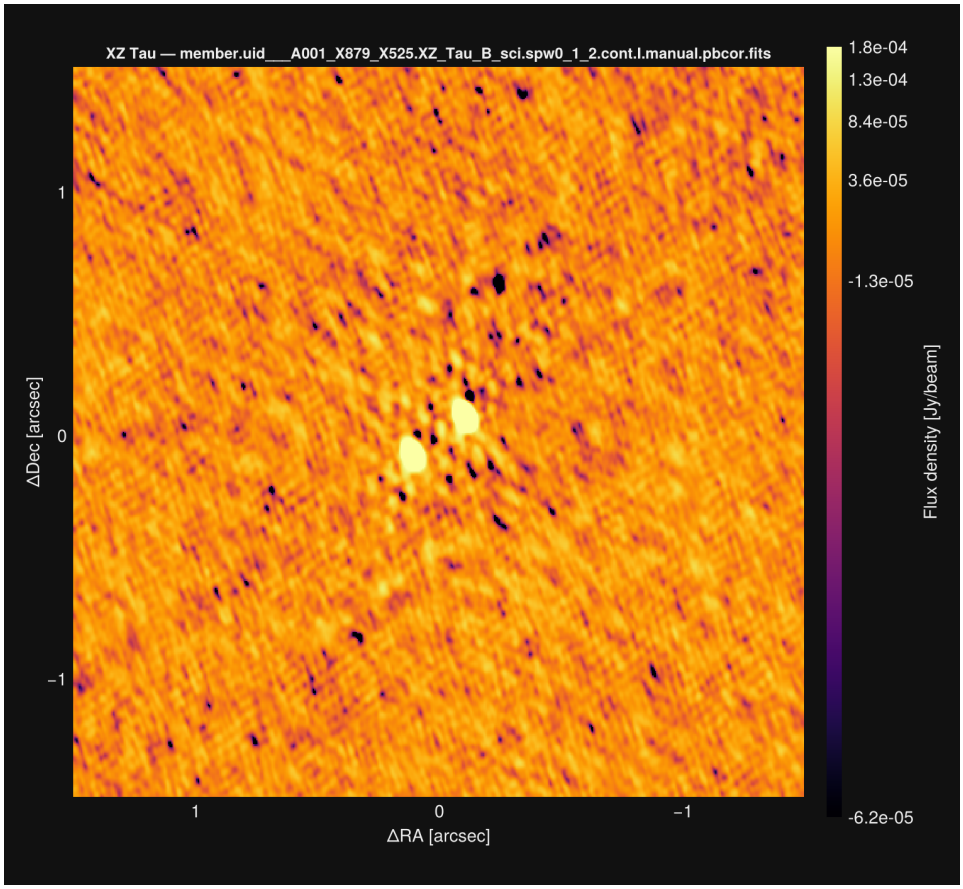
Gap-opening planet masses

Not evaluated: 31 au: no local maximum outward of the gap minimum at 30.5 au (the profile does not rise there)

Pipeline warnings and notes

- AZIMUTHAL SCATTER: in the bins that hold signal the scatter around the azimuthal mean is 1.5 times the mean itself (ordinary discs: 0.15-0.4). The emission is strongly non-axisymmetric (a horseshoe, arc or companion) or the inclination / PA is wrong, so the azimuthal mean is not a representative radial profile: the rings, gaps, fits and Fourier amplitudes derived from it are not reliable and their formal significance...
- the notebook fell back from observation #1 to #2 because the finer packages exceeded the size limit.

Continuum image



39. FT Tau

Barely resolved

Only barely resolved (90% of the flux within 2.9 beams of 16 au); overall size and trend are trustworthy, narrow rings are not. 0 ring(s) [au] and 1 gap(s) [54 au] detected at >= 3 sigma. Image holds 97% of the expected catalogue flux (consistent). Negative bowl present: missing short spacings bias sizes, contrasts and temperatures. A finer observation (0.0246 arcsec) exists but the package exceeded the download limit; finer structure may exist.

Target

Coordinates (J2000)	04:23:39.19 +24:56:14.3 (RA 65.91329, Dec +24.93729 deg, J2000)
Distance	127.0 pc
Class	T Tauri analogue
Versus TW Hya	Closest analogue: axisymmetric rings/gaps; Bosschaart+26: RD, R68 33 au
Catalogue	Bosschaart+26 (FTTau): class RD (ring disc); R68 33 au; F(1.33 mm) 90 mJy; M_dust 15 Earth masses; M* 0.31 Msun; distance 127 pc; this image: 68%-flux radius 35 au

Observation used

Project / band	2016.1.01164.S / Band 6 (225.5 GHz)
Beam	0.1293 x 0.1118 arcsec = 16.4 x 14.2 au
Archive resolution	0.1105 arcsec (14.0 au
Largest scale	1.66 arcsec (210 au)
Exposure / release	544 s (0.15 h) / 2019-02-02
Package	2016.1.01164.S_uid___A001_X885_X1f6_001_of_001.tar (0.06 GB)
FITS image	member.uid___A001_X885_X1f6.FT_Tau_sci.spw17_23_25_27_29.mfs.I.manual.pbcor.fits
Archive observations	7 total; finest 0.0246 arcsec; used #2 (fallback: finer packages too large)
Stellar parameters	Stellar parameters (Long+18 Table 2; verify): L* = 0.15 Lsun, M* = 0.34 Msun

Geometry and radial profile

Inclination / PA	35.5 / 121.8 deg
Geometry source	catalogue (literature) values.
Peak of mean profile	11 mJy/beam at r = 3.2 au
R50 / R90	29 / 48 au (1.7 / 2.9 beams)
Outer edge (>= 3 sigma)	67 au
Integrated flux	87.8 mJy (97% of catalogue expectation)
Image noise / dyn. range	0.0615 mJy/beam / 93
Shape	the emission is centrally peaked (the maximum lies at 3 au, within ~1.5 beams of the star), like TW Hya: a smooth, steeply falling profile on which any rings are shoulders and dips.

Substructure (baseline-subtracted profile)

type	radius	residual	modulation	significance
gap	r= 54.0 au	resid=-0.171 mJy/b	modul=-26.7%	signif= -10.2 sigma SIGNIFICANT

Sequence G 54 au

Gaussian ring fit

Model / chi2_red tapered power law + 1 Gaussian ring(s); chi2_red = 2.00

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed
1	r=36.3+/-1.0	amp=1.15+/-0.13	S/N=8.6	FWHM_int=10.9 +/- 3.7	core

Brightness temperature and optical depth

type	radius	brightness T	dust T	ratio	optical depth	scale height
gap	r= 54.0 au	T_b= 4.0 K	T_d= 10.5 K	T_b/T_d= 0.38	tau= 0.48	H/r=0.082

Azimuthal asymmetry

Not tested: disk spans too few beams (needs >= 4).

Gap-opening planet masses

Not evaluated: 54 au: no local maximum outward of the gap minimum at 60.3 au (the profile does not rise there)

Pipeline warnings and notes

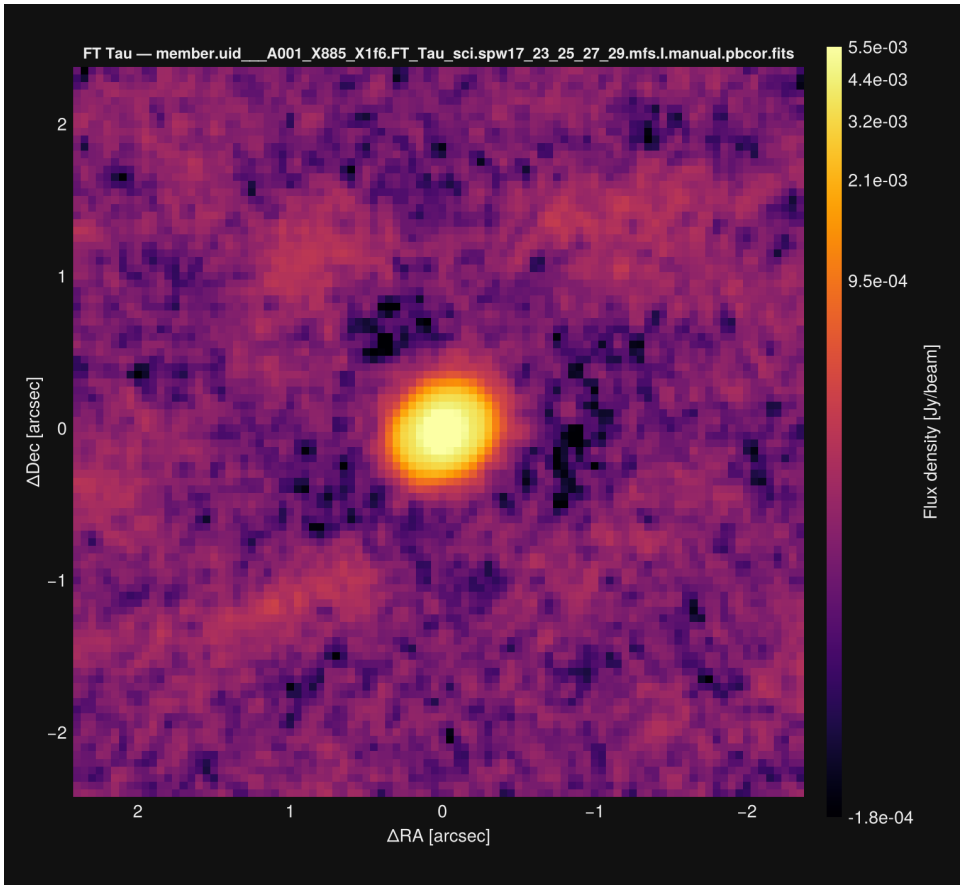
- the smooth disk collapsed to the unresolved core (characteristic radius 29 au < 3 beams): the extended emission is shared among the ring

components, so overlapping neighbours (shoulders) have degenerate widths and amplitudes; trust the isolated rings, not the blended ones.

- Negative bowl: the azimuthal mean is significantly negative over r = 92-137 au (8 bins, down to -3.9 sigma, -0.0727 mJy/beam = -0.7% of the peak, at 92 au), which real emission cannot produce. It is the signature of missing short spacings (spatial filtering): the extended emission is partly resolved out, so sizes, contrasts, asymmetries and brightness temperatures are biased and the smooth baseline is underestimated.

- the notebook fell back from observation #1 to #2 because the finer packages exceeded the size limit.

Continuum image



40. RX J1852.3-3700

Barely resolved

Only barely resolved (90% of the flux within 2.9 beams of 27 au); overall size and trend are trustworthy, narrow rings are not. 0 ring(s) [au] and 1 gap(s) [69 au] detected at >= 3 sigma. A finer observation (0.0908 arcsec) exists but the package exceeded the download limit; finer structure may exist.

Target

Coordinates (J2000)	18:52:17.30 -37:00:11.9 (RA 283.07209, Dec -37.00332 deg, J2000)
Distance	146.0 pc
Class	T Tauri + cavity
Versus TW Hya	Ring, not centrally peaked; gaps = cavity physics

Observation used

Project / band	2015.1.01083.S / Band 7 (343.5 GHz)
Beam	0.1835 x 0.1414 arcsec = 26.8 x 20.6 au
Archive resolution	0.1629 arcsec (23.8 au
Largest scale	3.94 arcsec (575 au)
Exposure / release	363 s (0.10 h) / 2019-04-16
Package	2015.1.01083.S_uid___A001_X340_X1d5_001_of_001.tar (0.02 GB)
FITS image	member.uid___A001_X340_X1d5.ari_l._RXJ1852.3-3700__sci.spw0_1_2_3_343512MHz.12m.cont.I.pbcor.fits
Archive observations	12 total; finest 0.0908 arcsec; used #5 (fallback: finer packages too large)
Stellar parameters	Stellar parameters: none stored for this target (pass lstar= / mstar=).

Geometry and radial profile

Inclination / PA	32.5 / 114.4 deg (UNRELIABLE: disk barely resolved)
Geometry source	FITTED FROM THE IMAGE, but the disk is barely resolved (90% of its flux within 4 beams), so the fitted inclination and PA are UNRELIABLE (the +-5 deg accuracy holds only for resolved disks). Give incl_deg / pa_deg from the literature.
Peak of mean profile	8.9 mJy/beam at r = 47.4 au
R50 / R90	55 / 77 au (2.0 / 2.9 beams)
Outer edge (>= 3 sigma)	91 au
Integrated flux	151.3 mJy (n/a of catalogue expectation)
Image noise / dyn. range	0.155 mJy/beam / 25
Shape	the azimuthal mean peaks at 47 au (1.8 beams) and the innermost bin holds 2% of the peak. A peak this close to the centre, with a dip of this size, is at the resolution limit: it is consistent with a centrally peaked disk smoothed by the 27 au beam, and a cavity smaller than ~54 au cannot be established.

Substructure (baseline-subtracted profile)

type	radius	residual		modulation	significance	
gap	r= 69.3 au	resid=-0.713	mJy/b	modul=-15.9%	signif= -10.8 sigma	SIGNIFICANT

Sequence	G 69 au
Published features	none stored in the catalogue

Gaussian ring fit

Model / chi2_red tapered power law + 1 Gaussian ring(s); chi2_red = 0.58

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed
1	r=51.3+/-0.2	amp=8.84+/-0.12	S/N=74.4	FWHM_int=23.0 +/- 0.6	core

Brightness temperature and optical depth

T_d, T_b/T_d and tau not computed: no stellar luminosity stored for this target.

Azimuthal asymmetry

Not tested: disk spans too few beams (needs >= 4).

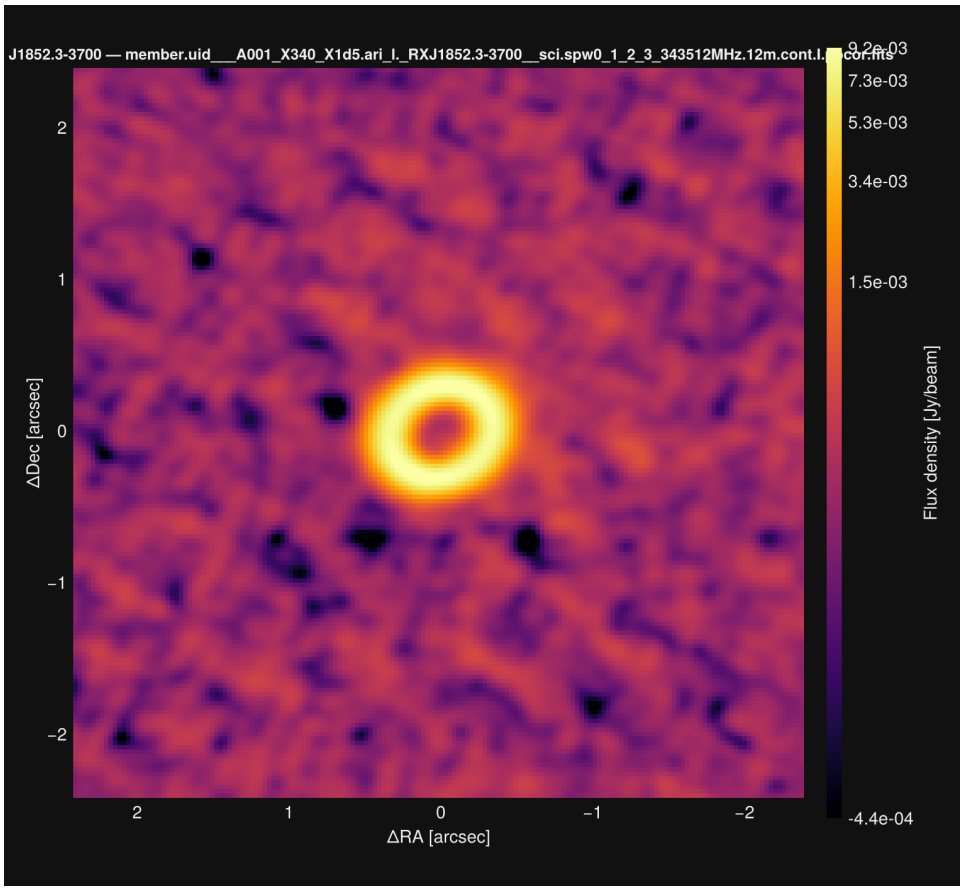
Gap-opening planet masses

Not available (needs L* and M*, which are not stored for this target; pass lstar= and mstar=).

Pipeline warnings and notes

- the notebook fell back from observation #1 to #5 because the finer packages exceeded the size limit.

Continuum image



41. EX Lup

Barely resolved

Only barely resolved (90% of the flux within 2.7 beams of 9 au); overall size and trend are trustworthy, narrow rings are not. 1 ring(s) [24 au] and 0 gap(s) [au] detected at >= 3 sigma. Image holds 21-32% of the expected catalogue flux. A finer observation (0.0390 arcsec) exists but the package exceeded the download limit; finer structure may exist.

Target

Coordinates (J2000)	16:03:05.49 -40:18:25.4 (RA 240.77288, Dec -40.30707 deg, J2000)
Distance	155.0 pc
Class	Outbursting
Versus TW Hya	EXor prototype; accretion-burst disk; Bosschaart+26: RD, R68 34 au
Catalogue	Bosschaart+26 (EXLup): class RD (ring disc); R68 34 au; F(1.33 mm) 20 mJy; M_dust 5.1 Earth masses; M* 0.53 Msun; distance 158 pc; this image: 68%-flux radius 25 au

Observation used

Project / band	2017.1.00388.S / Band 4 (153.0 GHz)
Beam	0.0589 x 0.0268 arcsec = 9.1 x 4.2 au
Archive resolution	0.0436 arcsec (6.8 au
Largest scale	0.80 arcsec (125 au)
Exposure / release	308 s (0.09 h) / 2019-02-21
Package	2017.1.00388.S_uid___A001_X1288_Xb74_001_of_001.tar (0.37 GB)
FITS image	member.uid___A001_X1288_Xb74.EX_Lup_sci.spw25_27_29_31.cont.I.manual.image.pbcor.fits
Archive observations	20 total; finest 0.0390 arcsec; used #2 (fallback: finer packages too large)
Stellar parameters	Stellar parameters (Bosschaart+26 catalogue stellar mass; verify): L* = ? Lsun, M* = 0.53 Msun

Geometry and radial profile

Inclination / PA	0.0 / 0.0 deg
Geometry source	the image fit gave i = 85.0 deg, PA = 91.9 deg but explains only 13% of the pixel variance (not credible), so the disk was treated as FACE-ON (geometry guard).
Peak of mean profile	0.217 mJy/beam at r = 2.3 au
R50 / R90	21 / 25 au (2.2 / 2.7 beams)
Outer edge (>= 3 sigma)	2 au
Integrated flux	2.0 mJy (21-32% of catalogue expectation)
Image noise / dyn. range	0.0499 mJy/beam / 15
Shape	the emission is centrally peaked (the maximum lies at 2 au, within ~1.5 beams of the star), like TW Hya: a smooth, steeply falling profile on which any rings are shoulders and dips.

Substructure (baseline-subtracted profile)

type	radius	residual	modulation	significance	
ring	r= 24.4 au	resid=0.0582 mJy/b	modul=+429.5%	signif= +4.7 sigma	SIGNIFICANT
gap	r= 46.8 au	resid=-0.0102 mJy/b	modul=n/a	signif= -2.1 sigma	noise-level
ring	r= 53.1 au	resid=0.00815 mJy/b	modul=n/a	signif= +1.0 sigma	noise-level
gap	r= 61.0 au	resid=-0.00527 mJy/b	modul=n/a	signif= -1.4 sigma	noise-level
ring	r= 66.9 au	resid=0.00563 mJy/b	modul=n/a	signif= +0.7 sigma	noise-level
gap	r= 78.3 au	resid=-0.0055 mJy/b	modul=n/a	signif= -2.0 sigma	noise-level
ring	r= 85.3 au	resid=0.00837 mJy/b	modul=n/a	signif= +1.3 sigma	noise-level
gap	r= 97.7 au	resid=-0.00475 mJy/b	modul=n/a	signif= -1.5 sigma	noise-level

Sequence	R 24 au -> G 47 au -> R 53 au -> G 61 au -> R 67 au -> G 78 au -> R 85 au -> G 98 au
Published features	none stored in the catalogue

Gaussian ring fit

Model / chi2_red tapered power law + 1 Gaussian ring(s); chi2_red = 0.31

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed
1	r=24.3+/-0.8	amp=0.0707+/-0.015	S/N=4.6	FWHM_int=5.6 +/- 3.0	det

Brightness temperature and optical depth

T_d, T_b/T_d and tau not computed: no stellar luminosity stored for this target.

Azimuthal asymmetry

ring radius	m=1 amplitude	m=2 amplitude
r= 24.3 au	m=1: 2.5% (S/N 0.1)	m=2: 21.1% (S/N 1.1)

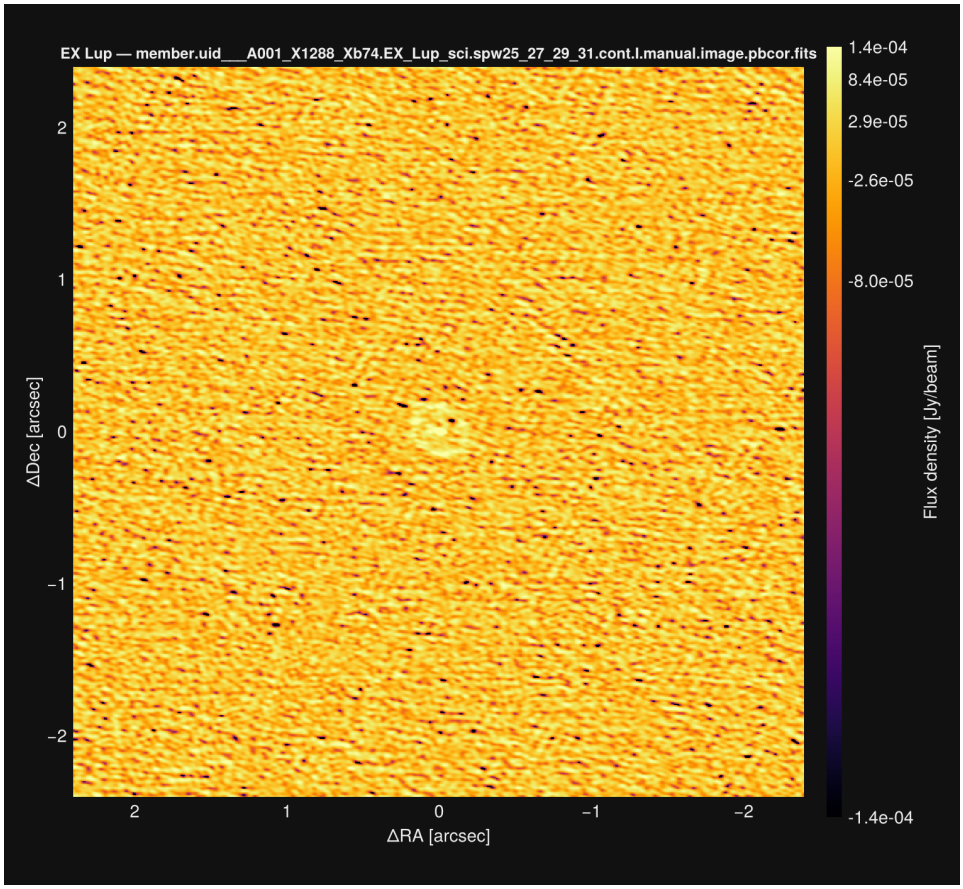
Gap-opening planet masses

Not available (needs L* and M*, which are not stored for this target; pass lstar= and mstar=).

Pipeline warnings and notes

- the smooth disk collapsed to the unresolved core (characteristic radius 9 au < 3 beams): the extended emission is shared among the ring components, so overlapping neighbours (shoulders) have degenerate widths and amplitudes; trust the isolated rings, not the blended ones.
- at 24 au the local azimuthal-mean flux is below 2x the image rms per beam (0.0499 mJy/beam): the ring is detected only by averaging over many beams, and the S/N and false-alarm values above assume white noise at the beam scale. Correlated residuals (sidelobes, large-scale noise, weak background sources) are not in that noise and can make a dipole or m=2 pattern of this size, so these asymmetries are unconfirmed; an '...
- the feature(s) at 24 au are shoulders (the raw-profile peak finder finds no peak or dip within 1 beam): an amplitude measured against a shoulder baseline can be large without a real azimuthal structure; the percentages of the local flux are the safer numbers.
- the image is spatially filtered (flux check above): arc contrasts and asymmetry amplitudes describe only the emission that survived the missing short spacings.
- the notebook fell back from observation #1 to #2 because the finer packages exceeded the size limit.
- gap at 47 au and ring at 53 au are 6.4 au apart (< 1 beam): probably one blended feature or noise.
- ring at 53 au and gap at 61 au are 7.8 au apart (< 1 beam): probably one blended feature or noise.
- gap at 61 au and ring at 67 au are 5.9 au apart (< 1 beam): probably one blended feature or noise.
- gap at 78 au and ring at 85 au are 7.0 au apart (< 1 beam): probably one blended feature or noise.

Continuum image



42. WL 17

Barely resolved

Only barely resolved (90% of the flux within 2.1 beams of 15 au); overall size and trend are trustworthy, narrow rings are not. 1 ring(s) [51 au] and 0 gap(s) [au] detected at >= 3 sigma. A finer observation (0.0310 arcsec) exists but the package exceeded the download limit; finer structure may exist.

Target

Coordinates (J2000)	16:27:06.78 -24:38:15.0 (RA 246.77823, Dec -24.63750 deg, J2000)
Distance	137.0 pc
Class	Young/embedded
Versus TW Hya	Class I; ring + cavity, embedded

Observation used

Project / band	2015.1.00761.S / Band 7 (349.6 GHz)
Beam	0.1087 x 0.1032 arcsec = 14.9 x 14.1 au
Archive resolution	0.1055 arcsec (14.4 au
Largest scale	2.06 arcsec (282 au)
Exposure / release	60 s (0.02 h) / 2017-10-21
Package	2015.1.00761.S_uid___A001_X2d8_X3dc_001_of_001.tar (12.74 GB)
FITS image	member.uid___A001_X2d8_X3dc.ari_l.WL_17_sci.spw0_1_2_3_4_349625MHz.12m.cont.I.pbcor.fits
Archive observations	6 total; finest 0.0310 arcsec; used #3 (fallback: finer packages too large)
Stellar parameters	Stellar parameters: none stored for this target (pass lstar= / mstar=).

Geometry and radial profile

Inclination / PA	38.8 / 63.8 deg (UNRELIABLE: disk barely resolved)
Geometry source	FITTED FROM THE IMAGE, but the disk is barely resolved (90% of its flux within 4 beams), so the fitted inclination and PA are UNRELIABLE (the +-5 deg accuracy holds only for resolved disks). Give incl_deg / pa_deg from the literature.
Peak of mean profile	17.9 mJy/beam at r = 10.3 au
R50 / R90	17 / 31 au (1.2 / 2.1 beams)
Outer edge (>= 3 sigma)	38 au
Integrated flux	132.8 mJy (n/a of catalogue expectation)
Image noise / dyn. range	0.513 mJy/beam / 25
Shape	the emission is centrally peaked (the maximum lies at 10 au, within ~1.5 beams of the star), like TW Hya: a smooth, steeply falling profile on which any rings are shoulders and dips.

Substructure (baseline-subtracted profile)

type	radius	residual		modulation	significance	
ring	r= 51.4 au	resid=0.56	mJy/b	modul=n/a	signif= +3.0 sigma	SIGNIFICANT

Sequence	R 51 au
Published features	none stored in the catalogue

Gaussian ring fit

Model / chi2_red tapered power law + 1 Gaussian ring(s); chi2_red = 0.23

#	radius [au]	amplitude [mJy/beam]	S/N	intrinsic FWHM [au]	seed
1	r=17.0+/-2.3	amp=5.43+/-14	S/N=0.4	FWHM_int=< 21.8 (unresolved)	core <3sig

Brightness temperature and optical depth

T_d, T_b/T_d and tau not computed: no stellar luminosity stored for this target.

Azimuthal asymmetry

Not tested: disk spans too few beams (needs >= 4).

Gap-opening planet masses

Not available (needs L* and M*, which are not stored for this target; pass lstar= and mstar=).

Pipeline warnings and notes

- the smooth disk collapsed to the unresolved core (characteristic radius 23 au < 3 beams): the extended emission is shared among the ring components, so overlapping neighbours (shoulders) have degenerate widths and amplitudes; trust the isolated rings, not the blended ones.

- the notebook fell back from observation #1 to #3 because the finer packages exceeded the size limit.

Continuum image